Notes on Complex Analysis

Slipper King

May 15, 2025

Source: `https://github.com/slipperking/complex-analysis-latex`

Contents

1 Prerequisites

1.1 Topological Preliminaries

The following definitions are subject to the assumption where the topological space is defined to be $X = \mathbb{C}^n$. This is satisfactory to the main purpose of our proceeding passage, but it is noteworthy that it can be generalized to more abstract sets.

Definition 1.1.1 *(Accumulation Point)*: A point $z \in \mathbb{C}^n$ is an *accumulation point* of X if for any open set U containing z, $(U \setminus \{z\}) \cap X \neq \emptyset$

Definition 1.1.2 *(Closure)*: For a set $X \in \mathbb{C}^n$, define the *closure* of X, or $\overline{X}$ to be the intersection of all closed sets containing X. In other words, it is the union of X and its accumulation points.

Definition 1.1.3 *(Interior)*: For a set $X \in \mathbb{C}^n$, the *interior* of X, denoted $\overset{\circ}{X}$, is the union of all open sets contained in X, or the set of points $z \in \mathbb{C}^n$ such that there exists an open neighborhood of z that is fully contained in X.

Definition 1.1.4 *(Compact Set)*: A set $X \in \mathbb{C}^n$ is compact iff X is closed and bounded.

Definition 1.1.5 *(Set Covering)*: A cover $\mathcal{C}$ of a set X is a collection of sets $\{U_n\}$ such that

$$\bigcup_{n \in \mathbb{N}} U_n \subseteq X.$$

A cover is *open* if every set in the collection is open.

Theorem 1.1.1 *(Bolzano–Weierstrass)*: Every infinite subset A of a compact set $X \subset \mathbb{C}^n$ has an accumulation point in X.

Proof: Since X is bounded, there exists a closed cube $Q \subset \mathbb{C}^n$ such that $A \subseteq X \subset Q$.

Bisect $Q_0 = Q$ into 2^{2n} congruent sub-cubes. Since A is infinite and the sub-cubes are finite in number, at least one of the sub-cubes contains infinitely many points of A, and choose one to be Q_1.

Bisect Q_1 into 2^{2n} sub-cubes, and choose a sub-cube $Q_2 \subset Q_1$ that contains infinitely many points of A. We then obtain the recursive sequence

$$Q_0 \supset Q_1 \supset Q_2 \supset \cdots$$

Because the side lengths shrink to zero and the cubes are nested, the intersection

$$\bigcap_{k=0}^{\infty} Q_k$$

consists of exactly one point. Call this point $z_\infty \in \mathbb{C}^n$.

For each k, Q_k contains infinitely many points of A. Because the side length of Q_k tends to zero, for any $\varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall k \geq N$, $Q_k \subset B^n(z_\infty, \varepsilon)$ where $B^n(a, r) \subset \mathbb{C}^n$ is the n-dimensional *ball* with radius r centered at $a = (a_1, a_2, ..., a_n) \in \mathbb{C}^n$, or

$$B^n(a, r) = \left\{ (z_1, z_2, ..., z_n) \in \mathbb{C}^n \mid \sum_{j=1}^{n} \left|z_j - a_j\right|^2 < r^2 \right\}.$$

Then, $B^n(z_\infty, \varepsilon)$ also contains infinitely many points of A. Therefore, z_∞ is an accumulation point of A.

We now show that $z_\infty \in X$. Suppose for contradiction that $z_\infty \notin X$. Since X is closed, $\mathbb{C}^n \setminus X$ is open, and $\exists \delta > 0$ such that

$$B^n(z_\infty, \delta) \subset \mathbb{C}^n \setminus X.$$

But then, for sufficiently large k, we have $Q_k \subset B^n(z_\infty, \delta)$, and hence $Q_k \cap X = \emptyset$. This contradicts the construction of Q_k, which ensures that Q_k contains infinitely many points of $A \subset X$. □

Theorem 1.1.2 *(Heine–Borel)*: A set $X \in \mathbb{C}^n$ is compact iff every open cover has a finite subcover.

Proof: We will first show that any set satisfying the condition is compact.

First we will show that X is bounded. Suppose that $\forall R > 0$, $\exists z \in X$ where $\|z\| > R$. Consider the collection of open sets

$$\mathcal{U} = \{B^n(0, k) \mid k \in \mathbb{N}\}.$$

$\mathcal{U}$ forms an open cover of X. Then by the assumption, there exists a finite subcover in $\mathcal{U}$, namely $\{B^n(0, k_1), ..., B^n(0, k_m)\}$ which covers X. Then,

$$X \subseteq \bigcup_{i=1}^{m} B^n(0, k_i) = B^n(0, \max\{k_1, ..., k_m\}).$$

By contradiction, X must be bounded.

X must also be a closed set. For the sake of contradiction, assume that there exists a point $z_0 \in \overline{X} \setminus X$. Since $z_0 \notin X$, the following open collection of sets covers X:

$$\mathcal{U} = \left\{ \mathbb{C}^n \setminus \overline{B^n\left(z_0, \frac{1}{k}\right)} \mid \forall k \in \mathbb{N} \right\}.$$

There then exists a finite subcover $\mathcal{C} = \left\{ \mathbb{C}^n \setminus \overline{B^n\left(z_0, \frac{1}{k_j}\right)} \mid j \in \mathbb{N}_{\leq m} \right\}$. Then,

$$X \subseteq \mathbb{C}^n \setminus \overline{B^n\left(z_0, \frac{1}{\max\{k_1, ..., k_m\}}\right)},$$

and that $X \cap \overline{B^n\left(z_0, \frac{1}{\max\{k_1,...,k_m\}}\right)} = \emptyset$. However, by the definition of the accumulation point, every open neighborhood of the accumulation point must intersect X. Therefore, by contradiction, X is closed.

We then prove the converse. By the assumption that X is bounded, $\exists R > 0$ such that X is contained within the closed cube

$$Q = \left\{ z \in \mathbb{C}^n \mid \max_{j \in \mathbb{N}_{\leq n}} \left|\Re e\left(z_j\right)\right| \leq R \wedge \max_{j \in \mathbb{N}_{\leq n}} \left|\Im m\left(z_j\right)\right| \leq R \right\}.$$

Assume that there exists an infinite open cover $\mathcal{U}$ of X without finite subcovering. Bisect $Q_0 = Q$ into 2^{2n} sub-cubes (for real and complex parts). Choose Q_1 such that $Q_1 \cup X$ has no finite subcover of $\mathcal{U}$. Under the previous assumptions, this is possible since if every sub-cube $\cap X$ had finite subcovering, then $Q_0 \cap X = X$ would have finite subcovering. Similarly, choose Q_2 by bisecting Q_1 similarly, and recursively obtain a sequence of cubes:

$$Q_0 \supset Q_1 \supset Q_2 \supset \cdots$$

Since the side length of each cube tends to 0, $\bigcap_{j=0}^{\infty} Q_j$ consists of a single point $z_\infty \in \mathbb{C}^n$. By the Bolzano–Weierstrass Theorem (Theorem 1.1.1), because $\forall j \in \mathbb{N}, Q_j \cap X \neq \emptyset$, select a point $z_j \in Q_j \cap X$, forming a sequence $\{z_k\} \in X$ convergent to $z_\infty \in X$ as X is closed. Therefore, $\exists U \in \mathcal{U}$ where $z_\infty \in U$. Since U is open, $\exists \varepsilon > 0$ such that $B^n(z_\infty, \varepsilon) \subseteq U$. $\exists N \in \mathbb{N}$ such that $\forall k > N$, $Q_k \subset B^n(z_\infty, \varepsilon)$. Then taking the intersection with X on both sides,

$$Q_k \cap X \subseteq B^n(z_\infty, \varepsilon) \cap X \subseteq U.$$

This contradicts the assumption that for every k, $Q_k \cap X$ has no finite subcovering, since U clearly covers $Q_k \cap X$, which is a single open set that covers a nonempty subset. Therefore by contradiction, every open cover has finite subcovering. □

Definition 1.1.6 *(Support of a Function)*: For a set X and a function $f : X \to \mathbb{C}$, the *support*, denoted by $\operatorname{supp}(f) = \overline{\{z \in X \mid f(z) \neq 0\}}$, is the closure of the set for which f is nonzero.

Remark: A notable classification of functions comes from the compactness of support—more specifically, its boundedness. Compactly supported functions in C^∞ are commonly referred to as *bump functions* (see @ sec:partitionsofunity).

1.2 Calculus

Since traditional complex analysis is the theory of calculus on complex functions, it is only natural that generalizations are made on classical formulas in calculus for complex functions.

It is well known that a function $f : (a, b) \to \mathbb{R}$ is differentiable at a point $x \in (a, b)$ if the limit

$$\lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

exists, and the value of this limit is the derivative of $f(x)$, denoted by $f'(x)$ or $\frac{\mathrm{d}f}{\mathrm{d}x}$. The value $\mathrm{d}f = f'(x)\,\mathrm{d}x$ is the differential of $f(x)$. Partition $[a, b]$ into $a = x_0 < x_1 < x_2 < \cdots < x_n = b$ such that the length of the intervals $[x_i, x_{i-1}]$ vanishes (we let the norm of the partition, or the size of the largest interval, tend to zero) as $n \to \infty$. If for any such partition, the sum

$$\sum_{i=1}^{n} f(\xi_i)(x_i - x_{i-1})$$

tends to the same value $\forall \xi_i \in [x_{i-1}, x_i]$ (as the length of the largest partition approaches 0), then the function can be roughly said to be integrable over $[a, b]$. The full details of Riemann integrability are simplified by the use of Darboux sums and will not be discussed here. The value of this sum is denoted by

$$\int_a^b f(x)\,\mathrm{d}x.$$

We will attempt to avoid notions involving Lebesgue integration. However, it is important to note that every Riemann integrable function is also Lebesgue integrable, and the two integrals are equal. Therefore, we will use Lebesgue integral theorems (where the resultant integral is Riemann integrable) when necessary without further mention of the Lebesgue integral itself.

The following theorems are the fundamental results of classical calculus:

Theorem 1.2.1 *(Fundamental Theorem of Calculus, Differential Form)*: Let $f(x)$ be a function continuous over $[a, b]$. For $x \in [a, b]$, define

$$\Phi(x) = \int_a^x f(t)\,\mathrm{d}t.$$

Then $\Phi(x)$ is differentiable over $[a, b]$, $\Phi'(x) = f(x)$, and $\mathrm{d}\Phi(x) = f(x)\,\mathrm{d}x$.

Theorem 1.2.2 *(Fundamental Theorem of Calculus, Integral Form)*: Let $\Phi(x)$ be a function differentiable over $[a, b]$. Let $f(x) = \Phi'(x)$ over $[a, b]$. Then,

$$\int_a^x f(t)\,\mathrm{d}t = \Phi(x) - \Phi(a).$$

The two forms of the theorem show that differentiation and integration are inverse operations to each other. Operations performed for differentiating oftentimes have a corresponding inverse operation that can be done for integrating. For instance,

$$\frac{\mathrm{d}f}{\mathrm{d}x} \pm \frac{\mathrm{d}g}{\mathrm{d}x}(f(x) \pm g(x)) = \frac{\mathrm{d}f}{\mathrm{d}x} \pm \frac{\mathrm{d}g}{\mathrm{d}x}$$

corresponds to

$$\int (f(x) \pm g(x))\,\mathrm{d}x = \int f(x)\,\mathrm{d}x \pm \int g(x)\,\mathrm{d}x,$$

and

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

corresponds to

$$\int f(x)g'(x)\,\mathrm{d}x = f(x)g(x) - \int f'(x)g(x)\,\mathrm{d}x,$$

and

$$\frac{\mathrm{d}f(g(x))}{\mathrm{d}x} = \frac{\mathrm{d}f(g)}{\mathrm{d}g} \cdot \frac{\mathrm{d}g}{\mathrm{d}x}$$

corresponds to

$$\int_a^b f(g(x))g'(x)\,\mathrm{d}x = \int_{g(a)}^{g(b)} f(u)\,\mathrm{d}u.$$

Another correspondence is the Mean Value Theorem:

Theorem 1.2.3 *(Mean Value Theorem, Differential Form)*: If $f(x)$ is differentiable over $[a, b]$, then $\exists c \in [a, b]$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

Theorem 1.2.4 *(Mean Value Theorem, Integral Form)*: If $f(x)$ is continuous over $[a, b]$, then $\exists \xi \in [a, b]$ such that

$$\int_a^b f(x)\,\mathrm{d}x = f(\xi)(b - a).$$

A curve is a one-dimensional manifold embedded within a higher dimensional space. They can be parameterized with a vector $\mathbf{F}(t) = (P(t), Q(t), R(t))$ of one parameter. In the complex plane, a curve is a complex-valued function $\gamma(t)$ for a real parameter $\alpha \leq t \leq \beta$. A curve is *closed* if $\gamma(\alpha) = \gamma(\beta)$. It is *smooth* if it is continuously differentiable, and its direction is defined to be the direction as t increases. If it is smooth everywhere except at a finite number of points, it is *piecewise smooth*. If it is of finite length, then the curve is said to be *rectifiable*. Piecewise smooth curves are rectifiable. A curve is *simple* if it is simple (non-self-intersecting), or if $\gamma(t_1) = \gamma(t_2)$ implies that $t_1 = t_2$. A simple closed curve is also called a *Jordan curve*.

Theorem 1.2.5 *(Jordan Curve Theorem)*: Let γ be a Jordan curve in $\mathbb{R}^2$. Then the set $\mathbb{R}^2 \setminus \gamma$ consists of exactly two connected subsets. One of them is the interior, denoted by $\mathrm{int}(\gamma)$, and is a bounded set, while the other is the exterior, denoted by $\mathrm{ext}(\gamma)$, which is unbounded. Both of the two sets share the common boundary γ.

The theorem above seems trivial, but its rigorous proof in topology is extremely complex. The theorem itself can also be stated on $\mathbb{C}$ instead of $\mathbb{R}^2$. For a region U, the boundary is denoted ∂U. If the region bounded by any closed curve in U also lies in U, then it is a *simply connected* region. A connected region that is not simply connected is multiply connected. A region bound by 2 non-intersecting Jordan curves is doubly connected, and a region bound by n non-intersecting Jordan curves is traditionally known as n-connected. Lastly, any closed curve can degenerate to a single point or slit.

Generalizations of the differential and integral exist for multivariate functions. The partial differentials of $f(x, y, z)$, $\frac{\partial f}{\partial x}\,\mathrm{d}x$, $\frac{\partial f}{\partial y}\,\mathrm{d}y$, and $\frac{\partial f}{\partial z}\,\mathrm{d}z$ sum up to form the total differential, denoted by $\mathrm{d}f$. An important result in multivariable calculus allows the calculation of the derivatives of a definite integral with respect to its parameter.

Theorem 1.2.6 *(Leibniz Integral Rule)*: Let $f(x, u)$ be continuous on $a \leq x \leq b$, $c \leq u \leq d$, and suppose $a \leq \alpha(u), \beta(u) \leq b$ are differentiable functions of $c \leq u \leq d$. If f is continuously differentiable with respect to u, then

$$\frac{\mathrm{d}}{\mathrm{d}u}\left(\int_{\alpha(u)}^{\beta(u)} f(x, u)\,\mathrm{d}x\right) = \int_{\alpha(u)}^{\beta(u)} \frac{\partial f}{\partial u}(x, u)\,\mathrm{d}x + \frac{\mathrm{d}\beta}{\mathrm{d}u} f(\beta(u), u) - \frac{\mathrm{d}\alpha}{\mathrm{d}u} f(\alpha(u), u).$$

Four main classical theorems exist, relating a function and its line integral in 2 and 3 dimensions, line and surface integrals in 2 and 3 dimensions, and the surface and volume integrals in 3 dimensions:

Theorem 1.2.7 *(Gradient Theorem)*: Let C be an oriented smooth curve in $\mathbb{R}^3$ with boundary points A to B. Then

$$f|_{\partial C} = f(B) - f(A) = \int_C \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y + \frac{\partial f}{\partial z}\,\mathrm{d}z.$$

Theorem 1.2.8 *(Green's Theorem)*: Let U be a positively oriented, multiply connected subset of $\mathbb{R}^2$ with a piecewise smooth oriented boundary ∂U. Suppose that $P(x, y), Q(x, y) \in C^1\left(\overline{U}\right)$. Then,

$$\oint_{\partial U} P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_U \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}x\,\mathrm{d}y.$$

Theorem 1.2.9 *(Stokes' Theorem)*: Suppose that $S \subset \mathbb{R}^3$ is a positively oriented surface with a positively oriented, piecewise smooth boundary curve ∂S. Suppose that $P(x, y, z), Q(x, y, z), R(x, y, z) \in C^1\left(\overline{S}\right)$. Then,

$$\oint_{\partial S} P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z$$
$$= \iint_S \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathrm{d}y\,\mathrm{d}z + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathrm{d}z\,\mathrm{d}x + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}x\,\mathrm{d}y.$$

Theorem 1.2.10 *(Gauss' Theorem)*: Suppose that $V \subset \mathbb{R}^3$ is a positively oriented region with a positively oriented, piecewise smooth boundary surface ∂V. Suppose that $P(x, y, z), Q(x, y, z), R(x, y, z) \in C^1\left(\overline{V}\right)$. Then,

$$\oint\!\!\oint_{\partial V} P\,\mathrm{d}y\,\mathrm{d}z + Q\,\mathrm{d}z\,\mathrm{d}x + R\,\mathrm{d}x\,\mathrm{d}y = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.$$

In 3-dimensional $\mathbb{R}^3$ space, define a scalar valued function to be a 0-form, a linear combination of $\mathrm{d}x$, $\mathrm{d}y$, and $\mathrm{d}z$ to be a 1-form, and a linear combination of $\mathrm{d}y \wedge \mathrm{d}z$, $\mathrm{d}z \wedge \mathrm{d}x$, and $\mathrm{d}x \wedge \mathrm{d}y$ to be a 2-form, and $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$ to be a 3-form, where $\wedge$ denotes an anti-commutative and associative product, where for any two differential forms ω_1 and ω_2

$$\omega_1 \wedge \omega_2 = -\omega_2 \wedge \omega_1.$$

Then consequently, for any differential form ω,

$$\omega \wedge \omega = 0.$$

We can generalize the operator d to increase the degree of a differential form. For instance,

$$\mathrm{d}f = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y + \frac{\partial f}{\partial z}\,\mathrm{d}z,$$

which is the definition of the total differential. For a 1-form in 3-dimensional space, $\omega_1 = P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z$, we can define the exterior derivative in a similar way:

$$\begin{aligned}
\mathrm{d}\omega_1 &= \mathrm{d}P \wedge \mathrm{d}x + \mathrm{d}Q \wedge \mathrm{d}y + \mathrm{d}R \wedge \mathrm{d}z \\
&= \left(\frac{\partial P}{\partial x}\,\mathrm{d}x + \frac{\partial P}{\partial y}\,\mathrm{d}y + \frac{\partial P}{\partial z}\,\mathrm{d}z\right) \wedge \mathrm{d}x \\
&\quad + \left(\frac{\partial Q}{\partial x}\,\mathrm{d}x + \frac{\partial Q}{\partial y}\,\mathrm{d}y + \frac{\partial Q}{\partial z}\,\mathrm{d}z\right) \wedge \mathrm{d}y \\
&\quad + \left(\frac{\partial R}{\partial x}\,\mathrm{d}x + \frac{\partial R}{\partial y}\,\mathrm{d}y + \frac{\partial R}{\partial z}\,\mathrm{d}z\right) \wedge \mathrm{d}z \\
&= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathrm{d}y \wedge \mathrm{d}z + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathrm{d}z \wedge \mathrm{d}x + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}x \wedge \mathrm{d}y.
\end{aligned}$$

Similarly, we can differentiate a 2-form $\omega = P\,\mathrm{d}y \wedge \mathrm{d}z + Q\,\mathrm{d}z \wedge \mathrm{d}x + R\,\mathrm{d}x \wedge \mathrm{d}y$ to get:

$$\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right)\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z.$$

The two results above resemble the curl and divergence of (P, Q, R). A differential form ω is *closed* if $\mathrm{d}\omega = 0$, and is *exact* if there exists η such that $\omega = \mathrm{d}\eta$.

Lemma 1.2.1 *(Poincaré)*: For any differential form ω on an open, contractible set $U \subseteq \mathbb{R}^n$, if ω is closed, then it is also exact.

It is true that for any set $U \subseteq \mathbb{R}^n$, regardless of contractibility, that for a differential form ω defined on U, $\mathrm{d}(\mathrm{d}\omega) = 0$. In other words, all exact differential forms are closed. (For a region U, we have $\partial\partial U = \emptyset$. This is one of many reasons for which the boundary operator is denoted by ∂, in analogy to d.)

The implications of this are important: if ω is a 0-form, then $\boldsymbol{\nabla} \times (\boldsymbol{\nabla}\omega) = 0$, and if ω is a 1-form, $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{v}) = 0$, where $\mathbf{v}$ is the vector of the coefficients of the basis differential forms of ω (there are no correlations for higher degree forms since in 3-dimensional space, the highest degree possible for any differential form is 3).

Then, the Fundamental Theorem of Calculus, the Gradient Theorem, Green's, Stokes', and Gauss' Theorems can be generalized into:

Theorem 1.2.11 *(Stokes–Cartan)*: For an oriented smooth n-dimensional compact manifold M with boundary ∂M, for a smooth differential $(n-1)$-form ω over $\overline{M}$,

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

Real analysis is the subject dedicated to rigorously defining concepts such as limits, continuity, integrability, convergence, etc. The most widely used definition of a finite limit of a function is the language of ε–δ, which states:

Definition 1.2.1 *(Epsilon–Delta)*: Let $f : U \to \mathbb{R}$ be a function defined over an open set $U \subseteq \mathbb{R}$ such that a is an accumulation point of U. We say that $\lim_{x \to a} f(x) = L$ if $\forall \varepsilon > 0, \exists \delta > 0$ such that for all $x \in U$ with $0 < |x - a| < \delta$, we have $|f(x) - L| < \varepsilon$.

Similarly, we define the *right-handed limit* $\lim_{x\to a^+} f(x) = L$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in U$ with $0 < x - a < \delta$, we have $|f(x) - L| < \varepsilon$.

Likewise, the *left-hand limit* $\lim_{x\to a^-} f(x) = L$ exists if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in U$ with $-\delta < x - a < 0$, we have $|f(x) - L| < \varepsilon$.

We also have the definition of the limit of a sequence:

Definition 1.2.2 *(Epsilon–N)*: Let $\{a_n\}_{n\in\mathbb{N}} \subset \mathbb{R}$ be a sequence. If $\exists a_\infty \in \mathbb{R}$ such that $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n > N$, $|a_n - a_\infty| < \varepsilon$, then $\{a_n\}$ *converges* to a_∞.

Theorem 1.2.12 *(Cauchy Criterion)*: Let $\{a_n\}_{n\in\mathbb{N}} \subset \mathbb{R}$ be a sequence. Then $\{a_n\}$ is convergent iff $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n, m > N$, $|a_n - a_m| < \varepsilon$.

Proof: Assume $\{a_n\}$ is convergent. Then $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n, m > N$, $|a_n - a_\infty| < \frac{\varepsilon}{2}$ and $|a_m - a_\infty| < \frac{\varepsilon}{2}$ for some $a_\infty \in \mathbb{R}$. It follows that

$$|a_n - a_m| \leq |a_n - a_\infty| + |a_m - a_\infty| = \varepsilon.$$

Conversely, $\{a_n\}$ is bounded (fixing N, $\forall n > N$, $|a_n - a_{N+1}| < \varepsilon$). By the Bolzano–Weierstrass Theorem (Theorem 1.1.1), $\{a_n\}_{n\in\mathbb{N}}$ has a subsequence $\left\{a_{n_k}\right\}_{k\in\mathbb{N}}$ that converges to a_∞. Therefore, $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ and $\exists M \in \mathbb{N}$ such that $\forall k > M$, $n_k > N$, and $\forall n > N$, $\left|a_n - a_{n_k}\right| < \frac{\varepsilon}{2}$ and $\left|a_{n_k} - a_\infty\right| < \frac{\varepsilon}{2}$. Then

$$|a_n - a_\infty| \leq \left|a_n - a_{n_k}\right| + \left|a_{n_k} - a_\infty\right| < \varepsilon.$$

Hence, $\{a_n\}$ converges to a_∞. □

Definition 1.2.3 *(Limit Superior)*: For a number sequence $\{a_n\} \subset \mathbb{R}$, if $\exists a \in \mathbb{R}$ such that:
1. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n > N$, $a_n < a + \varepsilon$,
2. $\forall \varepsilon > 0$, $\forall N \in \mathbb{N}$, $\exists n > N$ such that $a_n > a - \varepsilon$,

then the *superior limit* of $\{a_n\}$ is a, denoted by $\limsup_{n\to\infty} a_n = a$.

Definition 1.2.4 *(Limit Inferior)*: For a number sequence $\{a_n\} \subset \mathbb{R}$, if $\exists a \in \mathbb{R}$ such that:
1. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n > N$, $a_n > a - \varepsilon$,
2. $\forall \varepsilon > 0$, $\forall N \in \mathbb{N}$, $\exists n > N$ such that $a_n < a + \varepsilon$,

then the *inferior limit* of $\{a_n\}$ is a, denoted by $\liminf_{n\to\infty} a_n = a$.

Lemma 1.2.2: A number sequence $\{a_n\}$ is convergent iff $\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n$.

Proof: We first prove that $a = \lim_{n\to\infty} a_n$ implies $\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n = a$. By Definition 1.2.2, $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n > N$,

$$|a_n - a| < \varepsilon \Longleftrightarrow a - \varepsilon < a_n < a + \varepsilon.$$

Then from Definition 1.2.3 and Definition 1.2.4, we have that $\limsup_{n\to\infty} a_n \geq a$ and $\liminf_{n\to\infty} a_n \leq a$. By the second conditions, we get $\limsup_{n\to\infty} a_n \leq a$ and $\liminf_{n\to\infty} a_n \geq a$. Therefore,

$$\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

For the converse, assume $\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n$. Since $\exists N_1 \in \mathbb{N}$ such that $\forall n > N_1$, $a_n < a + \varepsilon$. $\exists N_2 \in \mathbb{N}$ such that $\forall n > N_2$, $a_n > a - \varepsilon$. Then $\forall n > \max\{N_1, N_2\}$, $|a_n - a| < \varepsilon$, as expected.
□

Definition 1.2.5 *(Continuity)*: A function $f : U \to \mathbb{R}$, defined on an open set $U \subseteq \mathbb{R}$ containing a point $a \in U$, is said to be continuous at a iff

$$\lim_{x \to a} f(x) = f(a).$$

It is important to note that in the case of multiple *explicit* variables, a distinction is made between (separate) continuity (where there are two δ's on which variable varies, and does not guarantee a single δ for when both variables vary simultaneously) and *joint* continuity (where a single δ controls both variables at once). To illustrate this, let (x_0, y_0) be fixed. The former is commonly written as

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall |x - x_0| < \delta, |f(x, y_0) - f(x_0, y_0)| < \varepsilon$$

in conjunction with

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall |y - y_0| < \delta, |f(x_0, y) - f(x_0, y_0)| < \varepsilon,$$

whereas the latter is expressed as

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall |(x - x_0, y - y_0)| < \delta, |f(x, y) - f(x_0, y_0)| < \varepsilon.$$

Theorem 1.2.13: Any continuous function on a compact set K is bounded on K.

Proof: Suppose for the sake of contradiction that $f : U \to \mathbb{R}$ is continuous and unbounded on compact K. Then for each $n \in \mathbb{N}$, there exists $x_n \in K$ such that $|f(x_n)| > n$. The sequence $\{x_n\}$ lies in K, which is compact, so by the Bolzano–Weierstrass Theorem (Theorem 1.1.1), $\{x_n\}$ has an accumulation point in K. In other words, there exists a convergent subsequence $\left\{x_{n_k}\right\}$ with $\lim_{k \to \infty} x_{n_k} \in K$.

Since f is continuous, $\lim_{k \to \infty} f\left(x_{n_k}\right) = f\left(\lim_{k \to \infty} x_{n_k}\right)$, which is well-defined because $\lim_{k \to \infty} x_{n_k} \in K$. However, this contradicts $\left|f\left(x_{n_k}\right)\right| > n_k \to \infty$, hence f must be bounded on K. □

Theorem 1.2.14 *(Extreme Value)*: A continuous function $f(x)$ defined on a compact set K attains its infimum and supremum in K.

Proof: Assume that f never attains its supremum M. Then, $f(x) < M$. Define the auxiliary function $\psi(x) = \frac{1}{M - f(x)}$, which is strictly positive and continuous as the denominator never reaches 0. By Theorem 1.2.13, $\psi(x)$ is bounded with some value of $\mu > 0$ satisfying $\psi(x) \leq \mu$. $f(x)$ also has the representation $M - \frac{1}{\psi(x)}$, and therefore,

$$f(x) \leq M - \frac{1}{\mu},$$

which means that M is not the supremum. Similarly, assume that f never attains its infimum m. Then $f(x) > m$. Let $\psi(x) = \frac{1}{f(x) - m}$, which is strictly positive and continuous as the denominator never reaches 0. By Theorem 1.2.13, $\psi(x)$ is bounded with some value of $\mu > 0$ satisfying $\psi(x) \leq \mu$. $f(x)$ also has the representation $m + \frac{1}{\psi(x)}$, and therefore,

$$f(x) \geq m + \frac{1}{\mu},$$

which means that m is not the infimum. □

Definition 1.2.6 *(Uniform Continuity)*: A function $f : U \to \mathbb{R}$, defined on a set $U \subseteq \mathbb{R}$, is uniformly continuous iff $\forall \varepsilon > 0, \exists \delta > 0$ such that $\forall x, y \in U$ where $|x - y| < \delta$, $|f(x) - f(y)| < \varepsilon$.

Example 1.2.1: The function $f(x) = \frac{1}{x}$ is not uniformly continuous over $(0, 1)$.

Proof: If $\exists \varepsilon > 0$ such that $\forall \delta > 0$, $\exists x, y \in (0, 1)$ satisfying both $|x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon$, then f is not uniformly continuous over $(0, 1)$.

Let $\varepsilon = 1$ and

$$x = \frac{1}{n}, \quad y = \frac{1}{n + 1}.$$

Then $\forall \delta > 0$, $\exists n > 1$ where $|x - y| < \delta$, since $\lim_{n \to \infty} |x - y| = 0$. Additionally, $|f(x) - f(y)| = 1 \geq \varepsilon$. This satisfies the negation, and thus, $f(x) = \frac{1}{x}$ is not uniformly continuous over $(0, 1)$. □

Theorem 1.2.15 *(Heine–Cantor)*: A continuous function on a compact set K is uniformly continuous on K.

Proof: Fix $x \in K$. Since f is continuous at x, for every $\varepsilon > 0$ there exists $\delta_x > 0$ such that for all $\zeta \in D(x, \delta_x) \cap K$,

$$|f(\zeta) - f(x)| < \frac{\varepsilon}{2}. \tag{1.2.1}$$

The collection of open balls $\left\{D\left(x, \frac{\delta_x}{2}\right)\right\}_{x \in K}$ forms an open cover of the compact set K. By Heine–Borel (Theorem 1.1.2), there is a finite subcover

$$\left\{D\left(x_k, \frac{\delta_{x_k}}{2}\right)\right\}_{k=1}^{n}.$$

Set

$$\delta = \min_{1 \leq k \leq n} \frac{\delta_{x_k}}{2}.$$

Now let $x, y \in K$ satisfy $|x - y| < \delta$. Then there exists an index $j \in \{1, \cdots, n\}$ such that $x \in D\left(x_j, \frac{\delta_{x_j}}{2}\right)$. Consequently,

$$|x_j - y| \leq |x_j - x| + |x - y| < \frac{\delta_{x_j}}{2} + \delta \leq \delta_{x_j}.$$

Applying (1.2.1) to the points x and y through x_j, we obtain

$$|f(x_j) - f(x)| < \frac{\varepsilon}{2}, \quad |f(x_j) - f(y)| < \frac{\varepsilon}{2}.$$

Therefore,

$$|f(x) - f(y)| \leq |f(x) - f(x_j)| + |f(x_j) - f(y)| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the uniform continuity of f on K follows. □

Definition 1.2.7: A function f is Lipschitz continuous over U if $\exists M \in \mathbb{R}_{\geq 0}$ such that $\forall x, y \in U$, $|f(x) - f(y)| \leq M|x - y|$. The smallest possible M satisfying the above condition is known as the Lipschitz constant.

Lipschitz continuity is an important concept in real analysis and the theory of differential equations. It is a strong form of uniform continuity.

Proposition 1.2.1: All Lipschitz continuous functions on U are uniformly continuous on U.

Proof: Let $M > 0$ be the Lipschitz constant. Then $\forall \varepsilon > 0$, let $\delta = \frac{\varepsilon}{M}$. It then follows that $\forall x, y \in U$ such that $|x - y| < \delta$, $|f(x) - f(y)| \leq M|x - y| < \varepsilon$. □

Proposition 1.2.2: A C^1 function on a compact set K is Lipschitz continuous on K.

Proof: Let $f : K \to \mathbb{R}$ be C^1. By Theorem 1.2.13, since K is compact and f' is continuous, $\exists M > 0$ such that $\forall x \in K$, $|f'(x)| \leq M$.

By the Mean Value Theorem, $\forall x, y \in K$, $\exists c$ between x and y such that $f(x) - f(y) = f'(c)(x - y)$. Then, $|f(x) - f(y)| = |f'(c)\|x - y| \leq M|x - y|$, which means f is Lipschitz continuous with Lipschitz constant less than or equal to M. □

2 Complex Prerequisites

2.1 Complex Differentiation

For $U \subseteq \mathbb{C}$ and a complex function $f : U \to \mathbb{C}$, $f(z)$ is *complex differentiable* at $z \in U$ if the following limit exists, regardless of the direction Δz approaches 0 from:

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}.$$

We can consider $f(z)$ to be a bivariate function $f(x, y)$ for $z = x + \mathrm{i}y$. Two main cases we are concerned with are when Δz approaches 0 from the real and imaginary axes:

$$\lim_{\Delta z \to 0, \Delta z \in \mathbb{R}} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0, \Delta z \in \mathbb{R}} \frac{f(z + \mathrm{i}\Delta z) - f(z)}{\mathrm{i}\Delta z}.$$

Expressing $f(z)$ as $f(x, y) = u(x, y) + \mathrm{i}v(x, y)$,

$$\lim_{\Delta z \to 0, \Delta z \in \mathbb{R}} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0, \Delta z \in \mathbb{R}} \frac{f(x + \Delta z, y) - f(x, y)}{\Delta z} = \frac{\partial u}{\partial x} + \mathrm{i}\frac{\partial v}{\partial x},$$

and

$$\lim_{\Delta z \to 0, \Delta z \in \mathbb{R}} \frac{f(z + \mathrm{i}\Delta z) - f(z)}{\mathrm{i}\Delta z} = -\mathrm{i} \lim_{\Delta z \to 0, \Delta z \in \mathbb{R}} \frac{f(x, y + \Delta z) - f(x, y)}{\Delta z} = \frac{\partial v}{\partial y} - \mathrm{i}\frac{\partial u}{\partial y}.$$

By comparing the real and imaginary parts, we obtain necessary conditions for complex differentiability:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \tag{2.1.1}$$

By multiplying the second equation by i and adding it to the first, we obtain the equivalent form

$$\frac{\partial f}{\partial x} = -\mathrm{i}\frac{\partial f}{\partial y}. \tag{2.1.2}$$

The identities (2.1.1) and (2.1.2) are known as the *Cauchy–Riemann equations*.

Although this condition is necessary, it is not sufficient. Consider the function

$$f(z) = \sqrt{|\mathfrak{Re}(z)\,\mathfrak{Im}(z)|}.$$

Let $z = x + \mathrm{i}y$, $x = \alpha t$, and $y = \beta t$. Then

$$\lim_{z\to 0}\frac{f(z)-f(0)}{z-0}=\lim_{z\to 0}\frac{f(z)}{z}=\lim_{t\to 0}\frac{\sqrt{|\alpha\beta t^2|}}{\alpha t+\mathrm{i}\beta t}=\frac{\sqrt{|\alpha\beta|}}{\alpha+\mathrm{i}\beta}.$$

The derivative along $\alpha = 1$, $\beta = 0$, or the real axis, vanishes. Along $\alpha = 0$, $\beta = 1$, or the imaginary axis, it also vanishes. However, the limit is different for any other pair of α and β, and hence for other directions of approach.

Definition 2.1.1 *(Holomorphy)*: A function $f : U \to \mathbb{C}$ is said to be *holomorphic* at $z_0 \in U$ if it is complex differentiable on a neighborhood of z_0. If $f(z)$ is holomorphic for every point in an open connected set U, then it is said to be holomorphic over U. A function is holomorphic over a compact set K if it is holomorphic on a neighborhood of K.

Weierstrass provided the following classification:

Definition 2.1.2: A function is *entire* if it is holomorphic over $\mathbb{C}$.

For the purpose of the following contents, a *region* or *domain* will denote a nonempty, open, connected subset of the complex plane.

Theorem 2.1.1: Let $U \subseteq \mathbb{C}$ be open, and let $f : U \to \mathbb{C}$ be a function. Then f is holomorphic on U iff $f \in C^1(U)$ and satisfies the Cauchy–Riemann equations.

Proof: The first part is to prove that any holomorphic function on U has continuous first-order partial derivatives in U. This requires an argument that will be covered later, specifically in @ sec:analyticityandholomorphy, which states that the complex derivative of any holomorphic function is also holomorphic over the region.

For the converse, let $f(z) = f(x, y) = u(x, y) + \mathrm{i}v(x, y)$. Assume that $u, v \in C^1(U)$ and satisfy the Cauchy–Riemann equations at $z_0 = x_0 + \mathrm{i}y_0$. Let

$$\alpha=\frac{\partial u}{\partial x}(x_0,y_0)=\frac{\partial v}{\partial y}(x_0,y_0),\quad \beta=\frac{\partial v}{\partial x}(x_0,y_0)=-\frac{\partial u}{\partial y}(x_0,y_0).$$

Because u and v are continuously differentiable, there are functions $\varepsilon_{u(\Delta x,\Delta y)}$ and $\varepsilon_{v(\Delta x,\Delta y)}$ such that $\varepsilon_u \to 0$ and $\varepsilon_v \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$, and

$$u(x,y)-u(x_0,y_0)=\alpha(x-x_0)-\beta(y-y_0)+\varepsilon_{u(\Delta x,\Delta y)}|\Delta z|,$$

$$v(x,y)-v(x_0,y_0)=\beta(x-x_0)+\alpha(y-y_0)+\varepsilon_{v(\Delta x,\Delta y)}|\Delta z|,$$

where $\Delta z = (x - x_0) + \mathrm{i}(y - y_0)$.

Then

$$\begin{aligned} f(z)-f(z_0)&=[\alpha(x-x_0)-\beta(y-y_0)]+\mathrm{i}[\beta(x-x_0)+\alpha(y-y_0)]\\ &\quad+\left[\varepsilon_{u(\Delta x,\Delta y)}+\mathrm{i}\varepsilon_{v(\Delta x,\Delta y)}\right]|\Delta z|\\ &=(\alpha+\mathrm{i}\beta)\Delta z+\left[\varepsilon_{u(\Delta x,\Delta y)}+\mathrm{i}\varepsilon_{v(\Delta x,\Delta y)}\right]|\Delta z|. \end{aligned}$$

Therefore,

$$\frac{f(z)-f(z_0)}{z-z_0}=\alpha+\mathrm{i}\beta+\left[\varepsilon_{u(\Delta x,\Delta y)}+\mathrm{i}\varepsilon_{v(\Delta x,\Delta y)}\right]\left(\frac{|\Delta z|}{\Delta z}\right).$$

Taking the limit as $\Delta z \to 0$, the last term vanishes because $\left|\frac{|\Delta z|}{\Delta z}\right| = 1$. Hence

$$\lim_{\Delta z \to 0} \frac{f(z) - f(z_0)}{z - z_0} = \alpha + \mathrm{i}\beta. \qquad \square$$

We will prove later in @ sec:analyticityandholomorphy that the complex derivative of a holomorphic function $f(z) = u(z) + \mathrm{i}v(z)$ is holomorphic. Under this assumption, $f(z)$ has continuous second-order partial derivatives, and therefore

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}, \quad \frac{\partial^2 v}{\partial x \partial y} = \frac{\partial^2 v}{\partial y \partial x},$$

and by the Cauchy–Riemann equations,

$$\frac{\partial^2 u}{\partial x \partial x} = \frac{\partial^2 v}{\partial y \partial x}, \quad \frac{\partial^2 u}{\partial y \partial y} = -\frac{\partial^2 v}{\partial x \partial y},$$

and

$$\frac{\partial^2 v}{\partial x \partial x} = -\frac{\partial^2 u}{\partial y \partial x}, \quad \frac{\partial^2 v}{\partial y \partial y} = \frac{\partial^2 u}{\partial x \partial y}.$$

Adding the equations,

$$\frac{\partial^2 u}{\partial x \partial x} + \frac{\partial^2 u}{\partial y \partial y} = 0, \quad \frac{\partial^2 v}{\partial x \partial x} + \frac{\partial^2 v}{\partial y \partial y} = 0.$$

This general type of equation is known as *Laplace's equation*, which is a basic example of an elliptic partial differential equation. Define the operator, the *Laplacian*, by

$$\Delta = \boldsymbol{\nabla} \cdot \boldsymbol{\nabla} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

A function u satisfying Laplace's equation $\Delta u = 0$ is a *harmonic function*. Thus, the real and imaginary parts of a holomorphic function are harmonic functions.

Letting $x = r\cos\theta$ and $y = r\sin\theta$, the Laplacian is equal to

$$\begin{aligned}
\Delta &= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \\
&= \frac{\partial}{\partial x}\left(\frac{\partial r}{\partial x}\frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x}\frac{\partial}{\partial \theta}\right) + \frac{\partial}{\partial y}\left(\frac{\partial r}{\partial y}\frac{\partial}{\partial r} + \frac{\partial \theta}{\partial y}\frac{\partial}{\partial \theta}\right) \\
&= \frac{\partial}{\partial x}\left(\frac{x}{r}\frac{\partial}{\partial r} - \frac{y}{r^2}\frac{\partial}{\partial \theta}\right) + \frac{\partial}{\partial y}\left(\frac{y}{r}\frac{\partial}{\partial r} + \frac{x}{r^2}\frac{\partial}{\partial \theta}\right) \\
&= \left(\cos\theta\frac{\partial}{\partial r} - \sin\frac{\theta}{r}\frac{\partial}{\partial \theta}\right)\left(\cos\theta\frac{\partial}{\partial r} - \sin\frac{\theta}{r}\frac{\partial}{\partial \theta}\right) \\
&+ \left(\sin\theta\frac{\partial}{\partial r} + \cos\frac{\theta}{r}\frac{\partial}{\partial \theta}\right)\left(\sin\theta\frac{\partial}{\partial r} + \cos\frac{\theta}{r}\frac{\partial}{\partial \theta}\right) \\
&= \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}.
\end{aligned} \tag{2.1.3}$$

Proposition 2.1.1: Let $U \subseteq \mathbb{C}$ be open and connected and let $f : U \to \mathbb{R}$ be holomorphic. Then f is constant over U.

Proof: Since $f(x,y) = u(x,y) + \mathrm{i}v(x,y)$ is real-valued, $v(x,y) \equiv 0$ on U. Then by the Cauchy–Riemann equations on U, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 0$. Similarly, $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 0$. Therefore, $f(z) = u(z)$ is constant. □

2.1.1 Wirtinger Derivatives

We have previously introduced the concept of expressing a complex function as a function of x and y. It can also be expressed in terms of z and $\overline{z}$, where $z = x + \mathrm{i}y$ and $\overline{z} = x - \mathrm{i}y$. Then $|z|^2 = z\overline{z}$, $x = \frac{z+\overline{z}}{2}$, and $y = \frac{z-\overline{z}}{2\mathrm{i}}$. By the rules of the derivative, it is only natural that we define

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x}\frac{\partial x}{\partial z} + \frac{\partial}{\partial y}\frac{\partial y}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathrm{i}\frac{\partial}{\partial y}\right) \tag{2.1.4}$$

and

$$\frac{\partial}{\partial \overline{z}} = \frac{\partial}{\partial x}\frac{\partial x}{\partial \overline{z}} + \frac{\partial}{\partial y}\frac{\partial y}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathrm{i}\frac{\partial}{\partial y}\right). \tag{2.1.5}$$

If (2.1.4) is set equal to 0, then it is the equivalent form of the homogeneous Cauchy–Riemann Equations. Then for a holomorphic function $f(z)$, the Wirtinger derivative $\frac{\partial f}{\partial z} = \frac{\mathrm{d}f}{\mathrm{d}z}$.

In terms of u and v, the two derivatives of a function $f(z)$ are equal to:

$$\frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial u}{\partial x} + \mathrm{i}\frac{\partial v}{\partial x} - \mathrm{i}\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\right),$$

and

$$\frac{\partial f}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial u}{\partial x} + \mathrm{i}\frac{\partial v}{\partial x} + \mathrm{i}\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y}\right).$$

If f is holomorphic,

$$\frac{\mathrm{d}f}{\mathrm{d}z} = \frac{\partial u}{\partial x} + \mathrm{i}\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} + \mathrm{i}\frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - \mathrm{i}\frac{\partial u}{\partial y} = \frac{\partial v}{\partial y} - \mathrm{i}\frac{\partial u}{\partial y}. \tag{2.1.6}$$

On the contrary, by the rules of the derivative,

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial z}\frac{\partial z}{\partial x} + \frac{\partial}{\partial \overline{z}}\frac{\partial \overline{z}}{\partial x} = \frac{\partial}{\partial z} + \frac{\partial}{\partial \overline{z}}$$

and

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial z}\frac{\partial z}{\partial y} + \frac{\partial}{\partial \overline{z}}\frac{\partial \overline{z}}{\partial y} = \mathrm{i}\frac{\partial}{\partial z} - \mathrm{i}\frac{\partial}{\partial \overline{z}}.$$

The Laplacian is equal to

$$\begin{aligned}\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} &= \left(\frac{\partial}{\partial z} + \frac{\partial}{\partial \overline{z}}\right)^2 + \left(\mathrm{i}\frac{\partial}{\partial z} - \mathrm{i}\frac{\partial}{\partial \overline{z}}\right)^2 \\ &= \frac{\partial^2}{\partial z^2} + \frac{\partial^2}{\partial \overline{z}^2} + 2\frac{\partial^2}{\partial z \partial \overline{z}} - \frac{\partial^2}{\partial z^2} - \frac{\partial^2}{\partial \overline{z}^2} + 2\frac{\partial^2}{\partial z \partial \overline{z}} \\ &= 4\frac{\partial^2}{\partial z \partial \overline{z}}. \end{aligned} \tag{2.1.7}$$

Under this definition, we can derive the chain rule:

Theorem 2.1.1.1 *(Chain Rule)*: Let $\Omega \subseteq \mathbb{C}$ be a region such that $g \in C^1(\Omega)$ and $f \in C^1(g(\Omega))$. Writing $w = g(z)$, it follows that

$$\frac{\partial}{\partial z}(f \circ g) = \left(\frac{\partial f}{\partial w} \circ g\right)\frac{\partial g}{\partial z} + \left(\frac{\partial f}{\partial \overline{w}} \circ g\right)\frac{\partial \overline{g}}{\partial z}$$
$$\frac{\partial}{\partial \overline{z}}(f \circ g) = \left(\frac{\partial f}{\partial w} \circ g\right)\frac{\partial g}{\partial \overline{z}} + \left(\frac{\partial f}{\partial \overline{w}} \circ g\right)\frac{\partial \overline{g}}{\partial \overline{z}}.$$

Proof: Write $z = x + \mathrm{i}y$. Let

$$g(z) = \xi(x, y) + \mathrm{i}\eta(x, y), \quad w = \xi + \mathrm{i}\eta$$

so that $w = g(z)$ with $\xi = \xi(x, y), \eta = \eta(x, y)$. Let f be regarded as a C^1 function of the real variables ξ, η; equivalently we may view f as $f(w, \overline{w})$ where $\overline{w} = \xi - \mathrm{i}\eta$. The composition is $h(z) = f \circ g(z) = f(\xi(x, y), \eta(x, y))$.

Using the real chain rule (provided by the continuous differentiability), we have

$$\frac{\partial h}{\partial x} = \frac{\partial f}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial f}{\partial \eta}\frac{\partial \eta}{\partial x}, \quad \frac{\partial h}{\partial y} = \frac{\partial f}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial f}{\partial \eta}\frac{\partial \eta}{\partial y}.$$

Hence,

$$\frac{\partial h}{\partial z} = \frac{1}{2}\left[\frac{\partial f}{\partial \xi}\left(\frac{\partial \xi}{\partial x} - \mathrm{i}\frac{\partial \xi}{\partial y}\right) + \frac{\partial f}{\partial \eta}\left(\frac{\partial \eta}{\partial x} - \mathrm{i}\frac{\partial \eta}{\partial y}\right)\right].$$

Now recall

$$\frac{\partial f}{\partial w} = \frac{1}{2}\left(\frac{\partial f}{\partial \xi} - \mathrm{i}\frac{\partial f}{\partial \eta}\right), \quad \frac{\partial f}{\partial \overline{w}} = \frac{1}{2}\left(\frac{\partial f}{\partial \xi} + \mathrm{i}\frac{\partial f}{\partial \eta}\right).$$

Thus,

$$\frac{\partial f}{\partial \xi} = \frac{\partial f}{\partial w} + \frac{\partial f}{\partial \overline{w}}, \quad \frac{\partial f}{\partial \eta} = \mathrm{i}\left(\frac{\partial f}{\partial w} - \frac{\partial f}{\partial \overline{w}}\right).$$

Then by substitution,

$$\begin{aligned}\frac{\partial h}{\partial z} &= \frac{1}{2}\left[\left(\frac{\partial f}{\partial w} + \frac{\partial f}{\partial \overline{w}}\right)\left(\frac{\partial \xi}{\partial x} - \mathrm{i}\frac{\partial \xi}{\partial y}\right) + \mathrm{i}\left(\frac{\partial f}{\partial w} - \frac{\partial f}{\partial \overline{w}}\right)\left(\frac{\partial \eta}{\partial x} - \mathrm{i}\frac{\partial \eta}{\partial y}\right)\right] \\ &= \frac{\partial f}{\partial w}\frac{1}{2}\left[\left(\frac{\partial \xi}{\partial x} - \mathrm{i}\frac{\partial \xi}{\partial y}\right) + \mathrm{i}\left(\frac{\partial \eta}{\partial x} - \mathrm{i}\frac{\partial \eta}{\partial y}\right)\right] + \frac{\partial f}{\partial \overline{w}}\frac{1}{2}\left[\left(\frac{\partial \xi}{\partial x} - \mathrm{i}\frac{\partial \xi}{\partial y}\right) - \mathrm{i}\left(\frac{\partial \eta}{\partial x} - \mathrm{i}\frac{\partial \eta}{\partial y}\right)\right].\end{aligned}$$

The terms in brackets equal $\frac{\partial g}{\partial z}$ and $\frac{\partial \overline{g}}{\partial z}$. Thus,

$$\frac{\partial h}{\partial z} = \left(\frac{\partial f}{\partial w} \circ g\right)\frac{\partial g}{\partial z} + \left(\frac{\partial f}{\partial \overline{w}} \circ g\right)\frac{\partial \overline{g}}{\partial z}.$$

A similar calculation using (2.1.5) gives

$$\frac{\partial}{\partial \overline{z}}(f \circ g) = \left(\frac{\partial f}{\partial w} \circ g\right)\frac{\partial g}{\partial \overline{z}} + \left(\frac{\partial f}{\partial \overline{w}} \circ g\right)\frac{\partial \overline{g}}{\partial \overline{z}}.$$

These are exactly the proclaimed identities. □

Last we have taking derivatives of conjugates:

Theorem 2.1.1.2: Let $f \in C^1(\Omega)$ where $\Omega \subseteq \mathbb{C}$ is a region. Then

$$\frac{\partial \overline{f}}{\partial z} = \overline{\frac{\partial f}{\partial \overline{z}}}, \quad \frac{\partial \overline{f}}{\partial \overline{z}} = \overline{\frac{\partial f}{\partial z}}.$$

Proof: Write $z = x + \mathrm{i}y$ and $f(z) = u(x, y) + \mathrm{i}v(x, y)$ with $u, v \in C^1(\Omega)$. Then $\overline{f(z)} = u(x, y) - \mathrm{i}v(x, y)$. We compute

$$\frac{\partial \overline{f}}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathrm{i}\frac{\partial}{\partial y}\right)(u - \mathrm{i}v) = \frac{1}{2}\left(u'_x - v'_y - \mathrm{i}\left(v'_x + u'_y\right)\right).$$

On the other hand,

$$\frac{\partial f}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathrm{i}\frac{\partial}{\partial y}\right)(u + \mathrm{i}v) = \frac{1}{2}\left(u'_x - v'_y + \mathrm{i}\left(v'_x + u'_y\right)\right).$$

Taking complex conjugates yields

$$\overline{\frac{\partial f}{\partial \overline{z}}} = \frac{1}{2}\left(u'_x - v'_y - \mathrm{i}\left(v'_x + u'_y\right)\right) = \frac{\partial \overline{f}}{\partial z}.$$

Similarly,

$$\frac{\partial \overline{f}}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathrm{i}\frac{\partial}{\partial y}\right)(u - \mathrm{i}v) = \frac{1}{2}\left(u'_x + v'_y + \mathrm{i}\left(u'_y - v'_x\right)\right),$$

while

$$\frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathrm{i}\frac{\partial}{\partial y}\right)(u + \mathrm{i}v) = \frac{1}{2}\left(u'_x + v'_y + \mathrm{i}\left(v'_x - u'_y\right)\right).$$

Taking complex conjugates gives

$$\overline{\frac{\partial f}{\partial z}} = \frac{1}{2}\left(u'_x + v'_y + \mathrm{i}\left(u'_y - v'_x\right)\right) = \frac{\partial \overline{f}}{\partial \overline{z}}.$$ □

2.2 Complex Power Series

Power series in real analysis can be generalized into complex series. Particularly, concepts such as uniform convergence are the same in complex analysis:

Definition 2.2.1 *(Uniform Convergence)*: For a set $U \subseteq \mathbb{C}$, a function sequence $\{f_n(z)\}$ *uniformly converges* to a function $f(z)$ on U iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\forall n > N, \forall z \in U, |f_n(z) - f(z)| < \varepsilon$.

Remark: The definition above is equivalent to the following definition.

For a set $U \subseteq \mathbb{C}$, a function sequence $\{f_n(z)\}$ uniformly converges to $f(z)$ iff

$$\lim_{n \to \infty} \sup_{z \in U} |f_n(z) - f(z)| = 0.$$

Informally, we will use the notation $f_n(z) \rightrightarrows f(z)$ to represent uniform convergence.

Theorem 2.2.1 *(Cauchy Criterion)*: For a set $U \subseteq \mathbb{C}$, a function sequence $\{f_n(z)\}$ uniformly converges to a function $f(z)$ iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\forall n, m > N, \forall z \in U, |f_n(z) - f_m(z)| < \varepsilon$.

Proof: If $f_n(z)$ uniformly converges to $f(z)$, then for $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n, m > N$ and $\forall z \in U$,

$$|f_m(z) - f(z)| < \frac{\varepsilon}{2}, \quad |f_n(z) - f(z)| < \frac{\varepsilon}{2}.$$

Then,

$$|f_m(z) - f_n(z)| \leq |f_n(z) - f(z)| + |f_m(z) - f(z)| < \varepsilon.$$

For the converse, refer to the analogous proof in Theorem 1.2.12. □

Function series are defined to be a sequence formed by the partial sums of function sequences. There are many ways to verify the uniform convergence of a function series. Perhaps the most widely known is the Weierstrass M-Test.

Theorem 2.2.2 *(Weierstrass M-Test)*: Let $U \subseteq \mathbb{C}$ be a region and $\{f_n\}$ be a function sequence on U. If $\exists \{M_n\} \subset \mathbb{R}_{\geq 0}$ such that $\forall n \in \mathbb{N}$, $\forall z \in U$, $|f_n(z)| \leq M_n$ and the series $\sum_{n=1}^{\infty} M_n$ converges, then the series $\sum_{n=1}^{\infty} f_n(z)$ converges uniformly and absolutely on U.

Proof: By the convergence of $\sum_{n=1}^{\infty} M_n$, $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall m \geq n > N$,

$$\left|M_m + M_{m-1} + \ldots + M_{n+1}\right| < \varepsilon.$$

Since M_j bounds $f_{j(z)}$, it follows that

$$\left|f_m(z) + f_{m-1}(z) + \ldots + f_{n+1}(z)\right| \leq \left|M_m + M_{m-1} + \ldots + M_{n+1}\right| < \varepsilon,$$

and the result follows from Theorem 2.2.1. □

The concept of uniform convergence is generalized to improper integrals with parameters, and the same theorems from series have a corresponding counterpart.

In both complex and real analysis, the concept of *power series*, a unique type of function series, is of trivial importance. Similar to real power series, complex series have the form

$$\sum_{n=0}^{\infty} a_n z^n,$$

where $\{a_n\}$ are constants.

Let $D(a, r) = B^1(a, r) = \{z \in \mathbb{C} \mid |z - a| < r\}$ denote the *open disk* centered at a with radius r. For simplicity, from now on we will have $\mathbb{D}$ denote the unit open disk, or $D(0, 1)$. We will now observe the convergence of power series.

Theorem 2.2.3 *(Abel's Theorem)*: For a power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$, there exists a constant $R \in \mathbb{R}_{\geq 0} \cup \{\infty\}$, known as the *radius of convergence* such that:

1. f absolutely converges on $D(0, R)$, and $\forall 0 \leq \rho < R$, uniformly converges on $\overline{D(0, \rho)}$.
2. $f(z)$ diverges when $|z| > R$.
3. f is holomorphic over $D(0, R)$ and $f'(z)$ can be obtained by termwise differentiation, or $f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$, which also has a convergence radius of R.

The disk $|z| < R$ is known as the *disk of convergence*, a direct generalization of the *interval of convergence* for real series. There are many ways to determine the radius of convergence:

Theorem 2.2.4 *(Cauchy–Hadamard)*: The radius of convergence of the power series in the form $\sum_{n=0}^{\infty} a_n z^n$ can be determined by

$$R = \frac{1}{\limsup_{n \to \infty}} \sqrt[n]{|a_n|}. \tag{2.2.1}$$

Of course, a convergence radius of 0 implies that the series is divergent everywhere except for possibly at 0, and a convergence radius of ∞ means that the series absolutely converges everywhere.

Proof: We will prove that the value in (2.2.1) satisfies the criteria in Theorem 2.2.3.

Assume $|z| < R$. Then $\forall \rho \in (|z|, R)$, and consequently, $\frac{1}{\rho} > \frac{1}{R}$. By Definition 1.2.3 and (2.2.1), $\exists N \in \mathbb{N}$ such that $\forall n > N$, $\sqrt[n]{|a_n|} < \frac{1}{\rho}$ and $|a_n| < \frac{1}{\rho^n}$. It follows that $|a_n z^n| < \frac{|z|^n}{\rho^n} < 1$ for all $n > N$. Then $\sum_{n=0}^{\infty} |a_n z^n|$ converges.

Let $\rho' \in (\rho, R)$. Similarly, $\exists N' \in \mathbb{N}$ such that $\forall n > N'$, $\sqrt[n]{|a_n|} < \frac{1}{\rho'}$, and $|a_n| < \frac{1}{\rho'^n}$. Then $|a_n z^n| < |a_n \rho^n| < \frac{\rho^n}{\rho'^n}$. By the Weierstrass M-Test (Theorem 2.2.2), $\sum_{n=0}^{\infty} |a_n z^n|$ is uniformly bounded for $n > N'$ by the convergent series $\sum_{n=0}^{\infty} a_n \rho^n$, and thus uniformly converges on $|z| < \rho$. This proves part 1.

Assume that $|z| > R$. For all $\rho \in (R, |z|)$, $\frac{1}{\rho} < \frac{1}{R}$. By Definition 1.2.3, $\forall N \in \mathbb{N}$, $\exists n_N > N$ such that $\sqrt[n_N]{\left|a_{n_N}\right|} > \frac{1}{\rho}$. It follows that $\left|a_{n_N} z^{n_N}\right| > \frac{|z^{n_N}|}{\rho^{n_N}} > 1$. Since $\forall N \in \mathbb{N}$,

$$\left|\sum_{k=0}^{n_N} a_k z^k - \sum_{k=0}^{n_N - 1} a_k z^k\right| > 1,$$

by the Cauchy Criterion (Theorem 1.2.12), $\sum_{n=0}^{\infty} a_n z^n$ is divergent. Thus, part 2 is satisfied.

To prove part 3, first observe that $\sum_{n=1}^{\infty} n a_n z^n$ and $\sum_{n=1}^{\infty} a_n z^n$ have the same convergence radius since $\limsup_{n\to\infty} \sqrt[n]{n} = 1$. For $z \in D(0, R)$, let $f(z) = S_n(z) + R_n(z)$, where

$$S_n(z) = \sum_{k=0}^{n-1} a_k z^k, \quad R_n(z) = \sum_{k=n}^{\infty} a_k z^k.$$

Let

$$f_1(z) = \lim_{n\to\infty} S_n'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}.$$

Let $\rho < R$ be positive and $|z_0| < \rho$. Then we aim to show that

$$\lim_{z\to z_0} \left(\frac{f(z) - f(z_0)}{z - z_0} - f_1(z)\right) = 0.$$

By analyzing the difference,

$$\begin{aligned} \frac{f(z) - f(z_0)}{z - z_0} - f_1(z) = &\left[\frac{S_n(z) - S_n(z_0)}{z - z_0} - S_n'(z)\right] \\ &+ S_n'(z) - f_1(z) + \frac{R_n(z) - R_n(z_0)}{z - z_0}. \end{aligned} \tag{2.2.2}$$

Since $S_n'(z) \to f_1(z)$ as $n \to \infty$, it follows that $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\forall n > N$, $|S_n'(z) - f_1(z)| < \frac{\varepsilon}{3}$. Since

$$\frac{R_n(z) - R_n(z_0)}{z - z_0} = \sum_{k=n}^{\infty} a_k \left(z^{k-1} + z^{k-2} z_0 + \dots + z_0^{k-1}\right)$$

for $z \neq z_0$, with $|z| < \rho < R$,

$$\left|\sum_{k=n}^{\infty} a_k \left(z^{k-1} + \dots + z_0^{k-1}\right)\right| \leq \sum_{k=n}^{\infty} |a_k| \left(\left|z^{k-1}\right| + \dots + \left|z_0^{k-1}\right|\right) < \sum_{k=n}^{\infty} |a_k| k \rho^{k-1}.$$

Since $\sum_{k=1}^{\infty} k|a_k|\rho^{k-1}$ is absolutely convergent, $\sum_{k=n}^{\infty}|a_k|k\rho^{k-1}$ is the remainder term of a convergent series. Then, $\exists N' \in \mathbb{N}$ such that $\forall n > N'$, $\sum_{k=n}^{\infty}|a_k|k\rho^{k-1} < \frac{\varepsilon}{3}$.

Finally, for a fixed $n > \max(N, N')$, $\exists\delta > 0$ such that $\forall z \in D(z_0, \delta) \setminus \{z_0\}$,

$$\left|\frac{S_n(z) - S_n(z_0)}{z - z_0} - S_n'(z)\right| < \frac{\varepsilon}{3}.$$

From (2.2.2), we get

$$\left|\frac{f(z) - f(z_0)}{z - z_0} - f_1(z)\right| < \varepsilon,$$

confirming part 3. □

Obviously, a substitution of $z = \zeta - a$ where $a \in \mathbb{C}$ translates the disk of convergence to $D(a, R)$. The subsequent results on uniform convergence hold for complex functions:

Theorem 2.2.5 *(Uniform Limit)*: Let $\{f_n(z)\}$ be continuous on $U \subseteq \mathbb{C}$ and uniformly convergent to $f(z)$. Then $f(z)$ is continuous on U.

Proof: By continuity, $\forall n \in \mathbb{N}$, $\forall z_0 \in U$, $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $\forall z \in D(z_0, \delta) \subseteq U$, $|f_n(z) - f_n(z_0)| < \frac{\varepsilon}{3}$. Additionally, $\exists N \in \mathbb{N}$ such that $\forall n > N$, $\forall z \in U$, $|f_n(z) - f(z)| < \frac{\varepsilon}{3}$. It follows that $|f_n(z_0) - f(z_0)| < \frac{\varepsilon}{3}$. By the triangle inequality,

$$|f(z) - f(z_0)| \leq |f(z) - f_n(z)| + |f_n(z) - f_n(z_0)| + |f_n(z_0) - f(z_0)| < \varepsilon$$

for all $z \in D(z_0, \delta)$. Then the continuity of f is satisfied. □

Lastly, the sufficient criteria to pass a limit through an integral:

Theorem 2.2.6: Let γ be a rectifiable curve on which the function sequence $\{f_n\}_{n\in\mathbb{N}}$ is continuous. If $\{f_n(z)\}$ uniformly converges to f, then

$$\lim_{n\to\infty} \int_\gamma f_n(z)\,\mathrm{d}z = \int_\gamma f(z)\,\mathrm{d}z.$$

Proof: Since $\{f_n(z)\}$ uniformly converges to $f(z)$ on γ, $\forall\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$|f_n(z) - f(z)| < \frac{\varepsilon}{\mathrm{length}(\gamma)}, \quad \forall z \in \gamma.$$

Since each f_n is continuous and γ is rectifiable, each integral $\int_\gamma f_n(z)\,\mathrm{d}z$ is convergent and well-defined.

Then $\forall n > N$,

$$\begin{aligned}\left|\int_\gamma f_n(z)\,\mathrm{d}z - \int_\gamma f(z)\,\mathrm{d}z\right| &= \left|\int_\gamma (f_n(z) - f(z))\,\mathrm{d}z\right| \\ &\leq \int_\gamma |f_n(z) - f(z)||\mathrm{d}z| \\ &< \int_\gamma \frac{\varepsilon}{\mathrm{length}(\gamma)}|\mathrm{d}z| = \varepsilon.\end{aligned}$$

Therefore,

$$\lim_{n\to\infty}\int_\gamma f_n(z)\,\mathrm{d}z = \int_\gamma f(z)\,\mathrm{d}z. \qquad \square$$

Remark: For a uniformly convergent series $\sum_{n=1}^{\infty} f_n(z)$, the commutation between the limit and the integral becomes a summation-integral switch:

$$\sum_{n=1}^{\infty}\int_\gamma f_n(z)\,\mathrm{d}z = \int_\gamma \sum_{n=1}^{\infty} f_n(z)\,\mathrm{d}z.$$

2.3 The Conformality of Holomorphic Maps

Let $f : U \to \mathbb{C}$ be a holomorphic function defined on an open and connected subset $U \subseteq \mathbb{C}$, and let $z_0 \in U$ be a point such that $f'(z_0) \neq 0$. Consider a differentiable curve $\gamma \in C^1([0,1])$ with $\gamma(0) = z_0$. The direction of the curve at z_0 is given by the argument of its derivative, namely $\operatorname{Arg}(\gamma'(0))$.

The image of γ under f, defined by $\sigma(t) = f(\gamma(t))$, is also a smooth curve passing through $w_0 = f(z_0)$. By the chain rule, the derivative of σ at $t = 0$ is given by

$$\sigma'(0) = f'(\gamma(0))\gamma'(0) = f'(z_0)\gamma'(0),$$

and hence

$$\operatorname{Arg}(\sigma'(0)) = \operatorname{Arg}(f'(z_0)\gamma'(0)) = \operatorname{Arg}(f'(z_0)) + \operatorname{Arg}(\gamma'(0)).$$

It follows that

$$\operatorname{Arg}(\sigma'(0)) - \operatorname{Arg}(\gamma'(0)) = \operatorname{Arg}(f'(z_0)).$$

This shows that the change in direction between a curve and its image under f is independent of the curve itself, depending only on the value of $f'(z_0)$.

Now consider two smooth curves $\gamma_1, \gamma_2 \in C^1([0,1])$ such that $\gamma_1(0) = \gamma_2(0) = z_0$, with respective images $\sigma_1(t) = f(\gamma_1(t))$ and $\sigma_2(t) = f(\gamma_2(t))$. Then

$$\operatorname{Arg}(\sigma_{1'}(0)) - \operatorname{Arg}(\gamma_{1'}(0)) = \operatorname{Arg}(f'(z_0)) = \operatorname{Arg}(\sigma_{2'}(0)) - \operatorname{Arg}(\gamma_{2'}(0)),$$

and by rearrangement,

$$\operatorname{Arg}(\sigma_{1'}(0)) - \operatorname{Arg}(\sigma_{2'}(0)) = \operatorname{Arg}(\gamma_{1'}(0)) - \operatorname{Arg}(\gamma_{2'}(0)).$$

This equality demonstrates that the angle between two smooth curves at z_0 is preserved under f, provided $f'(z_0) \neq 0$. In other words, holomorphic functions with non-vanishing derivatives preserve angles and orientation locally, a property known as *conformality*.

Furthermore, for any smooth curve $\gamma \in C^1([0,1])$ passing through z_0, the limit

$$\lim_{z\to z_0, z\in\gamma}\frac{|f(z) - f(z_0)|}{|z - z_0|} = |f'(z_0)|$$

shows that infinitesimal lengths are locally scaled by a factor of $|f'(z_0)|$ under the mapping f.

Therefore, at points where $|f'(z)| \neq 0$, the function f is conformal; it preserves angles but not necessarily lengths.

2.4 Elementary Functions

Univariate functions formed by compositions, sums, products, and powers of finitely many functions of the following form are known as *elementary functions*:

1. Power functions including polynomials, rational functions, and their inverses.
2. Trigonometric functions, hyperbolic functions, and their inverses.
3. Exponential functions and their inverses.

Power functions are easily extendable to the complex plane by simply changing the real variable to a complex variable. The other two classes of functions have to be redefined and reinterpreted for the complex plane. It is well known that the exponential function can be expanded as

$$\begin{aligned} \mathrm{e}^{x} &= \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \\ &= \frac{x^0}{0!\mathrm{i}^0} + \mathrm{i}\frac{x^1}{1!\mathrm{i}^1} - \frac{x^2}{2!\mathrm{i}^2} - \mathrm{i}\frac{x^3}{3!\mathrm{i}^3} + \ldots \\ &= \cos\left(\frac{x}{\mathrm{i}}\right) + \mathrm{i}\sin\left(\frac{x}{\mathrm{i}}\right). \end{aligned}$$

This is better written as

$$\mathrm{e}^{\mathrm{i}x} = \cos(x) + \mathrm{i}\sin(x), \tag{2.4.1}$$

which is the famous *Euler formula*. Then for any complex number $z = x + \mathrm{i}y$,

$$\mathrm{e}^{z} = \mathrm{e}^{x+\mathrm{i}y} = \mathrm{e}^{x}(\cos(y) + \mathrm{i}\sin(y)).$$

Then trigonometric functions and exponential functions can be written in terms of each other:

$$\begin{aligned} \sin(z) &= \frac{\mathrm{e}^{\mathrm{i}z} - \mathrm{e}^{-\mathrm{i}z}}{2\mathrm{i}}, \ \cos(z) = \frac{\mathrm{e}^{\mathrm{i}z} + \mathrm{e}^{-\mathrm{i}z}}{2}, \ \tan(z) = \frac{\mathrm{e}^{\mathrm{i}z} - \mathrm{e}^{-\mathrm{i}z}}{\mathrm{i}(\mathrm{e}^{\mathrm{i}z} + \mathrm{e}^{-\mathrm{i}z})} \\ \sinh(z) &= \frac{\mathrm{e}^{z} - \mathrm{e}^{-z}}{2}, \ \cosh(z) = \frac{\mathrm{e}^{z} + \mathrm{e}^{-z}}{2}, \ \tanh(z) = \frac{\mathrm{e}^{z} - \mathrm{e}^{-z}}{\mathrm{e}^{z} + \mathrm{e}^{-z}}. \end{aligned}$$

Hence, the following relationships are derived:

$$\sin(z) = -\mathrm{i}\sinh(\mathrm{i}z), \quad \cos(z) = \cosh(\mathrm{i}z), \quad \tan(z) = -\mathrm{i}\tanh(\mathrm{i}z).$$

The complex logarithm, denoted $w = \log(z)$, is the solution to $z = \mathrm{e}^{w}$. We can then define the inverse trigonometric and hyperbolic functions.

We can also define the power function for non-integer powers with $w = z^{\alpha} = \mathrm{e}^{\alpha\log(z)}$. Then power functions can all be written in terms of exponential functions and logarithms. Letting $x = \pi$ in (2.4.1) yields $\mathrm{e}^{\mathrm{i}\pi} = -1$. Furthermore, we can see that exponentiation with an imaginary number is a rotation:

Theorem 2.4.1 *(De Moivre)*: $\forall x \in \mathbb{R}, \forall n \in \mathbb{N}$,

$$(\cos(x) + \mathrm{i}\sin(x))^{n} = \cos(nx) + \mathrm{i}\sin(nx).$$

Since all elementary functions can be written in terms of exponential functions and complex logarithms, we will first study the exponential function.

1. The exponential function e^{z} never vanishes as $|\mathrm{e}^{z}| = \mathrm{e}^{x} > 0$.
2. Since $\mathrm{e}^{2\pi\mathrm{i}} = 1$, it is periodic over $2\pi\mathrm{i}$.
3. It is also an entire function with $(\mathrm{e}^{z})' = \mathrm{e}^{z}$.

Write $\mathrm{e}^{z} = \mathrm{e}^{x+\mathrm{i}y} = \mathrm{e}^{x}(\cos(y) + \mathrm{i}\sin(y))$ where $x, y \in \mathbb{R}$. Let $u(x,y) = \mathfrak{Re}(\mathrm{e}^{z}) = \mathrm{e}^{x}\cos(y)$ and $v(x,y) = \mathfrak{Im}(\mathrm{e}^{z}) = \mathrm{e}^{x}\sin(y)$. The first order derivatives are respectively

$$\frac{\partial u}{\partial x} = \mathrm{e}^{x}\cos(y), \quad \frac{\partial u}{\partial y} = -\mathrm{e}^{x}\sin(y),$$

and

$$\frac{\partial v}{\partial x} = \mathrm{e}^x \sin(y), \quad \frac{\partial v}{\partial y} = \mathrm{e}^x \cos(y),$$

and indeed, the condition described by Theorem 2.1.1 is satisfied.

4. For any two complex numbers z_1 and z_2, $\mathrm{e}^{z_1}\mathrm{e}^{z_2} = \mathrm{e}^{z_1+z_2}$.

In fact, most real exponentiation rules are identical to those in the complex number field. Previously we claimed the periodic properties of e^z. For $U \subseteq \mathbb{C}$, a holomorphic function $f : U \to \mathbb{C}$ is *univalent* over U if it is injective over U. This means that the solutions z_1 and z_2 satisfying $f(z_1) = f(z_2)$ will also always satisfy $z_1 = z_2$.

5. The function e^z is univalent over any horizontal strip of height 2π.

Let $z_1 = x_1 + \mathrm{i}y_1$ and $z_2 = x_2 + \mathrm{i}y_2$, with $x_1, y_1, x_2, y_2 \in \mathbb{R}$, and assume $\mathrm{e}^{z_1} = \mathrm{e}^{z_2}$. Then

$$\mathrm{e}^{x_1}\mathrm{e}^{\mathrm{i}y_1} = \mathrm{e}^{x_2}\mathrm{e}^{\mathrm{i}y_2}.$$

The moduli are equal, and therefore $x_1 = x_2$. By the periodic nature of exponentiation of imaginary numbers, $y_1 - y_2 = 2\pi k$, where $k \in \mathbb{Z}$. To satisfy univalence over a region U, we must exclude distinct points whose imaginary parts differ by an integer multiple of 2π. Thus, we may select U to be any horizontal strip

$$2\pi k \leq \mathfrak{Im}(z) < 2\pi(k+1)$$

or

$$2\pi k < \mathfrak{Im}(z) \leq 2\pi(k+1).$$

Similar to the exponential function, any belt region with thickness 2π is a region over which log is univalent.

Next we examine the complex logarithm.

1. From the periodicity of $z = \mathrm{e}^w$, log is a multi-valued function.
2. Let $z = r\mathrm{e}^{\mathrm{i}\theta}$ and $w = u + \mathrm{i}v$, where $r, \theta, u, v \in \mathbb{R}$. Then

   $$r\mathrm{e}^{\mathrm{i}\theta} = \mathrm{e}^{u+\mathrm{i}v},$$

   and $\mathrm{e}^u = r$, meaning that $u = \log(r)$ and $v = \theta + 2\pi k$, where $k \in \mathbb{Z}$. Then

   $$w = \log(r) + \mathrm{i}(\theta + 2\pi k),$$

   and using modulus-argument notation,

   $$\log(z) = \log|z| + \mathrm{i}\arg(z),$$

   where $\arg(z)$ is the multi-valued argument function. We denote the principal branch of the argument function by

   $$\mathrm{Arg} : \mathbb{C} \setminus \{0\} \to (-\pi, \pi].$$

   The principal branch of $\log(z)$, or $\mathrm{Log}(z)$, can be defined such that $\mathfrak{Im}(\mathrm{Log}(z)) \in (-\pi, \pi]$.

The functions sin and cos, through their exponential form, still satisfy the familiar properties such as their derivatives, periodicity of 2π, parity, sum and difference formulas, and the fundamental identities

$$\sin^2(z) + \cos^2(z) = 1, \quad \sin(z) = \cos\left(\frac{\pi}{2} - z\right).$$

However, due to the extension, some properties do not hold. For instance, $\sin(z)$ and $\cos(z)$ are unbounded, as along the imaginary axis they resemble their hyperbolic counterparts, which are unbounded along the real line.

We now examine the regions over which they are univalent. Consider

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2}.$$

Define the auxiliary functions

$$\xi(z) = iz, \quad \zeta(\xi) = e^{\xi}, \quad w(\zeta) = \frac{\zeta + \frac{1}{\zeta}}{2}.$$

Then

$$\cos(z) = (w \circ \zeta \circ \xi)(z).$$

ξ is clearly univalent on $\mathbb{C}$, as it is a linear map, specifically, a rotation by $\frac{\pi}{2}$ radians followed by scaling. The function ζ is univalent on any domain $U \subseteq \mathbb{C}$ such that for all $\xi_1, \xi_2 \in U$, $\xi_1 - \xi_2 \neq 2\pi i k$ for any $k \in \mathbb{Z}$. If $\xi_1 = iz_1$ and $\xi_2 = iz_2$, then this translates to $z_1 - z_2 \neq 2\pi k$ for $k \in \mathbb{Z}$. The function

$$w(\zeta) = \frac{\zeta + \frac{1}{\zeta}}{2}$$

is univalent on regions excluding pairs (ζ_1, ζ_2) such that $\zeta_1 = \frac{1}{\zeta_2}$. In terms of z, this condition becomes $e^{iz_1}e^{iz_2} \neq 1$, or equivalently, $z_1 + z_2 \neq 2\pi k$ for any $k \in \mathbb{Z}$.

Combining these constraints, we conclude that $\cos(z)$ is univalent on any vertical strip in the complex plane of width π, such as a region of the form

$$\{z \in \mathbb{C} \mid k\pi < \mathfrak{Re}(z) < (k+1)\pi, k \in \mathbb{Z}\}.$$

Let us now consider the specific region

$$\{z \in \mathbb{C} \mid 0 < \mathfrak{Re}(z) < \pi\},$$

and analyze how it is mapped under $\cos(z)$.

1. $\xi(z) = iz$ maps the region $\{z \in \mathbb{C} \mid 0 < \mathfrak{Re}(z) < \pi\}$ to $\{\xi \in \mathbb{C} \mid 0 < \mathfrak{Im}(\xi) < \pi\}$.
2. $\zeta(\xi) = e^{\xi}$ maps this region to the upper half-plane $\mathfrak{Im}(\zeta) > 0$ since $0 < \operatorname{Arg}(\zeta) < \pi$ and $0 < |\zeta|$.
3. $w(\zeta) = \frac{\zeta + \frac{1}{\zeta}}{2}$ maps $\mathfrak{Im}(\zeta) > 0$ to $\mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$.

Thus, the composition $\cos(z) = w \circ \zeta \circ \xi$ is univalent on the strip

$$\{z \in \mathbb{C} \mid 0 < \mathfrak{Re}(z) < \pi\},$$

and the image of this strip under cos is

$$\mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)).$$

We will now analyze the inverse cosine function, denoted $\arccos(z)$. Consider

$$z = \frac{e^{iw} + e^{-iw}}{2}.$$

Then

$$\begin{aligned}
\left(\mathrm{e}^{\mathrm{i}w}\right)^2 + 1 &= 2z\mathrm{e}^{\mathrm{i}w} \\
\mathrm{e}^{\mathrm{i}w} &= \frac{2z \pm \sqrt{4z^2 - 4}}{2} \\
w &= -\mathrm{i}\log\left(z \pm \sqrt{z^2 - 1}\right).
\end{aligned}$$

Then arccos is also a multi-valued function. We can also define

$$\arcsin(z) = \frac{\pi}{2} - \arccos(z).$$

Lastly, we will examine the power function. Let $\alpha = u + \mathrm{i}v$ where $u, v \in \mathbb{R}$. Then

$$z^\alpha = \exp(\alpha \log(z)) = \exp((u + \mathrm{i}v)(\log|z| + \mathrm{i}\arg(z))),$$

and in polar form,

$$z^\alpha = \exp(u \log|z| - v \arg(z)) \exp(\mathrm{i}(v \log|z| + u \arg(z))).$$

Let

$$r_k = \exp(u \log|z| - v \arg(z))$$

and

$$\theta_k = v \log|z| + u \arg(z).$$

Then $z^\alpha = r_k \mathrm{e}^{\mathrm{i}\theta_k}$, where $k \in \mathbb{Z}$. Analyzing the coefficient of v in the exponent of r_k, z^α is multi-valued if $v \neq 0$.

Then assuming $v = 0$, we have

$$z^\alpha = |z|^u \exp(\mathrm{i}u \arg(z)).$$

Doing casework on α,

1. If $\alpha = u \in \mathbb{Z}$, then u can be absorbed into k, and z^α is single-valued.
2. If $\alpha = u \in \mathbb{Q}$ with reduced fractional form $\frac{p}{q}$, where $p, q \in \mathbb{Z}$, $q > 0$, and $\gcd(p, q) = 1$, then the multi-valued function z^α is given by

$$z^\alpha = |z|^{\frac{p}{q}} \exp\left(\mathrm{i}\left(\frac{p}{q}\right)(\mathrm{Arg}(z) + 2\pi k)\right) = |z|^{\frac{p}{q}} \exp\left(\mathrm{i}\left(\frac{p}{q}\right)\mathrm{Arg}(z)\right) \exp\left(2\pi\mathrm{i}\left(\frac{p}{q}\right)k\right),$$

for $k \in \mathbb{Z}$. These values are periodic with period q, since

$$\exp\left(2\pi\mathrm{i}\left(\frac{p}{q}\right)(k+q)\right) = \exp\left(2\pi\mathrm{i}\left(\frac{p}{q}\right)k\right)\exp(2\pi\mathrm{i}p) = \exp\left(2\pi\mathrm{i}\left(\frac{p}{q}\right)k\right),$$

as $\exp(2\pi\mathrm{i}p) = 1$ for integer p. To prove there are exactly q distinct values, consider $k = 0, 1, \ldots, q - 1$. The exponential factors are $\exp\left(2\pi\mathrm{i}\left(\frac{p}{q}\right)k\right)$. These are distinct if, for $0 \leq j < k \leq q - 1$,

$$\exp\left(2\pi\mathrm{i}\left(\frac{p}{q}\right)j\right) \neq \exp\left(2\pi\mathrm{i}\left(\frac{p}{q}\right)k\right),$$

which holds unless $\left(\frac{p}{q}\right)(k-j) \in \mathbb{Z}$, or equivalently unless q divides $p(k-j)$. Since $\gcd(p,q)=1$, q must divide $k-j$, but $|k-j|<q$ and $k-j \neq 0$, a contradiction. Thus, z^α has exactly q distinct values.

1. If $\alpha = u \in \mathbb{R} \setminus \mathbb{Q}$, then z^α is infinite-valued.

Lastly, there exist series representations of functions using power functions, namely Taylor series, and trigonometric functions, namely Fourier series. There does not exist another distinct representation using exponential functions, as trigonometric functions can be written in terms of them.

2.5 The Extended Complex Plane and its Spherical Representation

All complex numbers form a field that extends the real number field. A complex number $\alpha + \mathrm{i}\beta$ can be visualized on a rectangular plane as the point (α, β), with two axes: the real axis and the imaginary axis. It is well known that any complex number also has the polar form $r\mathrm{e}^{\mathrm{i}\theta} = r(\cos\theta + \mathrm{i}\sin\theta)$.

The point at infinity, ∞, extends $\mathbb{C}$ to

$$\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

The following arithmetic operations are defined: for all $a \in \mathbb{C}$,

$$a + \infty = \infty + a = \infty,$$

and for all $b \in \mathbb{C} \setminus \{0\}$,

$$b \cdot \infty = \infty \cdot b = \infty, \quad \frac{a}{\infty} = 0.$$

Let

$$S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1^2 + x_2^2 + x_3^2 = 1\}.$$

There exists a *stereographic projection* of S^2 onto $\hat{\mathbb{C}}$. For every point other than $(0,0,1)$, there is a corresponding complex number

$$z = \frac{x_1 + \mathrm{i}x_2}{1 - x_3}. \tag{2.5.1}$$

This correspondence between $\mathbb{C}$ and $S^2 \setminus \{(0,0,1)\}$ is injective. In fact, the inverse can be solved for:

$$|z|^2 = \frac{1 - x_3^2}{(1 - x_3)^2} = \frac{1 + x_3}{1 - x_3},$$

which results in

$$x_3 = \frac{|z|^2 - 1}{|z|^2 + 1},$$

and consequently,

$$x_1 = \mathfrak{Re}(z)(1 - x_3) = \frac{z + \overline{z}}{|z|^2 + 1},$$

$$x_2 = \mathfrak{Im}(z)(1 - x_3) = \frac{z - \overline{z}}{\mathrm{i}|z|^2 + \mathrm{i}}.$$

By letting ∞ correspond to $(0,0,1)$, the bijection is complete. The sphere S^2 is also called the *Riemann sphere*. The region given by the disk $|z| < 1$ corresponds to $x_3 < 0$, and the region $|z| > 1$ corresponds to $x_3 > 0$.

We will now give a geometric visualization of this projection. Let $z = x + \mathrm{i}y$. Then we obtain

$$x = \frac{x_1}{1 - x_3} \quad \text{and} \quad y = \frac{x_2}{1 - x_3}.$$

Therefore,

$$x : y : 1 = x_1 : x_2 : 1 - x_3.$$

It follows that the points $(0,0,0)$, $(x,y,1)$, and $(x_1, x_2, 1 - x_3)$ are collinear in $\mathbb{R}^3$. Under the linear map

$$\mathbf{v} \mapsto \mathbf{v}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} + (0,0,1),$$

we get that $(0,0,1)$, $(x,y,0)$, and (x_1,x_2,x_3) are collinear. In other words, this correspondence is a central projection with center $(0,0,1)$, projecting the points from $S^2 \setminus (0,0,1)$ onto $\mathbb{C}$. Let this center correspond to ∞. In this representation, $\infty \in \hat{\mathbb{C}}$ is no longer considered to be "special".

It is worth noting that in several geometric contexts, an alternative paradigm exists where we let S be the sphere centered at $\left(0,0,\frac{1}{2}\right)$ of diameter 1, and project points from the north pole $(0,0,1)$ onto the horizontal plane of tangency. Later in Nevanlinna theory, specifically in @ sec:nevanlinnatheory, we will observe that in some sense this is the more natural object to study. The corresponding equations are then

$$x_1 = \frac{\mathfrak{Re}(z)}{|z|^2 + 1}, \quad x_2 = \frac{\mathfrak{Im}(z)}{|z|^2 + 1}, \quad x_3 = \frac{|z|^2}{|z|^2 + 1}.$$

The forward projection remains unchanged. Lastly, we define the upper half-plane; for the following sections, let

$$\mathbb{H}^+ = \{z \in \mathbb{C} \mid \mathfrak{Im}(z) > 0\}$$

3 Complex Integration

3.1 The Cauchy–Goursat Theorem

It is important to know the differential 2-forms even for a single variable complex function. Consider $z = x + \mathrm{i}y$ and $\overline{z} = x - \mathrm{i}y$. We can then define their corresponding differentials:

$$\mathrm{d}z = \mathrm{d}x + \mathrm{i}\,\mathrm{d}y, \mathrm{d}\overline{z} = \mathrm{d}x - \mathrm{i}\,\mathrm{d}y.$$

The antisymmetric properties of differential forms still hold in complex space. By taking the wedge product of the two basis complex differential forms, we get

$$\begin{aligned} \mathrm{d}\overline{z} \wedge \mathrm{d}z &= (\,\mathrm{d}x - \mathrm{i}\,\mathrm{d}y) \wedge (\,\mathrm{d}x + \mathrm{i}\,\mathrm{d}y) \\ &= 2\mathrm{i}\,\mathrm{d}x \wedge \mathrm{d}y. \end{aligned}$$

Analogous to the real case, a 0-form is defined as a scalar-valued function in the form $f(z,\overline{z})$, a 1-form in the form $\omega_0\,\mathrm{d}z + \omega_1\,\mathrm{d}\overline{z}$, and a 2-form as $\omega_0\,\mathrm{d}z \wedge \mathrm{d}\overline{z}$. The exterior differential operator for this one-dimensional case is defined as $\partial + \overline{\partial}$, where $\partial = \mathrm{d}z \wedge \frac{\partial}{\partial z}$ and $\overline{\partial} = \mathrm{d}\overline{z} \wedge \frac{\partial}{\partial \overline{z}}$. Occasionally, one will informally use ∂ and $\overline{\partial}$ as an abbreviation for $\frac{\partial}{\partial z}$ and $\frac{\partial}{\partial \overline{z}}$ respectively.

Theorem 3.1.1 *(Lusin Area Theorem)*: For a region $U \subset \mathbb{C}$ and $f : U \to \mathbb{C}$ univalent, the area of the image $f(U)$ is equal to

$$\int_U |f'(z)|^2 \, \mathrm{d}A.$$

Proof: We aim to find

$$\int_{f(U)} \mathrm{d}A.$$

By the properties above,

$$\begin{aligned}\int_{f(U)} \mathrm{d}u \wedge \mathrm{d}v &= \frac{\mathrm{i}}{2} \int_{f(U)} \mathrm{d}w \wedge \mathrm{d}\overline{w} = \frac{\mathrm{i}}{2} \int_U \mathrm{d}f(z) \wedge \mathrm{d}\overline{f(z)} \\ &= \frac{\mathrm{i}}{2} \int_U [f'(z) \, \mathrm{d}z] \wedge \left[\overline{f'(z)} \, \mathrm{d}\overline{z}\right] = \int_U |f'(z)|^2 \, \mathrm{d}x \wedge \mathrm{d}y,\end{aligned}$$

as desired. □

Remark: The Jacobian determinant of u, v with respect to x, y, for a holomorphic function $f(z) = u(x, y) + \mathrm{i}v(x, y)$ is equal to

$$\begin{vmatrix} u'_x & u'_y \\ v'_x & v'_y \end{vmatrix} = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} = \frac{\partial u}{\partial x}^2 + \frac{\partial u}{\partial y}^2 = |f'(z)|^2$$

by (2.1.6).

Theorem 3.1.2 *(Green's Theorem, Complex Form)*: Let $U \subset \mathbb{C}$ be bounded with a piecewise smooth boundary ∂U. For two scalar functions $\omega_1 = \omega_1(z, \overline{z})$ and $\omega_2 = \omega_2(z, \overline{z})$ satisfying $\omega_1, \omega_2 \in C^1\left(\overline{U}\right)$, define the 1-form $\omega = \omega_1 \, \mathrm{d}z + \omega_2 \, \mathrm{d}\overline{z}$. Then,

$$\int_{\partial U} \omega = \int_U \mathrm{d}\omega. \tag{3.1.1}$$

Proof: For real-valued functions $\xi_1, \xi_2, \eta_1, \eta_2$, let

$$\omega_1 = \xi_1 + \mathrm{i}\eta_1 \quad \text{and let} \quad \omega_2 = \xi_2 + \mathrm{i}\eta_2.$$

Then,

$$\begin{aligned}\omega &= (\xi_1 + \mathrm{i}\eta_1) \, \mathrm{d}z + (\xi_2 + \mathrm{i}\eta_2) \, \mathrm{d}\overline{z} && (3.1.2) \\ &= (\xi_1 + \mathrm{i}\eta_1)(\mathrm{d}x + \mathrm{i}\,\mathrm{d}y) + (\xi_2 + \mathrm{i}\eta_2)(\mathrm{d}x - \mathrm{i}\,\mathrm{d}y) \\ &= \xi_1 \, \mathrm{d}x + \mathrm{i}\eta_1 \, \mathrm{d}x + \mathrm{i}\xi_1 \, \mathrm{d}y - \eta_1 \, \mathrm{d}y + \xi_2 \, \mathrm{d}x + \mathrm{i}\eta_2 \, \mathrm{d}x - \mathrm{i}\xi_2 \, \mathrm{d}y + \eta_2 \, \mathrm{d}y \\ &= [(\xi_1 + \xi_2) \, \mathrm{d}x + (\eta_2 - \eta_1) \, \mathrm{d}y] + \mathrm{i}[(\eta_1 + \eta_2) \, \mathrm{d}x + (\xi_1 - \xi_2) \, \mathrm{d}y] && (3.1.3)\end{aligned}$$

Each of $\xi_1, \xi_2, \eta_1, \eta_2$ are real-valued functions that can be represented with a domain of $\mathbb{R}^2$. We then apply the $\mathrm{d} = \partial + \overline{\partial}$ definition of the exterior derivative and relate it to (3.1.1). Starting from (3.1.2),

$$\begin{aligned}
\mathrm{d}\omega &= \left(\partial+\overline{\partial}\right)(\xi_1+\mathrm{i}\eta_1)\,\mathrm{d}z+\left(\partial+\overline{\partial}\right)(\xi_2+\mathrm{i}\eta_2)\,\mathrm{d}\overline{z} \\
&= \left(\frac{\partial\xi_1}{\partial\overline{z}}+\mathrm{i}\frac{\partial\eta_1}{\partial\overline{z}}\right)\mathrm{d}\overline{z}\wedge\mathrm{d}z+\left(\frac{\partial\xi_2}{\partial z}+\mathrm{i}\frac{\partial\eta_2}{\partial z}\right)\mathrm{d}z\wedge\mathrm{d}\overline{z} \\
&= 2\left(\mathrm{i}\frac{\partial\xi_1}{\partial\overline{z}}-\frac{\partial\eta_1}{\partial\overline{z}}-\mathrm{i}\frac{\partial\xi_2}{\partial z}+\frac{\partial\eta_2}{\partial z}\right)\mathrm{d}x\wedge\mathrm{d}y \\
&= \left(\mathrm{i}\frac{\partial\xi_1}{\partial x}-\frac{\partial\xi_1}{\partial y}-\frac{\partial\eta_1}{\partial x}-\mathrm{i}\frac{\partial\eta_1}{\partial y}-\mathrm{i}\frac{\partial\xi_2}{\partial x}-\frac{\partial\xi_2}{\partial y}+\frac{\partial\eta_2}{\partial x}-\mathrm{i}\frac{\partial\eta_2}{\partial y}\right)\mathrm{d}x\wedge\mathrm{d}y \\
&= \left(\frac{\partial\eta_2}{\partial x}-\frac{\partial\xi_1}{\partial y}-\frac{\partial\eta_1}{\partial x}-\frac{\partial\xi_2}{\partial y}\right)\mathrm{d}A+\mathrm{i}\left(\frac{\partial\xi_1}{\partial x}-\frac{\partial\eta_1}{\partial y}-\frac{\partial\xi_2}{\partial x}-\frac{\partial\eta_2}{\partial y}\right)\mathrm{d}A. \qquad (3.1.4)
\end{aligned}$$

From (3.1.3), we can apply Theorem 1.2.8. For the real component of ω, we obtain

$$\oint_{\partial U}(\xi_1+\xi_2)\,\mathrm{d}x+(\eta_2-\eta_1)\,\mathrm{d}y=\iint_U\left(\frac{\partial\eta_2}{\partial x}-\frac{\partial\xi_1}{\partial y}-\frac{\partial\eta_1}{\partial x}-\frac{\partial\xi_2}{\partial y}\right)\mathrm{d}x\,\mathrm{d}y,$$

and for the imaginary component,

$$\oint_{\partial U}(\eta_1+\eta_2)\,\mathrm{d}x+(\xi_1-\xi_2)\,\mathrm{d}y=\iint_U\left(\frac{\partial\xi_1}{\partial x}-\frac{\partial\eta_1}{\partial y}-\frac{\partial\xi_2}{\partial x}-\frac{\partial\eta_2}{\partial y}\right)\mathrm{d}x\,\mathrm{d}y,$$

and the integrands on the right side both match those of (3.1.4). □

The theorem above is only a specific case of the Stokes-Cartan Theorem (Theorem 1.2.11). However, it proves the validity of the treatment of the ∂ and $\overline{\partial}$ operators, and the generalization to forms with basis $\mathrm{d}z$ and $\mathrm{d}\overline{z}$.

Theorem 3.1.3 *(Cauchy–Pompeiu)*: Let $U\subset\mathbb{C}$ be bounded with a piecewise C^1 boundary ∂U. Let $f(z)\in C^1\left(\overline{U}\right)$. Then $\forall z\in U\setminus\partial U$,

$$f(z)=\frac{1}{2\pi\mathrm{i}}\left(\oint_{\partial U}\frac{f(\zeta)}{\zeta-z}\,\mathrm{d}\zeta-\int_U\frac{\partial f(\zeta)}{\partial\overline{\zeta}}\frac{\mathrm{d}\overline{\zeta}\wedge\mathrm{d}\zeta}{\zeta-z}\right).$$

Proof: Since $z\in U\setminus\partial U$, $\exists\varepsilon>0$ such that $D(z,\varepsilon)\subset U$. Consider the complex differential form

$$\frac{f(\zeta)\,\mathrm{d}\zeta}{\zeta-z}$$

with a singularity at $\zeta=z$. Consider the region $U\setminus D(z,\varepsilon)$. Since $f\in C^1\left(\overline{U}\right)$, by applying Green's Theorem (Theorem 3.1.2),

$$\int_{U\setminus D(z,\varepsilon)}\mathrm{d}\left(\frac{f(\zeta)\,\mathrm{d}\zeta}{\zeta-z}\right)=\oint_{\partial U}\frac{f(\zeta)\,\mathrm{d}\zeta}{\zeta-z}-\oint_{\partial D(z,\varepsilon)}\frac{f(\zeta)\,\mathrm{d}\zeta}{\zeta-z}. \qquad (3.1.5)$$

By properties of d, the expression is equal to

$$\begin{aligned}
\int_{U\setminus D(z,\varepsilon)}\left(\partial+\overline{\partial}\right)\left(\frac{f(\zeta)}{\zeta-z}\right)\wedge\mathrm{d}\zeta=\int_{U\setminus D(z,\varepsilon)}&\frac{\partial}{\partial\zeta}\left(\frac{f(\zeta)}{\zeta-z}\right)\mathrm{d}\zeta\wedge\mathrm{d}\zeta \\
&+\frac{\partial}{\partial\overline{\zeta}}\left(\frac{f(\zeta)}{\zeta-z}\right)\mathrm{d}\overline{\zeta}\wedge\mathrm{d}\zeta.
\end{aligned}$$

The first term in the integrand vanishes as it contains $\mathrm{d}\zeta\wedge\mathrm{d}\zeta$. The second term can be simplified using the fact that $\frac{\partial\frac{1}{\zeta-z}}{\partial\overline{\zeta}}=0$, leading to

$$\int_{U \setminus D(z,\varepsilon)} \frac{\partial f}{\partial \overline{\zeta}} \cdot \frac{\mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta}{\zeta - z}.$$

The rightmost term in (3.1.5) can be parameterized with $\zeta = z + \varepsilon \mathrm{e}^{\mathrm{i}t}$, $t \in [0, 2\pi]$. Then,

$$\begin{aligned}
\oint_{\partial D(z,\varepsilon)} \frac{f(\zeta)\,\mathrm{d}\zeta}{\zeta - z} &= \int_0^{2\pi} \frac{f(z + \varepsilon \mathrm{e}^{\mathrm{i}t})}{\varepsilon \mathrm{e}^{\mathrm{i}t}} \cdot \mathrm{i}\varepsilon \mathrm{e}^{\mathrm{i}t}\,\mathrm{d}t \\
&= \mathrm{i} \int_0^{2\pi} f(z + \varepsilon \mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t \\
&= \mathrm{i} \int_0^{2\pi} \left(f(z + \varepsilon \mathrm{e}^{\mathrm{i}t}) - f(z)\right)\mathrm{d}t + \mathrm{i} \int_0^{2\pi} f(z)\,\mathrm{d}t.
\end{aligned}$$

Because $f \in C^1\left(\overline{U}\right)$, by Proposition 1.2.2, f is Lipschitz continuous on $\overline{U}$, and $\exists M \in \mathbb{R}_{>0}$ such that $\forall z_0, z_1 \in \overline{U}$, $|f(z_1) - f(z_0)| \leq M|z_1 - z_0|$. On $\partial D(z, \varepsilon)$, we get that $\left|f(z + \varepsilon \mathrm{e}^{\mathrm{i}t}) - f(z)\right| \leq M\varepsilon$. Therefore,

$$\left|\int_0^{2\pi} \left(f(z + \varepsilon \mathrm{e}^{\mathrm{i}t}) - f(z)\right)\mathrm{d}t\right| \leq \int_0^{2\pi} \left|f(z + \varepsilon \mathrm{e}^{\mathrm{i}t}) - f(z)\right|\mathrm{d}t \leq 2M\pi\varepsilon,$$

which approaches 0 as $\varepsilon \to 0$. Taking this limit, we obtain

$$2\pi \mathrm{i} f(z) = \oint_{\partial U} \frac{f(\zeta)\,\mathrm{d}\zeta}{\zeta - z} - \int_U \frac{\partial f}{\partial \overline{\zeta}} \cdot \frac{\mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta}{\zeta - z} + \lim_{\varepsilon \to 0} \int_{D(z,\varepsilon)} \frac{\partial f}{\partial \overline{\zeta}} \cdot \frac{\mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta}{\zeta - z}. \tag{3.1.6}$$

We then aim to prove that

$$\lim_{\varepsilon \to 0} \int_{D(z,\varepsilon)} \frac{\partial f}{\partial \overline{\zeta}} \cdot \frac{\mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta}{\zeta - z} = 0. \tag{3.1.7}$$

Notice that since $f \in C^1\left(\overline{U}\right)$, by Theorem 1.2.13, $\exists M' \in \mathbb{R}_{>0}$ such that $\forall \zeta \in \overline{U}$, $\left|\frac{\partial f}{\partial \overline{\zeta}}\right| \leq M'$. Then,

$$\lim_{\varepsilon \to 0} \left|\int_{D(z,\varepsilon)} \frac{\partial f}{\partial \overline{\zeta}} \cdot \frac{\mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta}{\zeta - z}\right| \leq M' \lim_{\varepsilon \to 0} \left|\int_{D(z,\varepsilon)} \frac{1}{\zeta - z}\,\mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta\right|.$$

By a change of variables to a polar coordinate system centered at z, we obtain

$$M' \lim_{\varepsilon \to 0} \left|\int_{D(z,\varepsilon)} \frac{1}{r\mathrm{e}^{\mathrm{i}\theta}}\,\mathrm{d}\left(z + r\mathrm{e}^{-\mathrm{i}\theta}\right) \wedge \mathrm{d}\left(z + r\mathrm{e}^{\mathrm{i}\theta}\right)\right|,$$

and by expansion of the wedge product,

$$\begin{aligned}
M' \lim_{\varepsilon \to 0} \left|\int_{D(z,\varepsilon)} \frac{2\mathrm{i}}{\mathrm{e}^{\mathrm{i}\theta}}\,\mathrm{d}r \wedge \mathrm{d}\theta\right| &= 2M' \lim_{\varepsilon \to 0} \left|\int_{D(z,\varepsilon)} \frac{1}{\mathrm{e}^{\mathrm{i}\theta}}\,\mathrm{d}r \wedge \mathrm{d}\theta\right| \\
&= 2M' \lim_{\varepsilon \to 0} \left|\int_0^{2\pi} \int_0^{\varepsilon} \mathrm{e}^{-\mathrm{i}\theta}\,\mathrm{d}r\,\mathrm{d}\theta\right| \\
&= 0.
\end{aligned}$$

Then from rearranging (3.1.6), we obtain:

$$f(z)=\frac{1}{2\pi\mathrm{i}}\left(\oint_{\partial U}\frac{f(\zeta)\,\mathrm{d}\zeta}{\zeta-z}-\int_U\frac{\partial f}{\partial\overline{\zeta}}\cdot\frac{\mathrm{d}\overline{\zeta}\wedge\mathrm{d}\zeta}{\zeta-z}\right).\qquad\square$$

Corollary 3.1.3.1: Let $f:\mathbb{C}\to\mathbb{C}$ be a continuously differentiable, compactly supported function. Then

$$f(z)=-\frac{1}{\pi}\iint_{\mathbb{C}}\frac{\partial f}{\partial\overline{\zeta}}\frac{\mathrm{d}\xi\,\mathrm{d}\eta}{\zeta-z}$$

for all $z\in\mathbb{C}$ where $\zeta=\xi+\mathrm{i}\eta$.

Proof: Choose $R>0$ such that $D(0,R)\supset\operatorname{supp}(f)$. By the Cauchy–Pompeiu Theorem (Theorem 3.1.3), we have

$$f(z)=\frac{1}{\pi}\left(\frac{1}{2\mathrm{i}}\oint_{\partial D(0,R)}\frac{f(\zeta)\,\mathrm{d}\zeta}{\zeta-z}-\iint_{D(0,R)}\frac{\partial f}{\partial\overline{\zeta}}\frac{\mathrm{d}\xi\,\mathrm{d}\eta}{\zeta-z}\right).$$

Then the proof is complete given that f vanishes on $\partial D(0,R)$ and by letting $R\to\infty$. $\square$

In complex analysis, when integrating over a region that contains a singularity, it is common to exclude a small disk of radius ε around the singularity, perform the integration over the punctured region, and then take the limit as $\varepsilon\to0$. As in the proof above, the displayed estimate for the integral over the removed disk is still necessary in confirmation, although it is typically tacitly elided.

From the above result, we can directly obtain the following theorem:

Theorem 3.1.4 *(Cauchy's Integral Formula)*: Let $U\subset\mathbb{C}$ be an open region with a piecewise C^1 boundary ∂U, and let $f\in C^1\left(\overline{U}\right)$ be holomorphic on U. Then for all $z\in U$,

$$f(z)=\frac{1}{2\pi\mathrm{i}}\oint_{\partial U}\frac{f(\zeta)}{\zeta-z}\,\mathrm{d}\zeta.\tag{3.1.8}$$

Proof: By (2.1.5), for $f\left(\zeta,\overline{\zeta}\right)$, $\frac{\partial f}{\partial\overline{\zeta}}=0$. Applying the Cauchy–Pompeiu Theorem (Theorem 3.1.3), the area integral vanishes, and (3.1.8) consequently follows. $\square$

Theorem 3.1.5 *(Cauchy's Integral Theorem)*: Let $U\subset\mathbb{C}$ be an open region with piecewise C^1 boundary ∂U. For a function $f(z)\in C^1\left(\overline{U}\right)$ holomorphic over U,

$$\oint_{\partial U}f(\zeta)\,\mathrm{d}\zeta=0.$$

Proof: Let $\psi(z)=zf(z)$. Applying Theorem 3.1.4 on $\psi(\zeta)$ with $z=0$, we obtain

$$0=\frac{1}{2\pi\mathrm{i}}\oint_{\partial U}\frac{\psi(\zeta)}{\zeta}\,\mathrm{d}\zeta=\frac{1}{2\pi\mathrm{i}}\oint_{\partial U}f(\zeta)\,\mathrm{d}\zeta.$$

Alternatively, we can use Green's Theorem (Theorem 3.1.2) with $\omega=f(\zeta)\,\mathrm{d}\zeta$:

$$\oint_{\partial U}f(\zeta)\,\mathrm{d}\zeta=\oint_{\partial U}\omega=\int_U\mathrm{d}\omega=\int_U\frac{\partial f}{\partial\overline{\zeta}}\,\mathrm{d}\overline{\zeta}\wedge\mathrm{d}\zeta=0.\qquad\square$$

Theorem 3.1.6: For a compactly supported function $\psi(z)\in C^1(\mathbb{C})$, a solution satisfying $u(z)\in C^1(\mathbb{C})$ to the non-homogeneous Cauchy–Riemann equation

$$\frac{\partial u(z)}{\partial\overline{z}}=\psi(z)$$

is

$$u(z) = -\frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{\psi(\zeta)}{\zeta - z} \, \mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta. \tag{3.1.9}$$

Proof: Split $\mathbb{C}$ into $\mathbb{C} \setminus D(z, \varepsilon)$ and $\overline{D(z, \varepsilon)}$. For all $\varepsilon > 0$, the integral

$$-\frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C} \setminus D(z,\varepsilon)} \frac{\psi(\zeta)}{\zeta - z} \, \mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta$$

is continuous. Since $\psi(\zeta)$ is compactly supported over $\mathbb{C}$ and continuous, by Theorem 1.2.13, it is bounded. Then the limit

$$\lim_{\varepsilon \to 0} \left(-\frac{1}{2\pi \mathrm{i}} \int_{D(z,\varepsilon)} \frac{\psi(\zeta)}{\zeta - z} \, \mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta \right) = 0.$$

Therefore, (3.1.9) is continuous. A trivial substitution can be used to rewrite

$$u(z) = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{\psi(\zeta + z)}{\zeta} \, \mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}$$

Then,

$$\frac{u(z + \Delta z) - u(z)}{\Delta z} = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{\psi(\zeta + z + \Delta z) - \psi(\zeta + z)}{\Delta z \zeta} \, \mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}. \tag{3.1.10}$$

For a fixed z, the value of

$$\frac{\psi(\zeta + z + \Delta z) - \psi(\zeta + z)}{\Delta z}$$

tends to $\frac{\partial \psi(\zeta + z)}{\partial \zeta}$ as $\Delta z \to 0$. Because $\psi(\zeta) = \psi(\zeta + z)$ has compact support and is C^1, by Proposition 1.2.2, it is Lipschitz continuous for a constant M. Let $|\Delta z| < 1$ and let $K = \left\{ w \in \mathbb{C} : \inf_{\zeta \in \operatorname{supp} \phi} |w - \zeta| \leq 1 \right\}$. Then,

$$\left| \frac{\psi(\zeta + z + \Delta z) - \psi(\zeta + z)}{\Delta z} \right| \leq M,$$

and specifically, when $\zeta + z \neg \in K$,

$$\frac{\psi(\zeta + z + \Delta z) - \psi(\zeta + z)}{\Delta z} = 0.$$

As shown above, the integrand is uniformly bounded by M, which has a convergent integral of $\int_K M \, \mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}$, the limit $\Delta z \to 0$ may commute with the integral in (3.1.10). Let $\zeta = \xi + \mathrm{i}\eta$. From the real axis,

$$\frac{\partial u}{\partial x}(z) = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{1}{\zeta} \frac{\partial \psi}{\partial \xi}(\zeta + z) \, \mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta} = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{\partial \psi(\zeta)}{\partial \xi} \cdot \frac{1}{\zeta - z} \, \mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}. \tag{3.1.11}$$

From the imaginary axis,

$$\frac{\partial u}{\partial y}(z) = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{1}{\zeta} \frac{\partial \psi}{\partial \eta}(\zeta + z) \, \mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta} = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{\partial \psi(\zeta)}{\partial \eta} \cdot \frac{1}{\zeta - z} \, \mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}. \tag{3.1.12}$$

Since $\psi \in C^1(\mathbb{C})$ and has Lipschitz constant M, (3.1.11), (3.1.12) are both continuous (by the same argument for the continuity of $u(z)$). Thus, $u \in C^1(\mathbb{C})$. It follows from the two equations that

$$\frac{\partial u}{\partial \bar{z}} = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{\partial \psi}{\partial \bar{\zeta}} \cdot \frac{1}{\zeta - z} \,\mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta} = \frac{1}{2\pi \mathrm{i}} \int_{K} \frac{\partial \psi}{\partial \bar{\zeta}} \cdot \frac{1}{\zeta - z} \,\mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta}.$$

By Corollary 3.1.3.1,

$$\frac{\partial u}{\partial \bar{z}} = \psi(z).$$

□

Remark: In the first part, we established that a function $\psi(z) \in C^0(\mathbb{C})$ with compact support satisfies

$$u(z) = -\frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{\psi(\zeta)}{\zeta - z} \,\mathrm{d}\bar{\zeta} \wedge \mathrm{d}\zeta \in C^0(\mathbb{C}).$$

If $\psi(z) \in C^1(\mathbb{C})$, then the first order derivatives of $u(z)$ can be written in the same form ((3.1.11), (3.1.12)) since $\frac{\partial \psi}{\partial \xi}, \frac{\partial \psi}{\partial \eta} \in C^0(\mathbb{C})$ and are also compactly supported. Then they too are continuous functions, and $u(z) \in C^1(\mathbb{C})$.

Then using the same argument, In general, for $\psi(z) \in C^k(\mathbb{C})$, the same process can be used recursively to find that $u(z) \in C^k(\mathbb{C})$ as well.

If the support of $\psi(z)$ is the union of infinitely many or finitely many disjoint compact sets, then the integral in (3.1.9) can be split into a sum of integrals over each compact set, and the same argument applies to each term.

When Cauchy formalized Theorem 3.1.4, Theorem 3.1.5, he included the necessary condition that $f(z) \in C^1\left(\overline{U}\right)$. It was later shown that all such holomorphic functions had holomorphic derivatives, and this condition was thus later dropped by Goursat:

Lemma 3.1.1: Let $f : G \to \mathbb{C}$ be a continuous function defined for a region $G \subseteq \mathbb{C}$. Let $\Gamma \subset G$ be a rectifiable piecewise smooth curve. Then $\forall \varepsilon > 0$, there exists a polygonal chain $P \subset G$ inscribing Γ (each vertex lies on Γ) where

$$\left| \int_{\Gamma} f(z) \,\mathrm{d}z - \int_{P} f(z) \,\mathrm{d}z \right| < \varepsilon.$$

Proof: Because $f \in C^0(G)$, there is a compact set $D \subseteq G$ enclosing Γ and is the closure of some open set. By Theorem 1.2.15, $\forall \varepsilon > 0, \exists \delta > 0$ such that $\forall z', z'' \in D$ satisfying $|z'' - z'| < \delta$, $|f(z'') - f(z')| < \varepsilon$. Partition Γ into $n \in \mathbb{N}$ curves $\gamma_0, \gamma_1, ..., \gamma_{n-1}$ between points $z_0, z_1, ...z_n$ such that $\forall k \in \{0, 1, ..., n-1\}$ the length of γ_k is less than δ. For all $k \in \{0, 1, ..., n-1\}$, let l_k denote the straight line segment connecting z_k and z_{k+1}. The length of l_k is less than δ as well. Then let

$$P = \bigcup_{k=0}^{n-1} l_k.$$

Over the partition formed with γ_k, the integral

$$\int_{\Gamma} f(z) \,\mathrm{d}z$$

can be approximated with the Riemann sum

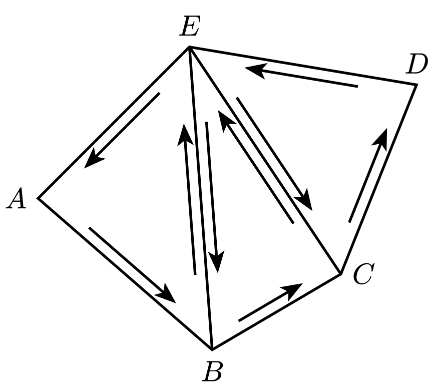

Figure 2: Closed triangulated polygonal chain

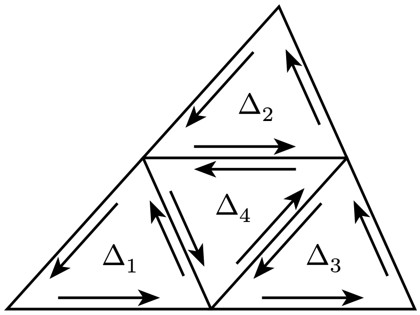

Figure 3: Quadrisection of int Δ

$$S = \sum_{k=0}^{n-1} f(z_k)\Delta z_k$$

where

$$\Delta z_k = z_{k+1} - z_k = \int_{\gamma_k} \mathrm{d}z = \int_{l_k} \mathrm{d}z.$$

Then the sum above can be written as

$$S = \sum_{k=0}^{n-1} \int_{\gamma_k} f(z_k)\,\mathrm{d}z = \sum_{k=0}^{n-1} \int_{l_k} f(z_k)\,\mathrm{d}z,$$

and it follows that

$$\left|\int_\Gamma f(z)\,\mathrm{d}z - S\right| = \left|\sum_{k=0}^{n-1} \int_{\gamma_k} [f(z) - f(z_k)]\,\mathrm{d}z\right| < \varepsilon \cdot \mathrm{length}(\Gamma)$$

and

$$\left|\int_P f(z)\,\mathrm{d}z - S\right| = \left|\sum_{k=0}^{n-1} \int_{l_k} [f(z) - f(z_k)]\,\mathrm{d}z\right| < \varepsilon \cdot \mathrm{length}(P) < \varepsilon \cdot \mathrm{length}(\Gamma)$$

where $\mathrm{length}(\Gamma)$ is the length of Γ and $\mathrm{length}(P)$ is the length of P. Then,

$$\left|\int_\Gamma f(z)\,\mathrm{d}z - \int_P f(z)\,\mathrm{d}z\right| \le \left|\int_\Gamma f(z)\,\mathrm{d}z - S\right| + \left|\int_P f(z)\,\mathrm{d}z - S\right| \le 2\varepsilon \cdot \mathrm{length}(\Gamma). \qquad \square$$

Lemma 3.1.2 *(Goursat)*: Given a holomorphic function $f(z)$ on a simply connected region $U \subseteq \mathbb{C}$, for any piecewise C^1 closed curve $\Gamma \subset U$,

$$\int_\Gamma f(\zeta)\,\mathrm{d}\zeta = 0. \tag{3.1.13}$$

Proof: By Lemma 3.1.1, $\forall \varepsilon > 0$, there is a polygonal chain P where

$$\left|\int_\Gamma f(z)\,\mathrm{d}z - \int_P f(z)\,\mathrm{d}z\right| < \varepsilon. \tag{3.1.14}$$

The statement we aim to prove is equivalent to proving that

$$\int_P f(z)\,\mathrm{d}z = 0. \tag{3.1.15}$$

Since P is a closed polygonal chain, we can triangulate the interior. For example, consider Figure 2, where

$$\begin{aligned}\oint_{\pentagon ABCDE} f(z)\,\mathrm{d}z &= \left(\int_{\overrightarrow{AB}} + \int_{\overrightarrow{BC}} + \int_{\overrightarrow{CD}} + \int_{\overrightarrow{DE}} + \int_{\overrightarrow{EA}}\right) f(z)\,\mathrm{d}z \\ &+ \left(\int_{\overrightarrow{BE}} + \int_{\overrightarrow{EB}} + \int_{\overrightarrow{CE}} + \int_{\overrightarrow{EC}}\right) f(z)\,\mathrm{d}z \\ &= \oint_{\Delta ABE} f(z)\,\mathrm{d}z + \oint_{\Delta BCE} f(z)\,\mathrm{d}z + \oint_{\Delta CDE} f(z)\,\mathrm{d}z.\end{aligned}$$

Thus, if the integral over every triangle in U vanishes, then (3.1.13) follows. Consider a triangle in U with boundary Δ. Then define M to be

$$M = \left|\oint_\Delta f(z)\,\mathrm{d}z\right|.$$

We can quadriinter the triangle bounded by Δ into four triangles with boundaries $\Delta_1, \Delta_2, \Delta_3, \Delta_4$ as in Figure 3. Then one of Δ_1, Δ_2, Δ_3, or Δ_4 (denote this to be Δ^1) satisfy

$$\left|\oint_{\Delta^1} f(z)\,\mathrm{d}z\right| \geq \frac{M}{4},$$

and recursively, choose

$$\left|\oint_{\Delta^2} f(z)\,\mathrm{d}z\right| \geq \frac{M}{4^2}, \ldots, \left|\oint_{\Delta^n} f(z)\,\mathrm{d}z\right| \geq \frac{M}{4^n}. \tag{3.1.16}$$

Let L denote the perimeter of Δ. Then, the perimeters of $\Delta^1, \Delta^2, \ldots$ respectively are $\frac{L}{2}, \frac{L}{2^2}, \ldots$. As $n \to \infty$, Δ_n shrinks to a single point z_0. Then, $\forall n \in \mathbb{N}$, $z_0 \in \Delta^n$.

By the definition of holomorphy, $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $\forall z \in D(z_0, \delta)$,

$$\left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| < \varepsilon,$$

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| < \varepsilon|z - z_0|,$$

and $\exists N \in \mathbb{N}$ such that $\forall n \in \mathbb{N}_{>N}$, $\Delta^n \subset D(z_0, \delta)$. By Theorem 3.1.5, since the functions $z \to 1$ and $z \to z$ are both entire,

$$\oint_{\Delta^n} \mathrm{d}z = 0, \oint_{\Delta^n} z\,\mathrm{d}z = 0.$$

Then

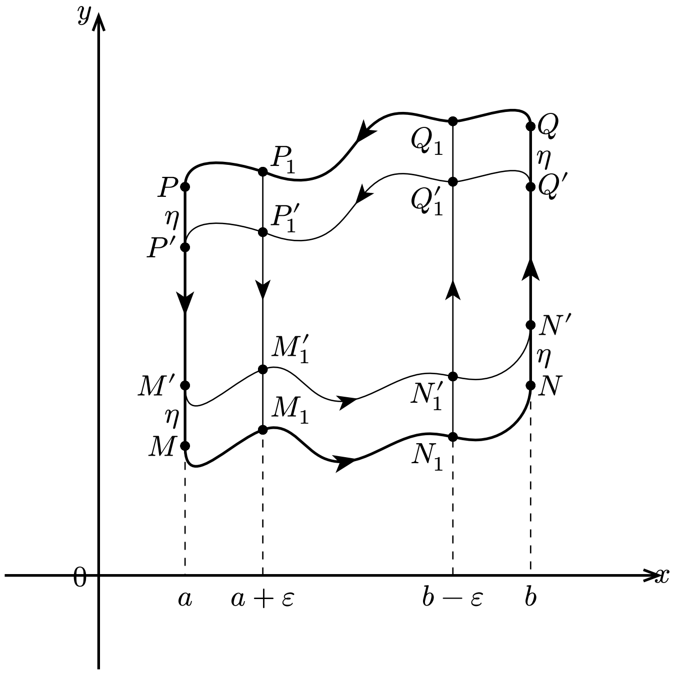

Figure 4: A simplified region containing two vertical lines and two continuous, rectifiable curves.

$$\begin{aligned}\oint_{\Delta^n} f(z)\,\mathrm{d}z &= \oint_{\Delta^n} f(z)\,\mathrm{d}z - f(z_0)\oint_{\Delta^n}\mathrm{d}z - f'(z_0)\left(\oint_{\Delta^n} z\,\mathrm{d}z - z_0\oint_{\Delta^n}\mathrm{d}z\right)\\ &= \oint_{\Delta^n}\left[f(z) - f(z_0) - f'(z_0)(z-z_0)\right]\mathrm{d}z.\end{aligned}$$

Because the distance between any two points in the interior of a triangle is always less than its perimeter, using the triangle inequality for complex integrals,

$$\left|\oint_{\Delta^n}\left[f(z) - f(z_0) - f'(z_0)(z-z_0)\right]\mathrm{d}z\right| \le \varepsilon\oint_{\Delta^n}|z-z_0||\mathrm{d}z| = \frac{\varepsilon L}{2^n}\oint_{\Delta^n}|\mathrm{d}z| = \frac{\varepsilon L^2}{4^n}.$$

Comparing the above equation with (3.1.16),

$$\frac{M}{4^n} < \varepsilon\frac{L}{4^n},\quad M < \varepsilon L.$$

Since Δ is rectifiable, L is finite, and letting $\varepsilon \to 0$, we find that $M \to 0$. Then, for every triangle in U, the integral vanishes, and (3.1.15), (3.1.14) follow. □

Theorem 3.1.7 *(Cauchy–Goursat)*: Let $U \subset \mathbb{C}$ be an open region bounded with boundary ∂U. Let $f : U \to \mathbb{C}$ be a holomorphic function continuous on $\overline{U}$. Then,

$$\oint_{\partial U} f(\zeta)\,\mathrm{d}\zeta = 0.$$

Proof: Since $\partial U \cap U = \emptyset$ and $f(z)$ is not necessarily holomorphic over $\overline{U}$, we cannot directly apply Lemma 3.1.2.

First assume U has the shape of $MNQP$ in Figure 4. That is, U consists of $x = a$, $x = b$ for $a < b$, and two rectifiable C^0 curves $\overrightarrow{MN} : y = \phi(x)$ and $\overrightarrow{QP} : y = \psi(x)$ such that $\phi(x) < \psi(x)$, $\forall a \le x \le b$.

For some $\varepsilon > 0$, $\eta > 0$, construct a new curve $M_1'N_1'Q_1'P_1' \in U$ to be the boundary of the region bounded by $P_1M_1 : x = a + \varepsilon$, $N_1Q_1 : x = b - \varepsilon$, $M'N' : \phi(x) + \eta$, and $Q'P' : \psi(x) - \eta$ such that $M_1'N_1'Q_1'P_1'$ remains simple. By Lemma 3.1.2,

$$\oint_{M_1'N_1'Q_1'P_1'} f(z)\,\mathrm{d}z = 0.$$

By Theorem 1.2.15, $f(z)$ is uniformly continuous over $\overline{U}$, and therefore $\forall \varepsilon' > 0$, we can choose $\eta > 0$ so that $\forall z \in \overrightarrow{M_1'N_1'}$, $|f(z) - f(z - \eta)| < \varepsilon'$ is satisfied. Letting $\eta \to 0$ with $\varepsilon' \to 0$ and fixing $\varepsilon > 0$, we get that

$$\begin{aligned}\left|\int_{\overrightarrow{M_1'N_1'}} f(z)\,\mathrm{d}z - \int_{\overrightarrow{M_1N_1}} f(z)\,\mathrm{d}z\right| &\leq \int_{\overrightarrow{M_1'N_1'}} |f(z) - f(z-\eta)||\mathrm{d}z| \\ &< \varepsilon' \int_{\overrightarrow{M_1'N_1'}} |\mathrm{d}z| \to 0,\end{aligned}$$

and consequently,

$$\int_{\overrightarrow{M_1'N_1'}} f(z)\,\mathrm{d}z \to \int_{\overrightarrow{M_1N_1}} f(z)\,\mathrm{d}z. \tag{3.1.17}$$

Under the same limit, we get

$$\int_{\overrightarrow{Q_1'P_1'}} f(z)\,\mathrm{d}z \to \int_{\overrightarrow{Q_1P_1}} f(z)\,\mathrm{d}z. \tag{3.1.18}$$

By the continuity of $f(z)$ over a compact set,

$$\int_{\overrightarrow{P_1'M_1'}} f(z)\,\mathrm{d}z \to \int_{\overrightarrow{P_1M_1}} f(z)\,\mathrm{d}z, \int_{\overrightarrow{N_1'Q_1'}} f(z)\,\mathrm{d}z \to \int_{\overrightarrow{N_1Q_1}} f(z)\,\mathrm{d}z. \tag{3.1.19}$$

Then letting $\varepsilon \to 0$, for the same reason as (3.1.19), (3.1.17), (3.1.18) yield

$$\int_{\overrightarrow{M_1N_1}} f(z)\,\mathrm{d}z \to \int_{\overrightarrow{MN}} f(z)\,\mathrm{d}z, \int_{\overrightarrow{Q_1P_1}} f(z)\,\mathrm{d}z \to \int_{\overrightarrow{QP}} f(z)\,\mathrm{d}z.$$

We are left to show the subsequent limits of the results from (3.1.19). For the left integral, let $y_\phi = \max(\phi(a), \phi(a+\varepsilon))$ and $y_\psi = \max(\psi(a), \psi(a+\varepsilon))$.

Then,

$$\int_{\overrightarrow{PM}} f(z)\,\mathrm{d}z = \mathrm{i}\int_{\psi(a)}^{\phi(a)} f(a + \mathrm{i}y)\,\mathrm{d}y = \mathrm{i}\left(\int_{\psi(a)}^{y_\phi} + \int_{y_\phi}^{y_\psi} + \int_{y_\psi}^{\phi(a)}\right) f(a + \mathrm{i}y)\,\mathrm{d}y.$$

Similarly,

$$\int_{\overrightarrow{P_1M_1}} f(z)\,\mathrm{d}z = \mathrm{i}\left(\int_{\psi(a+\varepsilon)}^{y_\phi} + \int_{y_\phi}^{y_\psi} + \int_{y_\psi}^{\phi(a+\varepsilon)}\right) f(a + \varepsilon + \mathrm{i}y)\,\mathrm{d}y.$$

The difference $\left(\int_{\overrightarrow{PM}} - \int_{\overrightarrow{P_1M_1}}\right) f(z)\,\mathrm{d}z$ between the two is then equal to

$$\mathrm{i}\int_{y_\phi}^{y_\psi}(f(a+\mathrm{i}y)-f(a+\varepsilon+\mathrm{i}y))\,\mathrm{d}y+\mathrm{i}\left(\int_{\psi(a)}^{y_\phi}+\int_{y_\psi}^{\phi(a)}\right)f(a+\mathrm{i}y)$$
$$-\mathrm{i}\left(\int_{\psi(a+\varepsilon)}^{y_\phi}+\int_{y_\psi}^{\phi(a+\varepsilon)}\right)f(a+\varepsilon+\mathrm{i}y).$$

The first term vanishes by uniform continuity, through the same argument used for $M_1'N_1' \to M_1N_1$, and the remaining four integrals all tend to 0 because they are taken over degenerating intervals. As $\varepsilon \to 0$, $y_\phi \to \phi(a)$ and $y_\psi \to \psi(a)$ because $\phi, \psi \in C^0$. Therefore,

$$\int_{\overrightarrow{P_1M_1}} f(z)\,\mathrm{d}z \to \int_{\overrightarrow{PM}} f(z)\,\mathrm{d}z,$$

and through similar logic,

$$\int_{\overrightarrow{N_1Q_1}} f(z)\,\mathrm{d}z \to \int_{\overrightarrow{NQ}} f(z)\,\mathrm{d}z.$$

Therefore,

$$\oint_{MNQP} f(z)\,\mathrm{d}z = 0.$$

Any open region $U \subset \mathbb{C}$ with a simple closed boundary can be broken up into smaller regions with the same form as $MNQP$ with finitely many auxiliary lines. Then the conclusion follows. □

Remark: The theorem is also valid for any multiply connected region (and its boundary will consist of multiple curves) as a multiply connected region is equal to the union of several simply connected regions with vertical auxiliary lines between.

Additionally, if $U \subset \mathbb{C}$ is simply connected and f is holomorphic on U, then for any two points $z, z_0 \in U$, the integral

$$\int_{z_0}^{z} f(\zeta)\,\mathrm{d}\zeta$$

is well-defined and independent of the path taken from z_0 to z. In this sense, holomorphic functions behave analogously to potential fields.

Theorem 3.1.8 *(Cauchy–Goursat)*: Let $U \subset \mathbb{C}$ be an open region bounded with a simple closed boundary ∂U, and let $f : U \to \mathbb{C}$ be a holomorphic function continuous on $\overline{U}$. Then for all $z \in U$,

$$f(z) = \frac{1}{2\pi\mathrm{i}}\oint_{\partial U}\frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta. \tag{3.1.20}$$

Proof: By the Cauchy–Goursat Theorem (Theorem 3.1.7),

$$\int_{\partial(U\setminus D(z,\varepsilon))}\frac{f(\zeta)}{\zeta-z}\,\mathrm{d}\zeta = \oint_{\partial U}\frac{f(\zeta)}{\zeta-z}\,\mathrm{d}\zeta - \oint_{\partial D(z,\varepsilon)}\frac{f(\zeta)}{\zeta-z}\,\mathrm{d}\zeta = 0.$$

From rearrangement,

$$\oint_{\partial U}\frac{f(\zeta)}{\zeta-z}\,\mathrm{d}\zeta = 2\pi\mathrm{i}f(z) + \mathrm{i}\int_0^{2\pi}\left[f(z+\varepsilon\mathrm{e}^{\mathrm{i}t}) - f(z)\right]\mathrm{d}t.$$

Since $f \in C^0(\partial D(z, \varepsilon))$, as $\varepsilon \to 0$,

$$\begin{aligned}\left|\int_0^{2\pi} [f(z+\varepsilon e^{it}) - f(z)]\, dt\right| &\le \int_0^{2\pi} \left|f(z+\varepsilon e^{it}) - f(z)\right| dt \\ &\le 2\pi \max_{t\in[0,2\pi]} \left|f(z+\varepsilon e^{it}) - f(z)\right| \to 0.\end{aligned}$$

By rearrangement,

$$f(z) = \frac{1}{2\pi i} \oint_{\partial U} \frac{f(\zeta)}{\zeta - z}\, d\zeta. \qquad \square$$

Remark: In the proof of Theorem 3.1.3, we used Lipschitz continuity for a smooth function, which was a stronger condition than necessary. The true necessity of smoothness was to be able to apply Green's Theorem (Theorem 3.1.2).

This profound theorem is extremely important and helpful in complex integration and essential in the evaluation of integrals, as demonstrated below.

Example 3.1.1: Evaluate the integral $\oint_{\partial D(0,2)} \frac{dz}{z^n - 1}$, where $n \in \mathbb{N}_{\ge 2}$.

Proof: Since $z^n - 1 = \prod_{k=0}^{n-1} (z - \omega_n^k)$, where $\omega_n^k = e^{2\pi i \frac{k}{n}}$, the integrand has singularities at every n-th root of unity. Then the integral is equal to:

$$\oint_{\partial D(0,2)} \frac{dz}{\prod_{j=0}^{n-1} (z - \omega_j)} = \oint_{\partial D(0,2)} \sum_{j=0}^{n-1} \frac{c_j}{z - \omega_j}\, dz, \qquad (3.1.21)$$

where c_j are the coefficients of the partial fraction decomposition. By the Cauchy–Goursat Formula (Theorem 3.1.8), (3.1.21) becomes:

$$\sum_{k=0}^{n-1} \oint_{\partial D(0,2)} \frac{c_k}{z - \omega_k}\, dz = 2\pi i \sum_{k=0}^{n-1} c_k.$$

Observe that $\sum_{k=0}^{n-1} c_k = \lim_{z\to\infty} \sum_{k=0}^{n-1} \frac{z c_k}{z - \omega_k} = \lim_{z\to\infty} \frac{z}{z^n - 1} = 0$ since $n \ge 2$. Therefore,

$$\oint_{\partial D(0,2)} \frac{dz}{z^n - 1} = 0. \qquad \square$$

We have also already seen the utility of parameterization via a polar transformation. Many useful identities in classical calculus can also be derived from concepts in its generalization:

Example 3.1.2: Prove that $\forall n \in \mathbb{N}$,

$$\int_0^{2\pi} \cos^{2n}\theta\, d\theta = 2\pi \prod_{k=1}^{n} \frac{2k-1}{2k}.$$

Proof: Consider the integral

$$\oint_{\partial \mathbb{D}} \left(z + \frac{1}{z}\right)^{2n} \frac{dz}{z}.$$

Letting $z = e^{i\theta}$, we get $\oint_{\partial \mathbb{D}} \left(e^{i\theta} + e^{-i\theta}\right)^{2n} e^{-i\theta}\, dz = 2^{2n} i \int_0^{2\pi} \cos^{2n}\theta\, d\theta$. Alternatively, we can expand the integrand and get

$$\oint_{\partial\mathbb{D}} \sum_{k=0}^{2n} \binom{2n}{k} z^{2k-2n} \frac{\mathrm{d}z}{z} = \sum_{k=0}^{2n} \oint_{\partial\mathbb{D}} \binom{2n}{k} z^{2k-2n-1}\,\mathrm{d}z.$$

When $2k-2n-1 \geq 0$, the integrand is holomorphic. The integral is then equal to

$$\binom{2n}{0}\oint_{\partial\mathbb{D}} z^{-2n-1}\,\mathrm{d}z + \binom{2n}{1}\oint_{\partial\mathbb{D}} z^{-2n+1}\,\mathrm{d}z + \cdots + \binom{2n}{n}\oint_{\partial\mathbb{D}} \frac{\mathrm{d}z}{z} = 2\pi\mathrm{i}\binom{2n}{n},$$

since all the higher order terms vanish:

$$\oint_{\partial\mathbb{D}} z^{2k-2n-1}\,\mathrm{d}z = \mathrm{i}\int_0^{2\pi} \mathrm{e}^{2\mathrm{i}\theta(k-n)}\,\mathrm{d}\theta = \begin{cases} 0 & \text{if } k < n, \\ 2\pi\mathrm{i} & \text{if } k = n. \end{cases}$$

Therefore,

$$2^{2n}\mathrm{i}\int_0^{2\pi} \cos^{2n}\theta\,\mathrm{d}\theta = 2\pi\mathrm{i}\binom{2n}{n} \iff \int_0^{2\pi} \cos^{2n}\theta\,\mathrm{d}\theta = \frac{2\pi(2n)!}{2^{2n}(n!)^2} = \frac{2\pi\prod_{k=1}^{2n} k}{\prod_{k=1}^{n}(2k)^2}.$$

From simple cancellation, we then have

$$2\pi\prod_{k=1}^{n} \frac{2k-1}{\prod_{k=1}^{n}}(2k) = 2\pi\prod_{k=1}^{n}\frac{2k-1}{2k}.$$

□

Example 3.1.3 *(Cauchy–Goursat Formula on the Exterior)*: Let $\gamma \subset \mathbb{C}$ be a simple closed curve, and suppose that $f : \mathrm{ext}(\gamma) \to \mathbb{C}$ is holomorphic and continuous on $\overline{\mathrm{ext}(\gamma)} = \mathbb{C} \setminus \mathrm{int}(\gamma)$, where int and ext respectively denote the interior and exterior as in Theorem 1.2.5.

1. If f has a removable singularity at ∞, or if $w = \lim_{z\to\infty} f(z)$ exists and is finite, then $\forall z \in \mathbb{C} \setminus \gamma$,

$$\frac{1}{2\pi\mathrm{i}}\oint_\gamma \frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta = \begin{cases} w & \text{if } z \in \mathrm{int}(\gamma), \\ w - f(z) & \text{if } z \in \mathrm{ext}(\gamma). \end{cases}$$

2. If γ encloses the origin, then $\forall z \in \mathbb{C} \setminus \gamma$,

$$\frac{1}{2\pi\mathrm{i}}\oint_\gamma z\frac{f(\zeta)}{z\zeta - \zeta^2}\,\mathrm{d}\zeta = \begin{cases} 0 & \text{if } z \in \mathrm{int}(\gamma), \\ f(z) & \text{if } z \in \mathrm{ext}(\gamma). \end{cases} \tag{3.1.22}$$

Proof:
1. By the compactness of γ, it can be completely contained within a sufficiently large disk centered at the origin ($\gamma \subset D(0,R)$). Then by applying Theorem 3.1.8 or Theorem 3.1.7 on the set $D(0,R) \cap \mathrm{ext}(\gamma) = D(0,R) \setminus \overline{\mathrm{int}(\gamma)}$, we get that

$$\frac{1}{2\pi\mathrm{i}}\oint_{\partial D(0,R)} \frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta = \frac{1}{2\pi\mathrm{i}}\oint_\gamma \frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta + \begin{cases} 0 & \text{if } z \in \mathrm{int}(\gamma), \\ f(z) & \text{if } z \in D(0,R) \cap \mathrm{ext}(\gamma). \end{cases}$$

By letting $R \to \infty$ and letting $\zeta = R\mathrm{e}^{\mathrm{i}\theta}$, we get that

$$\frac{1}{2\pi\mathrm{i}}\oint_\gamma \frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta = \frac{1}{2\pi}\lim_{R\to\infty}\int_0^{2\pi} \frac{f\left(R\mathrm{e}^{\mathrm{i}\theta}\right)}{1 - \frac{z}{R\mathrm{e}^{\mathrm{i}\theta}}}\,\mathrm{d}\theta - \begin{cases} 0 & \text{if } z \in \mathrm{int}(\gamma), \\ f(z) & \text{if } z \in \mathrm{ext}(\gamma). \end{cases}$$

By the continuity of f on $\partial D(0,R)$, it attains its maximum M. For sufficiently large R, $\left|1 - \frac{z}{R\mathrm{e}^{\mathrm{i}\theta}}\right|$ attains a positive minimum. Then the integrand is uniformly bounded in R and θ, and hence the order of the limit and the integral may be exchanged. Hence,

$$\frac{1}{2\pi\mathrm{i}}\oint_\gamma \frac{f(\zeta)}{\zeta - z}\,\mathrm{d}\zeta = \frac{1}{2\pi}\int_0^{2\pi} \frac{w}{1 - \lim_{R\to\infty}\frac{z}{R\mathrm{e}^{\mathrm{i}\theta}}}\,\mathrm{d}\theta - \begin{cases} 0 & \text{if } z \in \operatorname{int}(\gamma), \\ f(z) & \text{if } z \in \operatorname{ext}(\gamma) \end{cases}$$
$$= \begin{cases} w & \text{if } z \in \operatorname{int}(\gamma), \\ w - f(z) & \text{if } z \in \operatorname{ext}(\gamma) \end{cases}$$

as expected.

2. Under the partial fraction decomposition of (3.1.22), we get that

$$I = \oint_\gamma z\frac{f(\zeta)}{z\zeta - \zeta^2}\,\mathrm{d}\zeta = \oint_\gamma \left(\frac{f(\zeta)}{\zeta} - \frac{f(\zeta)}{\zeta - z}\right)\mathrm{d}\zeta \tag{3.1.23}$$
$$= \int_0^{2\pi}\left(f\left(R\mathrm{e}^{\mathrm{i}\theta}\right) - \frac{f\left(R\mathrm{e}^{\mathrm{i}\theta}\right)}{1 - \frac{z}{R\mathrm{e}^{\mathrm{i}\theta}}}\right)\mathrm{d}\theta + \begin{cases} 0 & \text{if } z \in \operatorname{int}(\gamma), \\ 2\pi\mathrm{i}f(z) & \text{if } z \in \operatorname{ext}(\gamma)\cap D(0,R) \end{cases} \tag{3.1.24}$$

when $\gamma \subset D(0, R)$. We will analyze the first integral as $R \to \infty$. By the triangle and reverse triangle inequalities,

$$\left|\int_0^{2\pi}\left(f\left(R\mathrm{e}^{\mathrm{i}\theta}\right) - \frac{f\left(R\mathrm{e}^{\mathrm{i}\theta}\right)}{1 - \frac{z}{R\mathrm{e}^{\mathrm{i}\theta}}}\right)\mathrm{d}\theta\right| \le \int_0^{2\pi}\left|\frac{z}{R\mathrm{e}^{\mathrm{i}\theta} - z}\right|\mathrm{d}\theta$$
$$\le \int_0^{2\pi}\frac{|z|}{R - |z|}\,\mathrm{d}\theta = \frac{2\pi|z|}{R - |z|} \to 0.$$

By substituting the result into (3.1.23), and letting $R \to \infty$, we get that

$$\frac{1}{2\pi\mathrm{i}}\oint_\gamma z\frac{f(\zeta)}{z\zeta - \zeta^2}\,\mathrm{d}\zeta = \begin{cases} 0 & \text{if } z \in \operatorname{int}(\gamma), \\ f(z) & \text{if } z \in \operatorname{ext}(\gamma) \end{cases}$$

as desired. □

3.2 Analyticity and Holomorphy

The Cauchy–Goursat Formula (Theorem 3.1.8) can be generalized into a result that directly relates complex integration and differentiation.

Theorem 3.2.1 *(Cauchy–Goursat)*: Let $U \subset \mathbb{C}$ be an open region bounded by a simple closed boundary ∂U, and let $f : U \to \mathbb{C}$ be holomorphic and continuous on $\overline{U}$. Then for every $z \in U$ and every $n \in \mathbb{N}$,

$$f^n(z) = \frac{n!}{2\pi\mathrm{i}}\oint_{\partial U}\frac{f(\zeta)}{(\zeta - z)^{n+1}}\,\mathrm{d}\zeta. \tag{3.2.1}$$

Additionally, since U is open, for every $z_0 \in U$ and every $r > 0$ such that $\overline{D(z_0, r)} \subset U$, f has the uniformly and absolutely convergent Taylor expansion

$$f(z) = \sum_{j=0}^{\infty} a_j(z - z_0)^j, \tag{3.2.2}$$

where

$$a_j = \frac{1}{2\pi\mathrm{i}}\oint_{\partial U}\frac{f(\zeta)}{(\zeta - z_0)^{j+1}}\,\mathrm{d}\zeta \tag{3.2.3}$$

on $\overline{D(z_0, r)}$.

Proof: Fix $z_0 \in U$, and choose $r > 0$ such that $\overline{D(z_0, r)} \subset U$. For $z \in D(z_0, r)$, by Theorem 3.1.8,

$$\begin{aligned} f(z) - f(z_0) &= \frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{\zeta - z} \, \mathrm{d}\zeta - \frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{\zeta - z_0} \, \mathrm{d}\zeta \\ &= \frac{z - z_0}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z)(\zeta - z_0)} \, \mathrm{d}\zeta. \end{aligned}$$

Dividing by $z - z_0$, we obtain

$$\frac{f(z) - f(z_0)}{z - z_0} = \frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z)(\zeta - z_0)} \, \mathrm{d}\zeta.$$

Hence,

$$\frac{f(z) - f(z_0)}{z - z_0} - \frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z_0)^2} \, \mathrm{d}\zeta = \frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{\zeta - z_0} \left(\frac{1}{\zeta - z} - \frac{1}{\zeta - z_0} \right) \mathrm{d}\zeta \tag{3.2.4}$$

$$= \frac{z - z_0}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z)(\zeta - z_0)^2} \, \mathrm{d}\zeta. \tag{3.2.5}$$

Let d be the distance from z_0 to ∂U. Then $0 < r < d$. Since $|z - z_0| < r$ and $|\zeta - z_0| \geq d$ for $\zeta \in \partial U$, we have $|\zeta - z| \geq d - r$. If $M = \max_{\zeta \in \partial U} |f(\zeta)|$, then the integrand in (3.2.4) is bounded above by $\frac{M}{d^2(d-r)}$. Therefore,

$$\left| \frac{z - z_0}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z)(\zeta - z_0)^2} \, \mathrm{d}\zeta \right| \leq \frac{|z - z_0|}{2\pi} \frac{M}{d^2(d - r)} \oint_{\partial U} |\mathrm{d}\zeta|.$$

As $z \to z_0$, the right-hand side tends to 0. Thus,

$$f'(z_0) = \frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z_0)^2} \, \mathrm{d}\zeta.$$

Now suppose

$$f^k(z) = \frac{k!}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z)^{k+1}} \, \mathrm{d}\zeta$$

for some $k \in \mathbb{N}$. Since $|z - z_0| < |\zeta - z_0|$ for $z \in D(z_0, r)$ and $\zeta \in \partial U$, we have the kernel expansion

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} \tag{3.2.6}$$

$$= \frac{1}{\zeta - z_0} \sum_{j=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^j. \tag{3.2.7}$$

Substituting into the inductive formula gives

$$
\begin{aligned}
f^k(z) &= \frac{k!}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z)^{k+1}} \, \mathrm{d}\zeta \\
&= \frac{k!}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} \left(\sum_{j=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^j \right)^{k+1} \mathrm{d}\zeta \\
&= f^k(z_0) + \frac{(k+1)!(z - z_0)}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z_0)^{k+2}} \, \mathrm{d}\zeta + \mathcal{O}\left(|z - z_0|^2 \right).
\end{aligned}
$$

Therefore,

$$
\frac{f^k(z) - f^k(z_0)}{z - z_0} = \frac{(k+1)!}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z_0)^{k+2}} \, \mathrm{d}\zeta + \mathcal{O}(|z - z_0|).
$$

Letting $z \to z_0$, we obtain

$$
f^{(k+1)}(z_0) = \frac{(k+1)!}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{(\zeta - z_0)^{k+2}} \, \mathrm{d}\zeta.
$$

By induction, (3.2.1) is valid for all $n \in \mathbb{N}$.

For the Taylor expansion, substitute (3.2.6) into Theorem 3.1.8:

$$
\begin{aligned}
f(z) &= \frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \frac{f(\zeta)}{\zeta - z_0} \sum_{j=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0} \right)^j \mathrm{d}\zeta \\
&= \frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \sum_{j=0}^{\infty} (z - z_0)^j \frac{f(\zeta)}{(\zeta - z_0)^{j+1}} \, \mathrm{d}\zeta.
\end{aligned}
$$

Since f is continuous on ∂U, it is bounded there by some constant M. Also, for $z \in \overline{D(z_0, r)}$ and $\zeta \in \partial U$,

$$
\left| (z - z_0)^j \frac{f(\zeta)}{(\zeta - z_0)^{j+1}} \right| \leq \frac{M r^j}{\inf_{\xi \in \partial U} |\xi - z_0|^{j+1}}.
$$

The majorant series

$$
\sum_{j=0}^{\infty} \frac{M r^j}{\inf_{\xi \in \partial U} |\xi - z_0|^{j+1}}
$$

converges, so by the Weierstrass M-Test (Theorem 2.2.2), the series is uniformly convergent. Hence we may interchange summation and integration:

$$
\begin{aligned}
\frac{1}{2\pi \mathrm{i}} \oint_{\partial U} \sum_{j=0}^{\infty} (z - z_0)^j \frac{f(\zeta)}{(\zeta - z_0)^{j+1}} \, \mathrm{d}\zeta &= \frac{1}{2\pi \mathrm{i}} \sum_{j=0}^{\infty} \oint_{\partial U} (z - z_0)^j \frac{f(\zeta)}{(\zeta - z_0)^{j+1}} \, \mathrm{d}\zeta \\
&= \sum_{j=0}^{\infty} a_j (z - z_0)^j.
\end{aligned}
$$

This proves (3.2.2) and (3.2.3). $\square$

Remark: We have shown that once the first complex derivative of a holomorphic function exists, all higher derivatives exist as well and remain holomorphic on the same region. Furthermore, every holomorphic function admits a convergent Taylor series expansion in a neighborhood of each point of its domain. This property is called *analyticity*.

Conversely, a power series is termwise differentiable inside its disk of convergence, so every analytic function is holomorphic. Thus analyticity and holomorphy are equivalent in complex analysis, which is a fundamental contrast with the real-variable setting.

The differentiation formula above can be viewed as a higher-order version of Theorem 3.1.8 and is often useful in evaluating integrals.

Example 3.2.1: A *Legendre polynomial* is defined by

$$P_{n(z)} = \frac{1}{2^n n!} \frac{\mathrm{d}^n \left(z^2 - 1\right)^n}{\mathrm{d}z^n}. \tag{3.2.8}$$

Prove the integral representation

$$P_{n(z)} = \frac{1}{2\pi \mathrm{i}} \oint_\gamma \frac{\left(\zeta^2 - 1\right)^n}{2^n (\zeta - z)^{n+1}} \,\mathrm{d}\zeta,$$

where γ is a simple closed curve enclosing z.

Proof: Apply Theorem 3.2.1 to the polynomial $\left(z^2 - 1\right)^n$. Then

$$P_{n(z)} = \frac{1}{2^n n!} \cdot \frac{n!}{2\pi \mathrm{i}} \oint_\gamma \frac{\left(\zeta^2 - 1\right)^n}{(\zeta - z)^{n+1}} \,\mathrm{d}\zeta = \frac{1}{2\pi \mathrm{i}} \oint_\gamma \frac{\left(\zeta^2 - 1\right)^n}{2^n (\zeta - z)^{n+1}} \,\mathrm{d}\zeta. \qquad \square$$

Theorem 3.2.2 *(Cauchy's Estimate)*: Let $f : U \to \mathbb{C}$ be holomorphic on $U \subseteq \mathbb{C}$. For every $z_0 \in U$ and every $R > 0$ such that $\overline{D(z_0, R)} \subseteq U$, and for every $n \in \mathbb{N}$,

$$|f^n(z_0)| \leq \frac{n! M}{R^n},$$

where

$$M = \max_{z \in \overline{D(z_0, R)}} |f(z)|.$$

Proof: By Theorem 3.2.1,

$$f^n(z_0) = \frac{n!}{2\pi \mathrm{i}} \oint_{\partial D(z_0, R)} \frac{f(\zeta)}{\left(\zeta - z_0\right)^{n+1}} \,\mathrm{d}\zeta.$$

Parameterize $\partial D(z_0, R)$ by $\zeta = z_0 + R\mathrm{e}^{\mathrm{i}\theta}$ for $0 \leq \theta \leq 2\pi$. Since $|f(\zeta)| \leq M$ on the circle,

$$\begin{aligned} |f^n(z_0)| &\leq \frac{n!}{2\pi} \int_0^{2\pi} \frac{M}{\left|\left(R\mathrm{e}^{\mathrm{i}\theta}\right)^{n+1}\right|} \left|\mathrm{i}R\mathrm{e}^{\mathrm{i}\theta}\right| \mathrm{d}\theta \\ &= \frac{n! M}{2\pi} \int_0^{2\pi} \frac{1}{R^n} \,\mathrm{d}\theta \qquad \square \\ &= \frac{n! M}{R^n}. \qquad \square \end{aligned}$$

The relationship between a holomorphic function and its derivatives is one of the strongest structural properties in complex analysis. Later, Theorem 3.2.5 will generalize this estimate substantially.

Example 3.2.2: Let f be entire and suppose that $\forall z \in \mathbb{C}, |f(z)| \leq M\mathrm{e}^{|z|}$. Prove that $|f(0)| \leq M$ and that for every $n \in \mathbb{N}$,

$$|f^n(0)| \leq Mn!\left(\frac{\mathrm{e}}{n}\right)^n.$$

Proof: The bound $|f(0)| \leq M$ follows immediately by setting $z = 0$.

For every $R > 0$, Theorem 3.2.2 gives

$$|f^n(0)| \leq Mn!\frac{\mathrm{e}^R}{R^n}.$$

Setting $R = n$ yields

$$|f^n(0)| \leq Mn!\left(\frac{\mathrm{e}}{n}\right)^n.$$

This is in fact the optimal choice of R. If $\varphi(R) = Mn!\frac{\mathrm{e}^R}{R^n}$, then

$$\varphi'(R) = Mn!\frac{\mathrm{e}^R R^n - n\mathrm{e}^R R^{n-1}}{R^{2n}} = 0$$

if and only if $R = n$. Moreover,

$$\varphi''(R) = Mn!\mathrm{e}^R\left(\frac{1}{R^n} - \frac{2n}{R^{n+1}} + \frac{n(n+1)}{R^{n+2}}\right)$$

so that

$$\varphi''(n) = M(n-1)!\frac{\mathrm{e}^n}{n^n} > 0.$$

Hence $R = n$ indeed minimizes the bound. □

Theorem 3.2.3 *(Liouville)*: Every bounded entire function is constant.

Proof: Let $f : \mathbb{C} \to \mathbb{C}$ be entire and bounded. Fix $z_0 \in \mathbb{C}$. For every $R > 0$, Theorem 3.2.2 gives

$$|f'(z_0)| \leq \frac{M}{R},$$

where $M = \sup_{z\in\mathbb{C}}|f(z)|$.

Letting $R \to \infty$, we obtain $f'(z_0) = 0$. Since z_0 was arbitrary, $f' \equiv 0$ on $\mathbb{C}$, so f is constant. □

Proof: Alternative proof.

Let $a, b \in \mathbb{C}$ be distinct. Since f is entire, for any $R > \max(|a|, |b|)$, the function

$$z \mapsto \frac{f(z)}{(z-a)(z-b)}$$

is holomorphic on the annular region between $\partial D(0,R)$ and the two small circles $\partial D(a,\varepsilon)$ and $\partial D(b,\varepsilon)$, where $\varepsilon > 0$ is chosen so that the closed disks around a and b are disjoint. By Theorem 3.1.7,

$$\oint_{\partial D(0,R)} \frac{f(z)}{(z-a)(z-b)}\,\mathrm{d}z = \oint_{\partial D(a,\varepsilon)} \frac{f(z)}{(z-a)(z-b)}\,\mathrm{d}z + \oint_{\partial D(b,\varepsilon)} \frac{f(z)}{(z-a)(z-b)}\,\mathrm{d}z.$$

Applying Theorem 3.1.8 to the two small circles yields

$$\oint_{\partial D(0,R)} \frac{f(z)}{(z-a)(z-b)}\,\mathrm{d}z = 2\pi\mathrm{i}\left(\frac{f(a)}{a-b} + \frac{f(b)}{b-a}\right) = 2\pi\mathrm{i}\frac{f(a)-f(b)}{a-b}.$$

On the other hand, if $|z| = R$, then $|z-a| \geq R - |a|$ and $|z-b| \geq R - |b|$. Hence

$$\left|\oint_{\partial D(0,R)} \frac{f(z)}{(z-a)(z-b)}\,\mathrm{d}z\right| \leq M \oint_{\partial D(0,R)} \frac{|\mathrm{d}z|}{|z-a||z-b|} \leq \frac{2\pi MR}{(R-|a|)(R-|b|)}.$$

As $R \to \infty$, the right-hand side tends to 0. Therefore,

$$2\pi\mathrm{i}\frac{f(a)-f(b)}{a-b} = 0.$$

Since a and b were arbitrary and distinct, $f(a) = f(b)$ for all $a, b \in \mathbb{C}$, so f is constant. □

Theorem 3.2.4 *(Morera)*: Let $U \subseteq \mathbb{C}$ and let $f : U \to \mathbb{C}$ be continuous. If for every oriented closed triangular contour $\gamma \subset U$,

$$\oint_\gamma f(\zeta)\,\mathrm{d}\zeta = 0,$$

then f is holomorphic on U.

Proof: Fix $z_0 \in U$. Since U is open, there exists $r > 0$ such that $\overline{D(z_0, r)} \subset U$. Define

$$F(z) = \int_{z_0}^{z} f(\zeta)\,\mathrm{d}\zeta,$$

where the path is the straight line segment from z_0 to z. This is well-defined for $z \in D(z_0, r)$ because the disk is convex.

If $z, z + \Delta z \in D(z_0, r)$, then the triangle with vertices z_0, z, and $z + \Delta z$ lies inside $D(z_0, r)$. By the hypothesis on triangular contours,

$$F(z+\Delta z) - F(z) = \int_{z}^{z+\Delta z} f(\zeta)\,\mathrm{d}\zeta.$$

Hence

$$\frac{F(z+\Delta z) - F(z)}{\Delta z} - f(z) = \frac{1}{\Delta z}\int_{z}^{z+\Delta z} (f(\zeta) - f(z))\,\mathrm{d}\zeta.$$

By continuity of f at z, for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|\zeta - z| < \delta \Rightarrow |f(\zeta) - f(z)| < \varepsilon.$$

Therefore, whenever $|\Delta z| < \delta$,

$$\left|\frac{1}{\Delta z}\int_{z}^{z+\Delta z} (f(\zeta) - f(z))\,\mathrm{d}\zeta\right| \leq \frac{1}{|\Delta z|}\int_{z}^{z+\Delta z} \varepsilon|\mathrm{d}\zeta| = \varepsilon.$$

Thus $F'(z) = f(z)$ on $D(z_0, r)$. In particular, f is holomorphic on $D(z_0, r)$. Since z_0 was arbitrary, f is holomorphic on U. □

Theorem 3.2.5: Let $U \subseteq \mathbb{C}$ be open, let $K \subset U$ be compact, and let $K \subset V \subseteq \mathbb{C}$ with V open and $\overline{V} \subset U$. Let f be holomorphic on U. Then there exists a sequence $\{c_n\} \subset \mathbb{R}$ depending only on K and V such that for every $n \in \mathbb{N}$,

$$\sup_{z \in K} |f^n(z)| \leq c_n \|f\|_{L^1(V)}, \tag{3.2.9}$$

where

$$\|f\|_{L^{p}(V)} = \left(\int_{V} |f(z)|^{p}\, \mathrm{d}x \wedge \mathrm{d}y\right)^{\frac{1}{p}}.$$

Proof: Let $\varphi \in C^{\infty(\mathbb{C})}$ satisfy $\operatorname{supp}(\varphi) \subset V$ and suppose that $\varphi \equiv 1$ on an open neighborhood W of K with $\overline{W} \subset V$. Since $f \in C^{\infty(U)}$, apply the Cauchy–Pompeiu Theorem (Theorem 3.1.3) to $f(z)\varphi(z)$:

$$f(z)\varphi(z) = \frac{1}{2\pi \mathrm{i}}\left(\oint_{\partial U} \frac{f(\zeta)\varphi(\zeta)}{\zeta - z}\, \mathrm{d}\zeta - \int_{U} \frac{\partial f(\zeta)\varphi(\zeta)}{\partial \overline{\zeta}} \cdot \frac{\mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta}{\zeta - z}\right).$$

Since $\operatorname{supp}(\varphi) \subset V \subset U$, the boundary term vanishes. Because f is holomorphic,

$$\frac{\partial f(\zeta)\varphi(\zeta)}{\partial \overline{\zeta}} = \frac{\partial \varphi(\zeta)}{\partial \overline{\zeta}} f(\zeta).$$

Therefore,

$$f(z)\varphi(z) = -\frac{1}{2\pi \mathrm{i}} \int_{U} \frac{\partial \varphi(\zeta)}{\partial \overline{\zeta}} f(\zeta) \cdot \frac{\mathrm{d}\overline{\zeta} \wedge \mathrm{d}\zeta}{\zeta - z}.$$

Let $K_1 = \operatorname{supp}\left(\frac{\partial \varphi}{\partial \overline{z}}\right)$. For $z \in K$, we have $\varphi(z) = 1$, so

$$f(z) = \frac{1}{2\pi \mathrm{i}} \int_{K_1} f(\zeta) \frac{\partial \varphi(\zeta)}{\partial \overline{\zeta}} \frac{\mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}}{\zeta - z}.$$

Since $K \subset W$ and $K_1 \cap W = \emptyset$, the distance between K and K_1 is positive. Hence there exists a constant $M > 0$ such that

$$\frac{1}{|\zeta - z|} \leq M, \quad \forall z \in K, \zeta \in K_1.$$

Differentiating under the integral sign is therefore justified, and for every $n \in \mathbb{N}$,

$$f^{n}(z) = \frac{n!}{2\pi \mathrm{i}} \int_{K_1} f(\zeta) \frac{\partial \varphi(\zeta)}{\partial \overline{\zeta}} \frac{\mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}}{(\zeta - z)^{n+1}}.$$

Taking absolute values gives

$$|f^{n}(z)| \leq \frac{n!}{2\pi} \int_{K_1} |f(\zeta)| \left|\frac{\partial \varphi(\zeta)}{\partial \overline{\zeta}}\right| \left|\frac{\mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}}{(\zeta - z)^{n+1}}\right|.$$

The factor

$$\frac{\left|\frac{\partial \varphi(\zeta)}{\partial \overline{\zeta}}\right|}{|\zeta - z|^{n+1}}$$

is bounded for all $z \in K$ and $\zeta \in K_1$ by some constant c_n' depending only on K, V, and n. Also,

$$\left|\mathrm{d}\zeta \wedge \mathrm{d}\overline{\zeta}\right| = 2|\mathrm{d}x \wedge \mathrm{d}y|.$$

Thus

$$|f^n(z)| \leq \frac{n! c_n'}{\pi} \int_{K_1} |f(\zeta)| |\mathrm{d}x \wedge \mathrm{d}y| \leq \frac{n! c_n'}{\pi} \int_V |f(\zeta)| |\mathrm{d}x \wedge \mathrm{d}y|.$$

Defining $c_n = n! \frac{c_n'}{\pi}$ proves (3.2.9).

The remaining issue is the existence of such a cutoff function φ, which will be supplied later by @ thm:bumpfunctionexistence. □

Corollary 3.2.5.1: Let $U \subseteq \mathbb{C}$ be open, let $K \subset U$ be compact, and let $K \subset V \subseteq \mathbb{C}$ with V open and $\overline{V} \subset U$. Then for every holomorphic function f on U, there exist constants $\{c_n\}$ independent of f and z such that

$$\sup_{z \in K} |f^n(z)| \leq c_n \sup_{z \in V} |f(z)|.$$

Proof: By (3.2.9),

$$\sup_{z \in K} |f^n(z)| \leq c_n \|f\|_{L^1(V)}.$$

Since

$$\|f\|_{L^1(V)} \leq \int_V \sup_{z \in V} |f(z)| |\mathrm{d}x \wedge \mathrm{d}y| = \left(\int_V |\mathrm{d}x \wedge \mathrm{d}y| \right) \sup_{z \in V} |f(z)|,$$

the result follows after absorbing $\int_V |\mathrm{d}x \wedge \mathrm{d}y|$ into the constant. □

For the next theorem we briefly introduce the notion of *analytic continuation.*

Definition 3.2.1 *(Analytic Continuation)*: Let $U \subseteq \mathbb{C}$ be open, and let $f : U \to \mathbb{C}$ be holomorphic. Let $V \subseteq \mathbb{C}$ be open with $U \subseteq V$. A function $F : V \to \mathbb{C}$ is an *analytic continuation* of f to V if:

1. F is holomorphic on V.
2. $F \equiv f$ on U.

The concept of analytic continuation and its consequences will be studied in more detail later. For now, we prove a classical removable singularity theorem that follows from the analyticity of holomorphic functions.

Theorem 3.2.6 *(Riemann)*: Let $D^*(z_0, r) = D(z_0, r) \setminus \{z_0\}$ be a punctured disk, and let $f : D^*(z_0, r) \to \mathbb{C}$ be holomorphic and bounded. Then f admits an analytic continuation to $D(z_0, r)$.

Proof: Since f is bounded on $D^*(z_0, r)$, there exists $M > 0$ such that $|f(z)| \leq M$ there. Define

$$g(z) = \begin{cases} (z - z_0)^2 f(z) & \text{if } z \in D^*(z_0, r) \\ 0 & \text{if } z = z_0. \end{cases}$$

Then g is holomorphic on $D^*(z_0, r)$, and

$$|g(z)| \leq M |z - z_0|^2$$

shows that $g(z) \to 0$ as $z \to z_0$. Moreover,

$$\left| \frac{g(z)}{z - z_0} \right| \leq M |z - z_0| \to 0,$$

so $g'(z_0) = 0$.

Define $g(z_0) = 0$. Then g is holomorphic on all of $D(z_0, r)$. By Theorem 3.2.1, g has a Taylor expansion at z_0,

$$g(z) = \sum_{j=2}^{\infty} a_j (z - z_0)^j.$$

Hence

$$\sum_{j=0}^{\infty} a_{j+2} (z - z_0)^j$$

defines a holomorphic function on $D(z_0, r)$. On the punctured disk this series equals

$$\frac{g(z)}{(z - z_0)^2} = f(z).$$

Therefore f extends holomorphically to $D(z_0, r)$. □

3.2.1 Topology, Partitions of Unity, and the Existence of Bump Functions

Definition 3.2.1.1 *(Topological Space)*: A *topological space* is a pair (X, τ), where X is a set and τ is a collection of subsets of X satisfying the following properties:

1. $\emptyset \in \tau$ and $X \in \tau$.
2. The union of any (possibly infinite) collection of sets in τ is also in τ.
3. The intersection of any finite collection of sets in τ is also in τ.

The collection τ is called a *topology* on X, and its elements are referred to as *open sets* under the topology τ.

Obviously the statement "let X be a topological space" itself has little meaning. However, when the topology is implicitly obvious or the space is describable without it, then it may be verbally elided.

The implied topology of a subspace A of (X, τ) is given by the intersection of each set in τ with A.

Definition 3.2.1.2: A subset A of a topological space X is *closed* iff $X \setminus A$ is open.

It is immediate from definition that the trivial sets X and $\emptyset$ are always closed. It is equally trivial from definition that the union of finitely many closed sets is closed, and the intersection of any collection of closed sets is closed.

If $\exists U \in \tau$ such that $x \in U$, then U is an (open) *neighborhood* of x. If $\forall x, y \in X$ (different) have disjoint neighborhoods, then X is a *Hausdorff space*.

The following discussions involved with topological spaces here will always be of Hausdorff spaces, although making such distinction is important for future extensibility.

Definition 3.2.1.3: A topological space X is *compact* iff every open cover has a finite subcover. For a topological space X, a set $A \subseteq X$ is *compact* iff every open cover has a finite subcover.

Proposition 3.2.1.1: Suppose X is a Hausdorff topological space and let $A \subseteq X$ be compact. Then A is closed in X.

Proof: Let $x \in X \setminus A$ be fixed. For each $a \in A$, since X is Hausdorff, there exist disjoint neighborhoods U_a and V_a with $x \in U_a$ and $a \in V_a$. The set

$$\bigcup_{a \in A} V_a \supseteq A$$

covers A, which by assumption, has a finite subcover

$$\bigcup_{k=1}^{n} V_{a_k} \supseteq A, \quad \forall k \in \mathbb{N}_{\leq n}, a_k \in A.$$

Moreover, the intersection

$$U_x = \bigcap_{k=1}^{n} U_{a_k}$$

is an open neighborhood of x and by construction, it is disjoint from the finite subcover. Since it is disjoint from a superset of A, it lies entirely in $X \setminus A$.

For each $x \in X \setminus A$, construct open U_x accordingly. Then we obtain

$$X \setminus A \subseteq \bigcup_{x \in X \setminus A} U_x \subseteq X \setminus A,$$

where the sandwiched union is open. Therefore, A is closed. □

Proposition 3.2.1.2: If X is a compact space and $A \subseteq X$ is closed, then A is compact.

Proof: Let $\mathcal{U}$ be an open cover of A in X. Since $X \setminus A$ is open, the set

$$\{U \cup (X \setminus A) \mid U \in \mathcal{U}\}$$

openly covers X. Then a finite subcover

$$\{U_k \cup (X \setminus A)\}_{k \in \mathbb{N}_{\leq n}}$$

exists and covers X. The refinement $\{U_k\}_{k \in \mathbb{N}_{\leq n}}$ then covers A. □

Definition 3.2.1.4: A point a is an accumulation point of a set A in a topological space X iff any open U with $a \in U$ implies that $U \cap A$ contains a point other than a.

Proposition 3.2.1.3: A set A in a topological space X is closed iff it contains all its accumulation points.

Proof: We first prove the forward implications under the assumption that A is closed. Since $X \setminus A$ is open, and suppose for contradiction, that $a \in X \setminus A$. Then for $a \in U = X \setminus A$ open, $U \cap A = \emptyset$ (and hence a cannot be an accumulation point by contradiction of definition).

Assume the converse assumption that A contains all its accumulation points. Let $x \in X \setminus A$ be arbitrary. By assumption, x is not an accumulation point of A. Hence, for some open set $U \supseteq \{x\}$, $U \cap A$ does not contain a point other than x (which it also cannot contain), implying that $U \cap A = \emptyset$, and hence $U \subseteq X \setminus A$.

For each $x \in X \setminus A$, we hence construct some open neighborhood fully contained in $X \setminus A$. Together, they must union (by the definition of a topology) to an open set, being $X \setminus A$. Therefore, A is closed.□

A topology allows the definition and general conceptualization of continuity, convergence, and connectivity in a general setting, without necessarily relying on a notion of distance (a metric).

Definition 3.2.1.5: A function $f : X \to Y$ between two topological spaces is said to be *continuous* if the *pre-image* of every open set in Y,

$$\{x \in X \mid f(x) \in Y\},$$

is an open set in X.

For the case of metric spaces, this generalizes the epsilon–delta notion of continuity.

Example 3.2.1.1: Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 \text{ if } x \geq 0 \\ 0 \text{ if } x < 0. \end{cases}$$

We equip both the domain and codomain with the standard topology on $\mathbb{R}$. Let $V = (.5, 1.5) \subseteq \mathbb{R}$. Then the pre-image of V is

$$f^{-1}(V) = \{x \in \mathbb{R} \mid f(x) \in V\} = \mathbb{R}_{\geq 0},$$

which is not an open set in the standard topology on $\mathbb{R}$. Thus, f is not continuous.

For two topological spaces X and Y, a function $f : X \to Y$ is a *homeomorphism* (also known as a *bicontinuous function*) if it is a bijection such that both f and f^{-1} are continuous. If such a function exists, then X and Y are *homeomorphic*.

The function $f : [0, 2\pi) \to S^1$ with $f(t) = (\cos(t), \sin(t))$ is indeed continuous, but the inverse $f^{-1}(x_1, x_2)$ is discontinuous at $(x_1, x_2) = (1, 0)$.

Proposition 3.2.1.4: Let $(X, \tau_1), (Y, \tau_2)$ be two topological spaces. Then for $f : X \to Y$, the following conditions are equivalent:

1. f is continuous.
2. If $A \subseteq Y$ is closed, then the pre-image $f^{-1}(A)$ is closed.
3. If $a \in X$ and $A \in \tau_2$ is an open neighborhood of $f(a)$ in Y, then there is some $U \in \tau_1$ that is a neighborhood of a such that $f(U) \subseteq A$.

Proof: We first show that 1 implies 2. By continuity, for $A \subseteq Y$ closed, $Y \setminus A \in \tau_2$, then

$$f^{-1}(Y \setminus A) = \{x \in X \mid f(x) \in Y \setminus A\} = X \setminus f^{-1}(A).$$

Assume the conditions of 2 for the converse. Let $U \in \tau_2$ be open, $Y \setminus U$ closed, then $f^{-1}(Y \setminus U)$ is closed. Similar logic shows

$$f^{-1}(Y \setminus U) = X \setminus f^{-1}(U),$$

which implies $f^{-1}(U)$ is open.

Next we aim to show that continuity implies 3. By assumption, the pre-image of any open $A \subseteq Y$ is $f^{-1}(A)$ and open and $a \in f^{-1}(A)$. The property is complete under $U = f^{-1}(A)$.

Assume the conditions of 3 for the converse. Let $A \in \tau_2$ be arbitrary. We aim to show that $f^{-1}(A) \in \tau_1$. If $f^{-1}(A) = \emptyset$, then the conclusion is satisfied trivially. Hence, assume that $\exists a \in f^{-1}(A)$. For any such a, there exists a neighborhood $U_a \in \tau_1$ such that $f(U_a) \subseteq A$. Hence $U_a \subseteq f^{-1}(A)$ for any $a \in f^{-1}(A)$. Therefore, we obtain

$$\bigcup_{a \in f^{-1}(A)} U_a \subseteq f^{-1}(A), \quad \bigcup_{a \in f^{-1}(A)} U_a \supseteq \bigcup_{a \in f^{-1}(A)} \{a\} = f^{-1}(A) \Rightarrow \bigcup_{a \in f^{-1}(A)} U_a = f^{-1}(A).$$

By the definition of topologies,

$$\bigcup_{a \in f^{-1}(A)} U_a \in \tau_1.$$

□

Definition 3.2.1.6 *(Basis for a Topology)*: Let X be a set. A *basis* for a topology on X is a collection $\mathfrak{B}$ of subsets of X satisfying

1. $\bigcup_{B \in \mathfrak{B}} B = X$.
2. For any $B_1, B_2 \in \mathfrak{B}$ and any point $x \in B_1 \cap B_2$, there exists a set $B_3 \in \mathfrak{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

The topology generated by $\mathfrak{B}$ is the collection of all unions of elements of $\mathfrak{B}$.

Definition 3.2.1.7: A *metric space* is a pair (X, d), where X is a set and d is a function from $X \times X$ to $\mathbb{R}_{\geq 0}$, called a *metric*, such that for all $x, y, z \in X$ the following properties hold:

1. $d(x, y) \geq 0$ and $d(x, y) = 0$ iff $x = y$ (positivity).
2. $d(x, y) = d(y, x)$ (symmetry).
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

This in turn implies the reverse triangle inequality:

$$d(x, z) \leq d(x, y) + d(y, z) \Rightarrow d(x, y) \geq d(x, z) - d(y, z),$$

and similarly,

$$d(y, z) \leq d(x, y) + d(x, z) \Rightarrow d(x, y) \geq d(y, z) - d(x, z).$$

Definition 3.2.1.8: Let (X, d) be a metric space. The *metric topology induced by* d is the topology τ_d generated by the basis

$$\{B(x, r) \mid x \in X, r > 0\}$$

comprising the balls

$$B(x, r) = \{y \in X \mid d(x, y) < r\}.$$

The pair (X, τ_d) is the *topological space induced by the metric* d.

We now justify a claim whose triviality we have taken for granted.

Proposition 3.2.1.5: Let (X, d) be a metric space under the induced metric topology. Then for any open set $U \subseteq X$, any point $x \in U$, there exists a ball $B(x, \delta)$ ($\delta > 0$) in U.

Proof: By definition, U lies in the topology for X and is the union of (possibly infinitely many) balls. There then exists some ball $B(x_0, \delta')$ in U that contains x. Let

$$\delta = \delta' - d(x_0, x).$$

Since $d(x_0, x) < \delta'$, for any $y \in B(x, \delta)$, we have

$$d(x_0, y) \leq d(x_0, x) + d(x, y) \leq \delta'.$$

Hence, the open ball $B(x, \delta)$ centered at x lies within $B(x_0, \delta')$. □

Theorem 3.2.1.1: Let $(X, d_x), (Y, d_y)$ be two metric spaces under the metric topology. Then a function $f : X \to Y$ is topologically continuous iff it is epsilon–delta continuous.

Proof: We first imply that topological continuity implies epsilon–delta continuity. For any $x \in X$, $\forall \varepsilon > 0$, the ball $B(f(x), \varepsilon)$ is an open set (it is in the basis) in Y. By 3 of Proposition 3.2.1.4, there is some open neighborhood U of x in X such that $f(U) \subset B(f(x), \varepsilon)$. By the previous proposition, there is a ball $B(x, \delta) \subseteq U$. This is equivalent to

$$\varepsilon > 0, x \in X \Rightarrow \exists \delta = \delta_x > 0 : y \in B(x, \delta) \Rightarrow f(y) \in B(f(x), \varepsilon).$$

Conversely, assume f is epsilon–delta continuous. Let $V \subseteq Y$ be open and $x \in f^{-1}(V)$. Since V is open in the metric topology, there exists $\varepsilon > 0$ such that

$$B(f(x), \varepsilon) \subseteq V.$$

By epsilon–delta continuity, there exists $\delta > 0$ such that

$$d_{x(x,y)} < \delta \Rightarrow d_{y(f(x),f(y))} < \varepsilon,$$

or that

$$y \in B(x, \delta) \Rightarrow f(y) \in B(f(x), \varepsilon) \subseteq V.$$

Thus, the ball $B(x, \delta_x)$ is an open neighborhood of x in X such that

$$B(x, \delta_x) \subseteq f^{-1}(V) \Rightarrow f^{-1}(V) \subseteq \bigcup_{x \in f^{-1}(V)} B(x, \delta_x) \subseteq f^{-1}(V).$$

Since the union of open sets is open, the pre-image of any open set is open, and hence f is topologically continuous. □

Theorem 3.2.1.2: Let X be a compact topological space and let Y be a Hausdorff space. If $f : X \to Y$ is a continuous bijection, then f is a homeomorphism.

Proof: If $A \subseteq X$ is compact, then the pre-images of any open cover $\mathcal{U}$ of $f(A)$ cover A. Hence, there is a finite subcover

$$\{f^{-1}(U_k) \mid U_k \in \mathcal{U}, k \in \mathbb{N}_{\leq n}\}$$

covering A. Then

$$\{U_k \mid U_k \in \mathcal{U}, k \in \mathbb{N}_{\leq n}\}$$

covers $f(A)$, and hence $f(A)$ is compact.

For any closed $C \subseteq X$, Proposition 3.2.1.2 implies C is compact. Hence, $f(C)$ is compact. By Proposition 3.2.1.1, $f(C)$ is closed. Hence, f maps closed sets to closed sets, and the pre-image of any closed set is closed under f^{-1}. Hence, Proposition 3.2.1.4 implies f^{-1} is continuous, thus f is a homeomorphism. □

It is worth noting some motivating examples for which the conclusion fails when certain hypotheses are not satisfied.

Example 3.2.1.2: Let $I = [0, 2\pi)$ be a topological space with a metric $|\cdot|$ under the standard topology (the subspace topology induced by the basis formed with open "balls" or symmetric intervals around each point). Equip the unit circle

$$S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$$

generated by the metric defined by arc-length (d_{S^1}). Then the continuous bijection $f : I \to S^1$ defined by $f(t) = \mathrm{e}^{\mathrm{i}t}$ is not a homeomorphism.

Proof: The non-continuity of $f^{-1} : S^1 \to I$ is easy to visually see, both topologically and by epsilon–delta. Topologically, select the *relatively* open interval $[0, \pi)$ in I. The pre-image of this set under f^{-1} is $\mathrm{e}^{\mathrm{i}[0,\pi)}$, which is clearly not an open set. This proves that f^{-1} is not continuous (by definition).

For continuity to hold by epsilon–delta, any ε would yield the existence of some δ such that $\forall a, b \in S^1$ with $d_{S^1}(a, b) < \delta$, $\left|f^{-1}(a) - f^{-1}(b)\right| < \varepsilon$.

Let $\varepsilon = \frac{\pi}{2}$. For any $0 < \delta < 2\pi$, the points

$$a = \mathrm{e}^{-\mathrm{i}\frac{\delta}{4}}, b = \mathrm{e}^{\mathrm{i}\frac{\delta}{4}}$$

satisfy

$$d_{S^1}(a, b) = \frac{\delta}{2} < \delta.$$

However,

$$f^{-1}(a) = 2\pi - \frac{\delta}{4}, \quad f^{-1}(b) = \frac{\delta}{4},$$

and

$$\left|f^{-1}(a) - f^{-1}(b)\right| = 2\pi - \frac{\delta}{2} > \pi > \varepsilon.$$

This contradicts the previous statement. □

We now provide a formal definition of the connectivity of sets:

Definition 3.2.1.9: A topological space X is *disconnected* if it can be written as the union of two nonempty disjoint open sets. Otherwise, it is *connected.*

In a topological space X, a subset can be open, closed (the complement of some open set), both (clopen), or neither. The only clopen sets that exist in any topological space X are $\emptyset$ and X iff X is connected. A technique pertinent to many future proofs relies on the following fact:

Theorem 3.2.1.3 *(Connectivity Argument)*: A topological space X is *connected* if and only if X and $\emptyset$ are the only clopen subsets of X.

Proof: Suppose X is connected and let $A \subseteq X$ be clopen. Then A and $X \setminus A$ are both open in X, disjoint, and their union is X. Thus, either $A = \emptyset$ or $X \setminus A = \emptyset$ (i.e. $A = X$).

Conversely, suppose X is disconnected. Then there exist nonempty open sets $U, V \subseteq X$ such that $U \cap V = \emptyset$ and $U \cup V = X$. Thus,

$$U = X \setminus V$$

and

$$V = X \setminus U$$

are both clopen, contradicting the assumption that X and $\emptyset$ are the only clopen subsets. Hence, X must be connected. □

Example 3.2.1.3: The topological space $\mathbb{R}$ under the standard topology has only two clopen sets: $\mathbb{R}$ and $\emptyset$.

Now consider

$$X = \bigcup_{n \in 2\mathbb{Z}} (n, n + 1),$$

equipped with the topology τ generated by the basis

$$\{(n, n + 1) \mid n \in 2\mathbb{Z}\}.$$

This space is disconnected. For instance, $(0,1) \subset X$ is open (as it is in τ) and closed (since its complement in X is

$$\bigcup_{n \in (2\mathbb{Z} \setminus \{0\})} (n, n+1) \in \tau).$$

In fact, every set in τ is clopen.

Proposition 3.2.1.6: The interval $[0,1]$ (under the subspace topology induced by $\mathbb{R}$) is connected.

Proof: Assume, for contradiction, that there exist two disjoint nonempty open sets $U, V \subset [0,1]$ such that

$$U \cup V = [0,1].$$

Without loss of generality, assume $0 \in U$ (otherwise switch U and V). Let

$$a = \inf V.$$

Since U, V are also closed in $[0,1]$, either $a \in V$ or a is an accumulation point. Either way, a is contained in V by Proposition 3.2.1.3. Assume that $a \neq 0$. Then by openness, there exists some $0 < \delta < a$ such that

$$(a - \delta, a) \subseteq (a - \delta, a + \delta) \cap [0,1] \subseteq V.$$

In particular,

$$a - \frac{\delta}{2} \in V,$$

which contradicts a being a lower bound of V.

Therefore, $a = 0$. However, since $[0, \delta)$ lies in U for some $\delta > 0$, $a \geq \delta > 0$. Thus, we arrive at a contradiction, and thus V is the empty set. This then shows that $[0,1]$ is connected. □

Connectivity intuitively means that a space cannot be split into two disjoint open subsets, but is not meaningful in terms of how points within the space relate to each other. In many geometric situations, the notion of *path-connectivity* requires that any two points be joined by a continuous path. We will see that this more concrete condition forces the space to be topologically connected.

Definition 3.2.1.10: A topological space X is said to be *path-connected* iff for any two points $a, b \in X$, there is a continuous function $f : [0,1] \to X$, where $[0,1]$ is equipped with the metric topology and $f(0) = a, f(1) = b$.

Theorem 3.2.1.4: A path-connected topological space X is connected.

Proof: Assume path-connectivity and suppose X is disconnected. Then two open nonempty disjoint components $U, V \subset X$ can be found. Let $u \in U, v \in V$ be two arbitrary points. Then there exists $f : [0,1] \to X$ such that $f(0) = u$, $f(1) = v$. By continuity, the pre-images of U and V, namely $f^{-1}(U)$ and $f^{-1}(V)$ respectively, are disjoint open subsets of $[0,1]$. Moreover, the pre-image are nonempty as they contain 0 and 1 respectively. This contradicts the connectivity of $[0,1]$ in Proposition 3.2.1.6. □

Definition 3.2.1.11 *(Exhaustion by Compact Sets)*: For a topological space X, an *exhaustion by compact sets* is a nested sequence of compact sets $\{K_n\}_{n \in \mathbb{N}} \subseteq X$ such that $K_n \subset \mathring{K}_{n+1}$ for all $n \in \mathbb{N}$ and

$$X = \bigcup_{n \in \mathbb{N}} K_n.$$

Lemma 3.2.1.1: Let $\Omega \subseteq \mathbb{C}$ be an open set and let $\mathfrak{B}$ be a basis for the topology on Ω. Then there exists a collection of sets $\{U_n\}_{n\in\mathbb{N}} \subseteq \mathfrak{B}$ such that

1. $\bigcup_{n\in\mathbb{N}} U_n = \Omega$.
2. For every compact $K \subset \Omega$, K intersects only finitely many sets in $\{U_n\}_{n\in\mathbb{N}}$.

Proof: Let $\{K_n\}_{n\in\mathbb{N}} \subset \Omega$ be an exhaustion by compact sets with $K_0 = \emptyset$ and $K_n \subseteq \overset{\circ}{K}_{n+1}$ for all $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, define

$$W_n = \overset{\circ}{K}_{n+1} \setminus K_{n-2}, \quad V_n = K_n \setminus \overset{\circ}{K}_{n-1},$$

where $K_{-1} = \emptyset$. Each W_n is open and each V_n is compact, with $V_n \subseteq W_n$ and

$$\bigcup_{n\in\mathbb{N}} V_n = \Omega.$$

For each $n \in \mathbb{N}$ and each $z \in V_n$, since W_n is open and contains z, there exists $U_{z,n} \in \mathfrak{B}$ such that

$$z \in U_{z,n} \subseteq W_n.$$

The collection

$$\{U_{z,n} \mid z \in V_n\}$$

is an open cover of the compact set V_n, so by Heine–Borel (Theorem 1.1.2) it admits a finite subcover, there exist finitely many points $z_{n,1}, ..., z_{n,k_n} \in V_n$ such that

$$V_n \subset \bigcup_{i=1}^{k_n} U_{z_{n,i},n} \subseteq W_n.$$

Enumerate all such $U_{z_{n,i},n}$ over $n \in \mathbb{N}$ and $i = 1, ..., k_n$ to obtain a countable collection $\{U_j\}_{j\in\mathbb{N}} \subseteq \mathfrak{B}$. Then

$$\bigcup_{j\in\mathbb{N}} U_j = \Omega,$$

proving 1.

For 2, let $K \subset \Omega$ be compact. There exists $N \in \mathbb{N}$ such that

$$K \subset \overset{\circ}{K}_N,$$

so K is disjoint from V_n for all $n > N + 1$. Since each V_n intersects only finitely many U_j, K intersects only finitely many U_j. Thus the collection is locally finite. □

Remark: The property of local finiteness of an open collection S in Ω is commonly stated as: for every $z \in \Omega$, there exists an open neighborhood of z that intersects only finitely many sets in S.

This is equivalent to 2 in Lemma 3.2.1.1. Indeed, if every point has such a neighborhood, then any compact $K \subset \Omega$ admits a finite subcover of these neighborhoods by Heine–Borel (Theorem 1.1.2), so K intersects finitely many sets in S. Conversely, for any $z \in \Omega$, take an open neighborhood V with $z \in V$ and with relatively compact closure in Ω; then $\overline{V}$ intersects finitely many sets in S, and so does V.

Theorem 3.2.1.5 *(Partition of Unity)*: Let $\Omega \subseteq \mathbb{C}$ be a nonempty open set and let $\{\Omega_k\}_{k\in\mathbb{N}}$ be an open cover of Ω. Then there exists a collection of bump functions $\{\alpha_j\}_{j\in\mathbb{N}} \subseteq C^{\infty(\mathbb{C})}$, each with compact support in Ω, satisfying:

1. For each $j \in \mathbb{N}$, there exists $k \in \mathbb{N}$ such that $\operatorname{supp}(\alpha_j) \subseteq \Omega_k$.
2. The collection $\{\operatorname{supp}(\alpha_j)\}_{j\in\mathbb{N}}$ is locally finite.
3. For each $j \in \mathbb{N}$, $0 \leq \alpha_j \leq 1$.
4. $\sum_{j=1}^{\infty} \alpha_j \equiv 1$ on Ω.

Then $\{\alpha_j\}_{j\in\mathbb{N}}$ is called a C^∞ partition of unity subordinate to $\{\Omega_k\}_{k\in\mathbb{N}}$.

Proof: For each $z \in \Omega$ there exists $r_z > 0$ and $k_z \in \mathbb{N}$ such that

$$\overline{D(z, r_z)} \subset \Omega_{k_z}.$$

The collection

$$\{D(z, r) \mid z \in \Omega \land 0 < r < r_z\}$$

is an open basis for Ω. By Lemma 3.2.1.1 there exists a locally finite open cover

$$\left\{D\left(z_j, r_{z_j}\right)\right\}_{j\in\mathbb{N}} \subseteq \mathfrak{B}$$

of Ω with

$$D\left(z_j, r_{z_j}\right) \subset \overline{D\left(z_j, r_{z_j}\right)} \subset \Omega_{k_{z_j}}, \quad \forall j \in \mathbb{N}.$$

Define the standard bump function

$$\theta(z) = \begin{cases} e^{\frac{1}{|z|^2-1}} & \text{if } |z| < 1 \\ 0 & \text{if } |z| \geq 1. \end{cases}$$

For $\varepsilon > 0$ let

$$\theta_{\varepsilon(z)} = \theta\left(\frac{z}{\varepsilon}\right),$$

which has support $\overline{D(0, \varepsilon)}$. Define

$$\beta_{j(z)} = \theta_{r_{z_j}}\left(z - z_j\right),$$

so

$$\operatorname{supp}(\beta_j) = \overline{D\left(z_j, r_{z_j}\right)} \subset \Omega_{k_{z_j}}.$$

By local finiteness of $\left\{D\left(z_j, r_{z_j}\right)\right\}_{j\in\mathbb{N}}$, for each $z \in \Omega$ there exists an open neighborhood V with $z \in V$ intersecting only finitely many $\overline{D\left(z_j, r_{z_j}\right)}$. Thus $\{\operatorname{supp}(\beta_j)\}_{j\in\mathbb{N}}$ is locally finite on Ω. Then the sum

$$S(z) = \sum_{j=1}^{\infty} \beta_{j(z)}$$

defined for $z \in \Omega$ involves only finitely many nonzero terms (by local finiteness) on a neighborhood of every point z. Hence $S \in C^{\infty(\Omega)}$ and $S(z) > 0$ (since $\left\{D\left(z_j, r_{z_j}\right)\right\}_{j\in\mathbb{N}}$ covers Ω). Define

$$\alpha_{j(z)} = \frac{\beta_{j(z)}}{S(z)}, \quad \forall j \in \mathbb{N}.$$

Each $\alpha_j \in C^{\infty(\mathbb{C})}$ has compact support in Ω, $0 \leq \alpha_j \leq 1$, the supports are locally finite, and

$$\sum_{j=1}^{\infty} \alpha_{j(z)} = 1$$

for all $z \in \Omega$. Moreover

$$\operatorname{supp}(\alpha_j) \subseteq \Omega_{k_{z_j}},$$

proving subordination. □

Theorem 3.2.1.6 (*Existence of Bump Functions*): Let $K \subset \mathbb{C}$ be compact and $V \subset \mathbb{C}$ an open neighborhood of K. Then there exists a compactly supported $\varphi \in C^{\infty(\mathbb{C})}$ such that

$$0 \leq \varphi(z) \leq 1 \quad \forall z \in \mathbb{C},$$

$\operatorname{supp}(\varphi) \subset V$, and $\varphi \equiv 1$ on some open neighborhood of K.

Proof: Let

$$V(K, \varepsilon) = \left\{ z \in \mathbb{C} \mid \inf_{\zeta \in K} |z - \zeta| < \varepsilon \right\}$$

denote the open ε-neighborhood of K. Since V is an open neighborhood of K, $\exists \varepsilon > 0$ such that

$$K \subset V(K, \varepsilon) \subset V(K, 2\varepsilon) \subset V,$$

where $A \subset B$ means that the closure of A is compact and contained in B.

Define the open sets

$$\Omega_1 = V(K, 2\varepsilon), \quad \Omega_2 = \mathbb{C} \setminus \overline{V(K, \varepsilon)}.$$

Then $\{\Omega_1, \Omega_2\}$ is an open cover of $\mathbb{C}$.

By the Partition of Unity Theorem (Theorem 3.2.1.5), there exist compactly supported functions $\{\alpha_j\}_{j \in \mathbb{N}} \subseteq C^{\infty(\mathbb{C})}$ forming a partition of unity subordinate to this cover. That is,

$$0 \leq \alpha_j \leq 1, \quad \operatorname{supp}(\alpha_j) \subseteq \Omega_{i_j} \quad \text{for some} \quad i_j \in \{1, 2\}, \quad \sum_{j=1}^{\infty} \alpha_j \equiv 1 \quad \text{on } \mathbb{C}.$$

Let

$$J = \{ j \in \mathbb{N} \mid \operatorname{supp}(\alpha_j) \subseteq \Omega_1 \}.$$

Define

$$\varphi(z) = \sum_{j \in J} \alpha_{j(z)}.$$

Then $\varphi \in C^{\infty(\mathbb{C})}$ is compactly supported within Ω_1, and since only finitely many α_j are nonzero on a neighborhood of each point, $\varphi \in C^{\infty(\mathbb{C})}$. Moreover,

$$\operatorname{supp}(\varphi) \subset \Omega_1 \subset V.$$

For $z \in V(K, \varepsilon)$, all functions with support in Ω_2 vanish at z, so

$$\varphi(z) = \sum_{j \in J} \alpha_{j(z)} = \sum_{j=1}^{\infty} \alpha_{j(z)} = 1.$$

Hence, $\varphi \equiv 1$ on the open neighborhood $V(K, \varepsilon)$ of K. Outside $V(K, 2\varepsilon)$, all terms with support in Ω_1 vanish, so $\varphi(z) = 0$. Finally, $0 \leq \varphi \leq 1$ everywhere by construction. Thus φ satisfies all required properties. □

3.3 Zeros of a Holomorphic Function

For a region $U \subseteq \mathbb{C}$ and a holomorphic function $f : U \to \mathbb{C}$, a point $z_0 \in U$ is a *zero* of f iff $f(z_0) = 0$. Furthermore, if f has the Taylor expansion at z_0 of

$$a_m(z - z_0)^m + a_{m+1}(z - z_0)^{m+1} + \cdots, \quad m \in \mathbb{N}, a_m \neq 0,$$

then the zero at z_0 has multiplicity m.

We will introduce a fundamental application of Liouville's Theorem (Theorem 3.2.3) below.

Theorem 3.3.1 *(Fundamental Theorem of Algebra)*: Every non-constant polynomial $p(z)$ with complex coefficients has at least one complex zero.

Proof: For the sake of contradiction, suppose that $p(z)$ has no complex zeros. Then the function $f(z) = \frac{1}{p(z)}$ is continuous and entire, because $p(z)$ has no zeros in $\mathbb{C}$. Moreover, as $z \to \infty$, $p(z) \to \infty$, so $f(z) \to 0$, and thus $f(z)$ is bounded. By Liouville's Theorem (Theorem 3.2.3), every bounded entire function is constant. Thus, $f(z)$ is constant, and so $p(z)$ must also be constant. By contradiction, $p(z)$ has at least one complex zero. □

Theorem 3.3.2: Let $U \subseteq \mathbb{C}$ be open and connected, and $f : U \to \mathbb{C}$ be holomorphic over U. Then if the set defined by

$$S = \{z \in U \mid f(z) = 0\}$$

has an accumulation point in U, then $f \equiv 0$ over U.

Proof: Let $\{z_n\}_{n \in \mathbb{N}}$ be a subset of S and assume it has an accumulation point z_∞ in U. Since f is holomorphic over U, $\exists \varepsilon > 0$ such that f is holomorphic over $D(z_\infty, \varepsilon) \subseteq U$. Then over this disk, f has the Taylor expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_\infty)^n. \tag{3.3.1}$$

By Definition 1.1.1, $\exists N \in \mathbb{N}$ such that $\forall n > N$, $z_n \in D(z_\infty, \varepsilon)$. Since z_n is a zero of f, $f(z_n) = 0$. Then, by the continuity of f,

$$\lim_{n \to \infty} f(z_n) = f\Big(\lim_{n \to \infty} z_n\Big) = f(z_\infty) = 0.$$

Using this result in comparison to (3.3.1), we get that $a_0 = 0$.

The function

$$f_1(z) = \frac{f(z)}{z - z_\infty}$$

has a Taylor expansion over $D(z_\infty, \varepsilon)$ of

$$f_1(z) = \sum_{n=0}^{\infty} a_{n+1} (z - z_\infty)^n.$$

Let $z = z_n \neq z_\infty$ for some $n > N$. Then f_1 vanishes, leaving

$$0 = a_1 + \mathcal{O}(z_n - z_\infty).$$

Letting $n \to \infty$, $z_n \to z_\infty$, and $a_1 = 0$. Define

$$f_2(z) = \frac{f_1(z)}{z - z_\infty}.$$

Then,

$$f_2(z) = \sum_{n=0}^{\infty} a_{n+2}(z - z_\infty)^n.$$

Similarly, $a_2 = 0$. Letting

$$f_{n(z)} = \frac{f(z)}{(z - z_\infty)^n},$$

the sequence $\{a_n\}_{n \in \mathbb{Z}_{\geq 0}}$ vanishes, and $f \equiv 0$ on $D(z_\infty, \varepsilon)$.

Let

$$\tilde{S} = \{z \in U \mid \forall n \in \mathbb{Z}_{\geq 0}, f^n(z) = 0\}.$$

For all $z \in D(z_\infty, \varepsilon)$, since $f(z)$ locally vanishes (and has vanishing derivatives as a consequence),

$$D(z_\infty, \varepsilon) \subseteq \tilde{S}.$$

Furthermore, for all $z' \in \tilde{S}$, $\exists \varepsilon' > 0$ such that $f(z)$ has a convergent Taylor series with vanishing coefficients on $D(z', \varepsilon') \subseteq U$. Then $f \equiv 0$ on $D(z', \varepsilon')$. Then for all $z \in D(z', \varepsilon')$, since f is constant at z, it also has vanishing derivatives. It follows that

$$D(z', \varepsilon') \subseteq \tilde{S}.$$

Since every point in $\tilde{S}$ has an open neighborhood also in $\tilde{S}$, $\tilde{S}$ is open.

It is evident that for all $k \in \mathbb{Z}_{\geq 0}$, f^k is continuous in U by the holomorphy of f. Let

$$S_k = \{z \in U \mid f^k(z) = 0\}.$$

For any sequence $\{\tilde{z}_n\} \in S_k$ converging to some $\tilde{z}_\infty \in U$, by the continuity of f,

$$\lim_{n\to\infty} f^k(\tilde{z}_n) = f^k\left(\lim_{n\to\infty} \tilde{z}_n\right) = f^k(\tilde{z}_\infty) = 0,$$

and therefore $\tilde{z}_\infty \in S_k$. Thus, S_k contains all of its accumulation points in U and is therefore closed in U (if $\tilde{z}_\infty \notin U$, then it is no longer relevant; we are concerned about it being closed within U). Since

$$\tilde{S} = \bigcap_{k \in \mathbb{Z}_{\geq 0}} S_k$$

and each of S_k is closed in U, $\tilde{S}$ is the intersection of closed sets and consequently closed.

Since $\tilde{S}$ is nonempty and clopen in the connected set U, $\tilde{S} = U$ (by Theorem 3.2.1.3). It follows that $f \equiv 0$ on U. □

Remark: This is a trivial property of holomorphic functions that allows for the uniqueness of analytic continuations. It is oftentimes stated in the form below:

Theorem 3.3.3 (*Identity Theorem*): Let $U \subseteq \mathbb{C}$ be open and connected, and define $f(z)$ and $g(z)$ to be two holomorphic functions on U. For a set $S \subseteq U$ with an accumulation point in U, if $f \equiv g$ on S, then $f \equiv g$ on U.

Proof: Let $h = f - g$ be holomorphic over U. Since S has an accumulation point in U, and $h \equiv 0$ over S, then by Theorem 3.3.2, $h \equiv 0$ over U. □

Theorem 3.3.4 (*Holomorphic Argument Principle*): Let $U \subseteq \mathbb{C}$ be a region and $f : U \to \mathbb{C}$ be holomorphic. Let $\gamma \subset U$ be a simple, closed, positively oriented curve that is null-homotopic in U. If f has no zeros on γ, then f has finitely many zeros in the region bounded by γ, and this number, counting multiplicities, is given by

$$k = \frac{1}{2\pi \mathrm{i}} \oint_\Gamma \frac{f'(z)}{f(z)} \,\mathrm{d}z.$$

Let Γ be the image of γ under the map $w = f(z)$. Then

$$k = \frac{1}{2\pi} \Delta_\Gamma \arg(w),$$

where $\Delta_\Gamma \arg(w)$ denotes the total change in argument of w as it traverses Γ.

Proof: Let $z_1, ..., z_n$ be the distinct zeros of f enclosed by γ with the respective multiplicities $k_1, ..., k_n$. Choose disjoint disks $D(z_j, \varepsilon_j)$ centered at each z_j with radii $\varepsilon_j > 0$, each contained in the interior of γ and avoiding γ. The function

$$\frac{f'(z)}{f(z)}$$

is holomorphic on the domain

$$\operatorname{int}(\gamma) \setminus \bigcup_{j=1}^{n} \overline{D(z_j, \varepsilon_j)},$$

where $\operatorname{int}(\gamma)$ denotes the interior relative to γ. The oriented boundary of this domain is

$$\gamma^+ \cup \bigcup_{j=1}^{n} \partial D(z_j, \varepsilon_j)^-.$$

By Cauchy–Goursat (Theorem 3.1.7),

$$\int_{\gamma^+ \cup \bigcup_{j=1}^{n} \partial D(z_j, \varepsilon_j)^-} \frac{f'(z)}{f(z)} \,\mathrm{d}z = 0,$$

which rearranges to

$$\oint_{\gamma^+} \frac{f'(z)}{f(z)} \,\mathrm{d}z = \sum_{j=1}^{n} \oint_{\partial D(z_j, \varepsilon_j)^+} \frac{f'(z)}{f(z)} \,\mathrm{d}z.$$

Near each z_j, express

$$f(z) = (z - z_j)^{k_j} h_{j(z)}$$

where h_j is holomorphic and non-vanishing on $D(z_j, \varepsilon_j)$. Differentiation yields

$$f'(z) = k_j(z - z_j)^{k_j - 1} h_{j(z)} + (z - z_j)^{k_j} h_{j'}(z),$$

and thus

$$\frac{f'(z)}{f(z)} = \frac{k_j}{z - z_j} + \frac{h_{j'}(z)}{h_{j(z)}}.$$

Since h_j is holomorphic and non-vanishing on $D(z_j, \varepsilon_j)$, the function

$$\frac{h_{j'}}{h_j}$$

is holomorphic there. By the Cauchy–Goursat Theorem,

$$\oint_{\partial D(z_j, \varepsilon_j)} \frac{h_{j'}(z)}{h_{j(z)}} \, \mathrm{d}z = 0.$$

The Cauchy–Goursat Formula (Theorem 3.1.8) gives

$$\oint_{\partial D(z_j, \varepsilon_j)} \frac{k_j}{z - z_j} \, \mathrm{d}z = 2\pi \mathrm{i} k_j.$$

Combining results,

$$\oint_\Gamma \frac{f'(z)}{f(z)} \, \mathrm{d}z = \sum_{j=1}^{n} 2\pi \mathrm{i} k_j = 2\pi \mathrm{i} k.$$

Finally, parameterize Γ by $w = f(z)$. Then $\mathrm{d}w = f'(z)\,\mathrm{d}z$, and

$$k = \frac{1}{2\pi \mathrm{i}} \oint_\Gamma \frac{\mathrm{d}w}{w} = \frac{1}{2\pi \mathrm{i}} \Delta_\Gamma \log(w) = \frac{1}{2\pi} \Delta_\Gamma \arg(w),$$

which proves the result. □

Thus, one defines the *winding index* to quantify how many times a closed curve winds counterclockwise around a given point in the complex plane. Formally, if $\gamma = \gamma([0,1])$ is a counterclockwise-oriented closed curve and z is a point satisfying $z \neg \in \gamma$, then

$$\mathrm{Ind}_\Gamma(z) = \frac{1}{2\pi \mathrm{i}} \oint_\gamma \frac{\mathrm{d}\zeta}{\zeta - z} = \frac{1}{2\pi \mathrm{i}} \int_0^1 \frac{\gamma'(t)\,\mathrm{d}t}{\gamma(t) - z}.$$

Theorem 3.3.5: Let $\{f_n(z)\}$ be a sequence of holomorphic functions on the open set $U \subseteq \mathbb{C}$ that uniformly converges to $f(z)$ on every compact subset of U. If $\forall n \in \mathbb{N}$, $f_n(z)$ has no zeros in U, then f is either identically 0 or has no zeros in U.

Proof: By the holomorphy of $f_n(z)$, for any simple closed rectifiable curve $\gamma \subset U$ (whose interior is a subset of U), by the Cauchy–Goursat Theorem (Theorem 3.1.7),

$$\oint_\Gamma f_n(\zeta) \, \mathrm{d}\zeta = 0.$$

Since γ is a subset of any compact subset of U, $\{f_n(\zeta)\}$ uniformly converges on γ, and by Theorem 2.2.6,

$$\lim_{n\to\infty}\oint_\Gamma f_n(\zeta)\,\mathrm{d}\zeta = \oint_\Gamma \lim_{n\to\infty} f_n(\zeta)\,\mathrm{d}\zeta = \oint_\Gamma f(\zeta)\,\mathrm{d}\zeta = 0. \tag{3.3.2}$$

Then by Morera's Theorem (Theorem 3.2.4), $f(z)$ is holomorphic, and $f'(z)$ is holomorphic. We aim to show that $f_n'(z) \rightrightarrows f'(z)$.

Let $K \subset U$ be arbitrary and compact and $V \supset K$ be open and relatively compact in U. Since $\{f_n'(z)\}$ is holomorphic, by Corollary 3.2.5.1, there exists a finite constant $c > 0$ such that

$$\lim_{n\to\infty}\sup_{z\in K}|f_n'(z) - f'(z)| \le c \lim_{n\to\infty}\sup_{z\in V}|f_n(z) - f(z)|.$$

By the definition of uniform convergence, the right-hand side approaches 0, and $\{f_n'(z)\}$ is then uniformly convergent to $f'(z)$ by the same reasoning.

Through the proof of Theorem 3.3.2, if $f \neg \equiv 0$ over U, then the zeros of f do not have an accumulation point in U and are therefore discrete. In this case, let $\gamma \subset U$ be a curve that does not pass through the zeros of f. Since each function in the sequence f_n does not contain zeros in U, by the Argument Principle (Theorem 3.3.4),

$$\lim_{n\to\infty}\oint_\Gamma \frac{f_n'(z)}{f_n(z)}\,\mathrm{d}z = 0. \tag{3.3.3}$$

Since f and f' are holomorphic over γ, by Theorem 1.2.13, there exists a finite value $M > 0$ such that $\forall z \in \gamma$, $\max\{|f(z)|, |f'(z)|\} < M$.

Since γ does not pass through the zeros of f, $\exists \lambda > 0$ such that $\forall z \in \gamma$, $|f(z)| > \lambda$. By the uniform convergence of $\{f_n(z)\}$, $\exists N \in \mathbb{N}$ such that

$$|f_n(z) - f(z)| < \frac{\lambda}{2}, \quad \forall n > N, \forall z \in \gamma.$$

Then $|f_n(z)| > \frac{\lambda}{2}$ on γ. Hence, $\frac{1}{f_n(z)}$ and its limit are uniformly bounded;

$$\left|\frac{1}{f(z)}\right| < \frac{1}{\lambda}, \quad \left|\frac{1}{f_n(z)}\right| < \frac{2}{\lambda}, \quad \forall z \in \gamma, \forall n > N.$$

$$\begin{aligned}
\left|\frac{f'}{f} - \frac{f_n'}{f_n}\right| &= \left|\frac{f' f_n - f_n' f}{f_n f}\right| \\
&< 2\frac{|f' f_n - f' f| + |f' f - f_n' f|}{\lambda^2} \\
&< \frac{2M}{\lambda^2} \cdot (|f_n - f| + |f' - f_n'|).
\end{aligned}$$

$$\begin{aligned}
\sup_{z\in\gamma}\left|\frac{f'(z)}{f(z)} - \frac{f_n'(z)}{f_n(z)}\right| &\le \frac{2M}{\lambda^2}\left(\sup_{z\in\gamma}|f_n(z) - f(z)| + \sup_{z\in\gamma}|f'(z) - f_n'(z)|\right) \\
\lim_{n\to\infty}\sup_{z\in\gamma}\left|\frac{f'(z)}{f(z)} - \frac{f_n'(z)}{f_n(z)}\right| &\le \frac{2M}{\lambda^2}\left(\lim_{n\to\infty}\sup_{z\in\gamma}|f_n(z) - f(z)| + |f'(z) - f_n'(z)|\right) \\
&= 0.
\end{aligned}$$

Therefore, $\frac{f'(z)}{f(z)}$ is uniformly convergent on γ. By Theorem 2.2.6, we can pass the limit through the integral in (3.3.3). Then,

$$\lim_{n\to\infty} \oint_\gamma \frac{f_n'(z)}{f_n(z)}\,\mathrm{d}z = \oint_\gamma \frac{f'(z)}{f(z)}\,\mathrm{d}z = 0.$$

By the Argument Principle (Theorem 3.3.4), $f(z)$ has no zeros in the interior of γ. Since γ was arbitrarily chosen, either $f(z) \equiv 0$ on U or has no zeros in U. □

Theorem 3.3.6 *(Rouché)*: Let $U \subseteq \mathbb{C}$ be open and f, g be two holomorphic functions over U. Let $\gamma \subset U$ be a simple, closed, rectifiable curve, and for all $z \in \gamma$

$$|f(z) - g(z)| < |f(z)|. \tag{3.3.4}$$

Then f and g have the same number of zeros enclosed by γ and do not vanish on γ.

Proof: It is obvious that $g(z)$ has no zeros on γ. Otherwise, $\exists z_0 \in \gamma$ such that $g(z_0) = 0$, implying that $|f(z_0)| < |f(z_0)|$ which is impossible. Similarly, $f(z)$ has no zeros on γ, since $|g(z)| < 0$ is an impossibility.

Let k_f and k_g denote the number of zeros of f and g enclosed by γ, respectively. By the Argument Principle (Theorem 3.3.4),

$$\begin{aligned} k_g - k_f &= \oint_\Gamma \frac{g'(z)}{g(z)}\,\mathrm{d}z - \oint_\gamma \frac{f'(z)}{f(z)}\,\mathrm{d}z \\ &= \oint_\gamma \frac{g'(z)f(z) - f'(z)g(z)}{g(z)f(z)}\,\mathrm{d}z = \oint_\gamma \frac{\left(\frac{g(z)}{f(z)}\right)'}{\frac{g(z)}{f(z)}}\,\mathrm{d}z. \end{aligned}$$

Let $w = h(z) = \frac{g(z)}{f(z)}$ with $\Gamma = h(\gamma)$. Then,

$$k_g - k_f = \oint_\Gamma \frac{\mathrm{d}w}{w}.$$

From (3.3.4), by dividing both sides by $f(z)$, we obtain $|w - 1| < 1$. Then Γ lies in the open disk $D(1, 1)$, which will never intersect or enclose 0. Then by Lemma 3.1.2,

$$k_g - k_f = \oint_\Gamma \frac{\mathrm{d}w}{w} = 0,$$

as desired. □

By the Fundamental Theorem of Algebra (Theorem 3.3.1), any polynomial in the form $p(z) = \sum_{k=0}^{n} a_k z^k$ ($n \in \mathbb{N}$, $a_n \neq 0$, $a_k \in \mathbb{C}$ where $k = 1, \ldots n$) has at least one complex zero. Consider the function $q(z) = a_n z^n$, with a zero at $z = 0$ with multiplicity n. By Rouché's Theorem (Theorem 3.3.6), since $\exists R \in \mathbb{R}$ such that $|q(z) - p(z)| = \left|\sum_{k=0}^{n-1} a_k z^k\right| < |a_n z^n|$ over $|z| = R$, p and q have the same number of zeros, counting multiplicity.

Theorem 3.3.7: Let $U \subseteq \mathbb{C}$ be open and connected, and $f(z)$ be holomorphic and non-constant on U.

If $z_0 \in U$ and $w_0 = f(z_0)$, and the multiplicity of the zero at z_0 of $f - w_0$ is m, then for all $\rho > 0$ such that $f - w_0$ is non-vanishing on $\overline{D(z_0, \rho)} \setminus \{z_0\}$, $\exists \delta > 0$ such that $\forall \xi \in D(w_0, \delta)$, $f - \xi$ has m zeros in $D(z_0, \rho)$, counting multiplicity.

Proof: The zero at z_0 is isolated by Theorem 3.3.2. Furthermore, $|f - w_0|$ is continuous on $\partial D(z_0, \rho)$ and attains a positive infimum δ. In other words, on this set, $|f - w_0| \geq \delta$. Hence, $\forall \xi \in D(w_0, \delta)$, we have $|\xi - w_0| < \delta \leq |f(z) - w_0|$ for any $z \in \partial D(z_0, \rho)$.

By Rouché's Theorem, since $|(f(z) - w_0) - (f(z) - \xi)| < |f(z) - w_0|$, it follows that $f - \xi$ and $f - w_0$ have the same number of zeros in $D(z_0, \rho)$. □

We also have the following generalization of Theorem 3.3.5, which is a heuristic restatement of Theorem 3.3.7:

Theorem 3.3.8 *(Hurwitz)*: Let $U \subseteq \mathbb{C}$ be an open and connected set, and suppose $\{f_n(z)\}_{n\in\mathbb{N}}$ is a holomorphic function sequence that uniformly converges to a non-constant function $f(z)$ on all compact sets of U.

If $z_0 \in U$ and $w_0 = f(z_0)$, and the multiplicity of the zero at z_0 of $f - w_0$ is m, then for all $\rho > 0$ such that $f - w_0$ is non-vanishing on $\overline{D(z_0, \rho)} \setminus \{z_0\}$, $\exists N \in \mathbb{N}$ such that $\forall n > N$, $f_n - w_0$ has m zeros in $D(z_0, \rho)$, counting multiplicity.

Proof: The zero at z_0 is isolated by Theorem 3.3.2. Furthermore, $|f - w_0|$ is continuous on $\partial D(z_0, \rho)$ and attains a positive infimum δ. In other words, on this set, $|f - w_0| \geq \delta$. By uniform convergence, $\exists N \in \mathbb{N}$ such that $\forall n > N$, we have $|f(z) - f_n(z)| < \delta \leq |f(z) - w_0|$ for any $z \in \partial D(z_0, \rho)$.

By Rouché's Theorem (Theorem 3.3.6), since

$$|(f(z) - w_0) - (f_n(z) - w_0)| < |f(z) - w_0|,$$

it follows that $f_n - w_0$ and $f - w_0$ have the same number of zeros in $D(z_0, \rho)$. □

3.4 Further Properties of Holomorphic Functions

A useful corollary of Theorem 3.1.8 is the Maximum Modulus Principle.

Before the theorem, we first introduce the mean-value property of holomorphic functions.

Lemma 3.4.1: Let $U \subseteq \mathbb{C}$ be open and simply connected, and let $f : U \to \mathbb{C}$ be holomorphic. Then $\forall z \in U$ and $\forall \varepsilon > 0$ such that $\overline{D(z, \varepsilon)} \subset U$, $f(z)$ is the average of $f(\zeta)$ where $\zeta \in D(z, \varepsilon)$ is uniform. In other words,

$$f(z) = \frac{1}{2\pi\varepsilon} \oint_{\partial D(z,\varepsilon)} f(\zeta) |\mathrm{d}\zeta|.$$

Proof: By the Cauchy–Goursat Formula (Theorem 3.1.8),

$$f(z) = \frac{1}{2\pi \mathrm{i}} \oint_{\partial D(z,\varepsilon)} \frac{f(\zeta)}{\zeta - z} \,\mathrm{d}\zeta = \frac{1}{2\pi} \int_0^{2\pi} f\left(z + \varepsilon \mathrm{e}^{\mathrm{i}t}\right) \mathrm{d}t.$$

Observe that

$$\begin{aligned} f(z) = \frac{1}{2\pi\varepsilon} \oint_{\partial D(z,\varepsilon)} f(\zeta) |\mathrm{d}\zeta| &= \frac{1}{2\pi\varepsilon} \int_0^{2\pi} f\left(z + \varepsilon \mathrm{e}^{\mathrm{i}t}\right) \left|\mathrm{i}\varepsilon \mathrm{e}^{\mathrm{i}t} \,\mathrm{d}t\right| \\ &= \frac{1}{2\pi} \int_0^{2\pi} f\left(z + \varepsilon \mathrm{e}^{\mathrm{i}t}\right) \mathrm{d}t, \end{aligned}$$

and the conclusion follows. □

Since the real and imaginary parts of holomorphic functions are real-valued harmonic functions, they also satisfy the mean-value property. Furthermore, if a real continuous function satisfies the mean-value property, it is harmonic (to be proved in @ thm:meanvaluepropertysolutionsareharmonic). This equivalence allows for the alternative definition of harmonic functions.

Theorem 3.4.1 *(Maximum Modulus Principle)*: Let $f(z)$ be holomorphic on an open connected region $U \subseteq \mathbb{C}$. If $\exists z_0 \in U$ and an open neighborhood $V \subseteq U$ of z_0 such that $\forall z \in V$, $|f(z_0)| \geq |f(z)|$, then f is a constant function on U.

Proof: Assume that z_0 exists. We will first prove that the set

$$S = \{z \mid f(z) = f(z_0), z \in V\}$$

is all of V. This is equivalent to proving that S is nonempty, open, and closed in V.

Since $z_0 \in S$, the first condition is satisfied (nonemptiness). For any sequence $\{z_n\} \in S$ converging to some $z_\infty \in V$, by the continuity of f,

$$\lim_{n\to\infty} f(z_n) = f\left(\lim_{n\to\infty} z_n\right) = f(z_\infty) = f(z_0),$$

and $z_\infty \in S$. Thus, S contains all of its accumulation points in V and is therefore closed (if $z_\infty \neg \in V$, then it is no longer relevant; we are concerned with its relative closedness in V).

Since $S \subseteq V$ and V are both open, $\forall z \in S$, $\exists \lambda > 0$ such that $D(z, \lambda) \subseteq V$. By Lemma 3.4.1, $\forall 0 < \varepsilon < \lambda$,

$$\begin{aligned}|f(z)| &= \left|\frac{1}{2\pi}\int_0^{2\pi} f(z + \varepsilon e^{it})\,dt\right| \leq \frac{1}{2\pi}\int_0^{2\pi} \left|f(z + \varepsilon e^{it})\right| dt \\ &\leq \frac{1}{2\pi}\int_0^{2\pi} |f(z)|\,dt = |f(z)|.\end{aligned}$$

It follows that all inequalities above are equalities, or that

$$\begin{aligned}|f(z)| &= \left|\frac{1}{2\pi}\int_0^{2\pi} f(z + \varepsilon e^{it})\,dt\right| = \frac{1}{2\pi}\int_0^{2\pi} \left|f(z + \varepsilon e^{it})\right| dt \\ &= \frac{1}{2\pi}\int_0^{2\pi} |f(z)|\,dt = |f(z)|.\end{aligned}$$

From the equality of the last two integrals,

$$\int_0^{2\pi} \left[|f(z)| - \left|f(z + \varepsilon e^{it})\right|\right] dt = 0.$$

Since this integrand is strictly non-negative, we have equality. Thus, $\forall z \in S$, $\exists \lambda > 0$ such that $D(z, \lambda) \subseteq S$. In other words, every $z \in S$ has an open neighborhood that also lies in S. Therefore, S is open and $S = V$ as it is a nonempty clopen subset. Since V is nonempty and open, it has an accumulation point in U. It follows that $f(z) \equiv f(z_0)$ over U by the Identity Theorem (Theorem 3.3.3). □

Remark: If f is holomorphic and non-constant on an open region $U \subseteq \mathbb{C}$, then for any compact set $K \subset U$, the maximum of f in K lies on ∂K. Otherwise, f would attain a maximum at some $z \in \mathring{K}$, and contradict the statement of Theorem 3.4.1 under the assumption of being non-constant.

A similar theorem exists for real-valued harmonic functions. The proof follows in the same way as the one for holomorphic functions. We will state it formally below.

Theorem 3.4.2: Let $U \subseteq \mathbb{C}$ be open and connected and let $f : U \to \mathbb{R}$ be harmonic. Suppose that $\exists z_0 \in U$ and a neighborhood $V \subseteq U$ of z_0 such that either

$$f(z) \geq f(z_0) \quad \forall z \in V \quad \text{or} \quad f(z_0) \geq f(z) \quad \forall z \in V.$$

Then f is constant on U.

By nature of the proof, the result holds for any continuous function satisfying the mean-value property.

3.5 The Group of Holomorphic Automorphisms on the Unit Disk

The following important result can be directly obtained from the Maximum Modulus Principle.

Lemma 3.5.1 *(Schwarz)*: If $f : \mathbb{D} \to \mathbb{D}$ is holomorphic and $f(0) = 0$, then

$$|f(z)| \leq |z|, \quad |f'(0)| \leq 1.$$

Any one of the inequalities becomes equalities iff $f(z)$ is in the form of $z\mathrm{e}^{\mathrm{i}\tau}$, where $\tau \in \mathbb{R}$. In other words, f is a pure rotation.

Proof: Define the auxiliary function

$$g(z) = \begin{cases} \frac{f(z)}{z} & \text{if } z \neq 0 \\ f'(0) & \text{if } z = 0. \end{cases}$$

Because $\lim_{z\to 0} \frac{f(z)}{z} = f'(0)$, $g(z)$ is holomorphic on $\mathbb{D}$. Since f is an automorphism on the open disk, $\forall |z| < 1$, $|f(z)| < 1$. By the Maximum Modulus Principle (Theorem 3.4.1), $\forall 0 < \varepsilon < 1$, $\forall z \in D(0, \varepsilon)$,

$$|g(z)| \leq \max_{z_\varepsilon \in \partial D(0,\varepsilon)} \frac{|f(z_\varepsilon)|}{\varepsilon} < \frac{1}{\varepsilon}.$$

As $\varepsilon \to 1^-$, we obtain that $\forall z \in \mathbb{D}$, $|g(z)| \leq 1$, or that $|f(z)| \leq |z|$. Let $z = 0$. Then we get $|g(0)| = |f'(0)| \leq 1$.

For the sake of the equality condition, assume $|f(z)| = |z|$. Then $|g(z)| \equiv 1$ on the unit open disk. By Theorem 3.4.1, $g(z) = \mathrm{e}^{\mathrm{i}\tau}$ where $\tau \in \mathbb{R}$ and $f(z) = z\mathrm{e}^{\mathrm{i}\tau}$ on $\mathbb{D}$.

Next, assume only that $|f'(0)| = 1$. It follows that $|g(0)| = 1$. Since $|g(z)| \leq 1$ for all $z \in \mathbb{D}$, it follows from Theorem 3.4.1 that g is constant with magnitude 1, or in the form of $\mathrm{e}^{\mathrm{i}\tau}$, where $\tau \in \mathbb{R}$ is a constant. Consequently, $f(z) = z\mathrm{e}^{\mathrm{i}\tau}$. □

To discuss the main topic of this section, we will first introduce the concept of a *group*.

Definition 3.5.1 *(Group)*: A group is a nonempty set G and a binary operation (we will denote this as $*$) satisfying the four *group axioms*:

1. Closure: $\forall a, b \in G$, $a * b \in G$.
2. Associativity: $\forall a, b, c \in G$, $(a * b) * c = a * (b * c)$.
3. Identity Element: $\exists e \in G$ such that $\forall a \in G$, $a * e = e * a = a$. Note that e is unique; if $e, f \in G$ were both identity elements, then $e * f = f * e = e = f$, and are equal.
4. Inverse Element: $\forall a \in G$, $\exists a^{-1} \in G$ such that $a * a^{-1} = e = a^{-1} * a$, where e is the identity element. Note that a^{-1} is unique. Assume b, c were both inverses of a. Then, $b = b * e = b * (a * c) = (b * a) * c = c$, and are equal.

A *subgroup* H of G is a subset of G that is also a group under the same operation as G. This relationship is denoted by $H \leq G$ or $H < G$ for *proper subgroups.*

Group operations are not necessarily commutative. In the case that they are, (specifically if $a, b \in G \Rightarrow a * b = b * a$), then G is an *abelian group.*

If $U \subseteq \mathbb{C}$ is connected and $f : U \to U$ is holomorphic on U and bijective, f is a *holomorphic automorphism* on U. The *group of holomorphic automorphisms* on U is denoted by $\operatorname{Aut}(U)$, which is the set of all holomorphic automorphisms such as f, with the operation of composition $(f \circ g)$.

First we will show that $\forall a \in \mathbb{D}$,

$$\varphi_{a(z)} = \frac{z-a}{1-\overline{a}z} \in \operatorname{Aut}(\mathbb{D}). \tag{3.5.1}$$

Firstly, the function is holomorphic on $\mathbb{D}$ as $|z| \leq 1$, $|\overline{a}| < 1$, the denominator never vanishes. Additionally, $\varphi_{a(a)} = 0$.

First, we will observe the image of $\partial\mathbb{D}$. Let $|z| = 1$. Then,

$$\left|\varphi_{a(z)}\right| = \left|\frac{1}{z}\right|\left|\frac{z-a}{\frac{1}{z}-\overline{a}}\right| = \left|\frac{z-a}{\overline{z}-\overline{a}}\right| = 1.$$

Therefore, the image of $\partial\mathbb{D}$ lies on $\partial\mathbb{D}$, and since f is holomorphic and non-constant, by the Maximum Modulus Principle (Theorem 3.4.1), for any $|z| < 1$, $\left|\varphi_{a(z)}\right| < 1$. Therefore, f maps $\mathbb{D}$ to $\mathbb{D}$. We next aim to show that $f : \mathbb{D} \to \mathbb{D}$ is bijective.

Let us first confirm injectivity. For all $z_1, z_2 \in \mathbb{D}$, we will observe when

$$\frac{z_1 - a}{1 - \overline{a}z_1} = \frac{z_2 - a}{1 - \overline{a}z_2}$$

is satisfied. It follows that

$$\begin{gathered} (z_1 - a)(1 - \overline{a}z_2) = (z_2 - a)(1 - \overline{a}z_1), \\ z_1 - a - \overline{a}z_1z_2 + |a|^2 z_2 = z_2 - a - \overline{a}z_1z_2 + |a|^2 z_1. \end{gathered}$$

Then,

$$|a|^2(z_2 - z_1) = z_2 - z_1 \Longleftrightarrow (|a|^2 - 1)(z_2 - z_1) = 0.$$

Since $|a| < 1$, then $|a|^2 - 1 \neq 0$, and we get $z_2 - z_1 = 0$. This proves the univalence of $\varphi_{a(z)}$.

Next, we will solve for the inverse of φ_a. Let $z = \varphi_{a(w)} = \frac{w-a}{1-\overline{a}w}$. Then,

$$z - \overline{a}zw = w - a \Longleftrightarrow w = \frac{z+a}{1+\overline{a}z}. \tag{3.5.2}$$

It follows that $\varphi_{-a} = (\varphi_a)^{-1}$. Thus φ_a is surjective and a bijective automorphism. It follows that (3.5.1) is true. Functions in the form of φ_a (where $a \in \mathbb{D}$) are known as *Möbius transformations*, and the group of all such transformations is known as the *Möbius transformation group on the unit disk*, which is a subgroup of $\operatorname{Aut}(\mathbb{D})$. Functions in the form of $\rho_{\tau(z)} = ze^{\mathrm{i}\tau}$, where $\tau \in \mathbb{R}$ is constant, form a group known as the *rotation group*, which is also a subgroup of $\operatorname{Aut}(\mathbb{D})$.

Theorem 3.5.1 *(The Holomorphic Automorphism Group on $\mathbb{D}$)*: $\forall f \in \operatorname{Aut}(\mathbb{D})$, f is a composition between a Möbius transformation and a rotation. In other words, $\exists |a| < 1$ and $\exists \tau \in \mathbb{R}$ such that

$$f(z) = \varphi_a \circ \rho_{\tau(z)}.$$

Moreover, all such functions are in $\operatorname{Aut}(\mathbb{D})$.

Proof: Define the auxiliary function

$$\psi(z) = \varphi_{f(0)} \circ f(z).$$

It follows that $\psi \in \mathrm{Aut}(\mathbb{D})$. Furthermore,

$$\psi(0) = \varphi_{f(0)} \circ f(0) = 0.$$

By the Schwarz Lemma (Lemma 3.5.1), $|\psi'(0)| \leq 1$. Since $\psi^{-1} \in \mathrm{Aut}(\mathbb{D})$ with $\psi^{-1}(0) = 0$, $\left|\left(\psi^{-1}\right)'(0)\right| \leq 1$. Then,

$$\left|\left(\psi^{-1}\right)'(0)\right| = \left|\frac{1}{\psi'\left(\psi^{-1}(0)\right)}\right| = \left|\frac{1}{\psi'(0)}\right| \leq 1.$$

Then, $|\psi'(0)| = 1$, and by the equality statement of Lemma 3.5.1,

$$\psi(z) = z\mathrm{e}^{\mathrm{i}\tau} = \rho_{\tau(z)}$$

for some constant $\tau \in \mathbb{R}$, and

$$f(z) = \varphi_{f(0)}^{-1} \circ \rho_{\tau(z)}.$$

By (3.5.2), it follows that

$$f(z) = \varphi_{-f(0)} \circ \rho_{\tau(z)}.$$

□

As a direct consequence of Theorem 3.5.1, we have the following result:

Lemma 3.5.2 *(Schwarz–Pick)*: Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic. For all $z_1, z_2 \in \mathbb{D}$, let $w_1 = f(z_1)$ and $w_2 = f(z_2)$. Then,

$$\left|\frac{w_1 - w_2}{1 - w_1\overline{w_2}}\right| \leq \left|\frac{z_1 - z_2}{1 - z_1\overline{z_2}}\right|. \tag{3.5.3}$$

and

$$\frac{|\mathrm{d}w|}{1 - |w|^2} \leq \frac{|\mathrm{d}z|}{1 - |z|^2}. \tag{3.5.4}$$

The equalities hold iff $f \in \mathrm{Aut}(\mathbb{D})$.

Proof: Let

$$\varphi_{-z_1}(z) = \frac{z + z_1}{1 + \overline{z_1}z} \in \mathrm{Aut}(\mathbb{D}), \quad \varphi_{w_1}(z) = \frac{z - w_1}{1 - \overline{w_1}z} \in \mathrm{Aut}(\mathbb{D}).$$

It follows that

$$\varphi_{w_1} \circ f \circ \varphi_{-z_1}(0) = \varphi_{w_1}(w_1) = 0.$$

Then by the Schwarz Lemma (Lemma 3.5.1), for $z \in \mathbb{D}$,

$$\left|\varphi_{w_1} \circ f \circ \varphi_{-z_1}(z)\right| \leq |z|.$$

Let $z_2 = \varphi_{-z_1}(z)$. Then,

$$\left|\varphi_{w_1} \circ f(z_2)\right| \leq \left|\varphi_{z_1}(z_2)\right| \Longleftrightarrow \left|\varphi_{w_1}(w_2)\right| \leq \left|\varphi_{z_1}(z_2)\right|,$$

confirming (3.5.3). By the second statement of the Schwarz Lemma (Lemma 3.5.1), $\left|\left(\varphi_{w_1} \circ f \circ \varphi_{-z_1}\right)'(0)\right| \leq 1$.

By the chain rule,

$$\left|\varphi_{(w_1)'}(w_1) f'(z_1) \varphi_{(-z_1)'}(0)\right| \leq 1.$$

Let us now calculate the derivatives of φ_{w_1} and φ_{-z_1}. By the quotient rule,

$$\varphi'_{w_1}(z) = \frac{1 - \overline{w_1} w_1}{(1 - \overline{w_1} z)^2}, \quad \varphi'_{w_1}(w_1) = \frac{1}{1 - \overline{w_1} w_1},$$

and

$$\varphi'_{-z_1}(z) = \frac{1 - \overline{z_1} z_1}{(1 + \overline{z_1} z)^2}, \quad \varphi'_{-z_1}(0) = 1 - \overline{z_1} z_1.$$

Since both derivatives are positive,

$$|f'(z_1)| \leq \frac{1 - \overline{w_1} w_1}{1 - \overline{z_1} z_1}.$$

Since $z_1 \in \mathbb{D}$ is arbitrary, it follows that

$$\left|\frac{\mathrm{d}w}{\mathrm{d}z}\right| \leq \frac{1 - \overline{w} w}{1 - \overline{z} z} \Longleftrightarrow \frac{|\mathrm{d}w|}{1 - \overline{w} w} \leq \frac{|\mathrm{d}z|}{1 - \overline{z} z}. \tag{3.5.5}$$

By the Schwarz Lemma (Lemma 3.5.1), under the equality condition that

$$\left|\varphi_{(w_1)'}(w_1) f'(z_1) \varphi_{(-z_1)'}(0)\right| = 1,$$

we have that

$$\varphi_{w_1} \circ f \circ \varphi_{-z_1} = \mathrm{e}^{\mathrm{i}\tau},$$

where $\tau \in \mathbb{R}$ is constant. It follows that

$$f = \varphi_{-w_1} \circ \mathrm{e}^{\mathrm{i}\tau} \circ \varphi_{z_1} \in \mathrm{Aut}(\mathbb{D}).$$

□

Remark: In @ sec:differentialgeometry, we will introduce the *hyperbolic metric* on $\mathbb{D}$, defined as

$$\mathrm{d}s^2 = \frac{4|\mathrm{d}z|^2}{(1 - |z|^2)^2}.$$

From (3.5.5), we get that the hyperbolic metric does not increase under a holomorphic mapping of $\mathbb{D}$ to itself. This metric is invariant (the equality condition) under all functions in $\mathrm{Aut}(\mathbb{D})$. This gives a geometric explanation for Lemma 3.5.1.

3.6 Alternative Integral Formulas

As in the Cauchy Integral Formula (Theorem 3.1.8), we can write holomorphic functions in terms of an integral representation. We define the *Cauchy kernel* to be

$$H(\zeta, z) = \frac{1}{2\pi\mathrm{i}(\zeta - z)}.$$

Then Theorem 3.1.8 can be written as

$$f(z) = \oint_{\partial U} f(\zeta) H(\zeta, z)\, \mathrm{d}\zeta.$$

There also exist other integral formulas for functions, varying in the kernel of the expression.

Let $\Phi : \mathbb{D} \to \mathbb{R}$ be harmonic such that Φ is continuous on $\overline{\mathbb{D}}$. By the mean-value property introduced in Lemma 3.4.1, we have

$$\Phi(0) = \frac{1}{2\pi} \int_0^{2\pi} \Phi(\rho \mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t,$$

where $0 < \rho < 1$. By the uniform continuity of Φ on $\overline{\mathbb{D}}$ (Theorem 1.2.15), $\forall \varepsilon > 0, \exists \delta > 0$ such that for all $\rho \in \left(\frac{1}{2}, 1\right)$ satisfying $1 - \rho < \delta$ and all $t \in [0, 2\pi]$,

$$\left|\Phi(\mathrm{e}^{\mathrm{i}t}) - \Phi(\rho \mathrm{e}^{\mathrm{i}t})\right| < \varepsilon.$$

It then follows that

$$\left|\frac{1}{2\pi} \int_0^{2\pi} \Phi(\mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t - \frac{1}{2\pi} \int_0^{2\pi} \Phi(\rho \mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t\right| < \varepsilon.$$

Hence,

$$\lim_{\rho \to 1^-} \frac{1}{2\pi} \int_0^{2\pi} \Phi(\rho \mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t = \frac{1}{2\pi} \int_0^{2\pi} \Phi(\mathrm{e}^{\mathrm{i}t})\,\mathrm{d}t = \Phi(0). \tag{3.6.1}$$

Let $z \in \mathbb{D}$ and notice that

$$\varphi_z(\zeta) = \frac{\zeta - z}{1 - \overline{z}\zeta} \in \operatorname{Aut}(\mathbb{D})$$

maps $\partial\mathbb{D}$ to $\partial\mathbb{D}$ bijectively. Let u be harmonic on $\mathbb{D}$ and continuous on $\overline{\mathbb{D}}$. Then $u \circ \varphi_{-z}$ is also harmonic on $\mathbb{D}$, and by (3.6.1),

$$u(z) = u \circ \varphi_{-z}(0) = \frac{1}{2\pi} \int_0^{2\pi} u \circ \varphi_{-z}(\mathrm{e}^{\mathrm{i}\psi})\,\mathrm{d}\psi.$$

By the univalence of φ_z, let $\mathrm{e}^{\mathrm{i}\psi} = \varphi_z(\mathrm{e}^{\mathrm{i}t})$. It follows that

$$\mathrm{i}\mathrm{e}^{\mathrm{i}\psi}\,\mathrm{d}\psi = \mathrm{i}\frac{1 - \overline{z}z}{\left(1 - \overline{z}\mathrm{e}^{\mathrm{i}t}\right)^2}\mathrm{e}^{\mathrm{i}t}\,\mathrm{d}t \tag{3.6.2}$$

$$\mathrm{d}\psi = \frac{1 - \overline{z}z}{\left(1 - \overline{z}\mathrm{e}^{\mathrm{i}t}\right)^2}\frac{1 - \overline{z}\mathrm{e}^{\mathrm{i}t}}{\mathrm{e}^{\mathrm{i}t} - z}\mathrm{e}^{\mathrm{i}t}\,\mathrm{d}t \tag{3.6.3}$$

$$= \frac{1 - |z|^2}{|1 - \overline{z}\mathrm{e}^{\mathrm{i}t}|^2}\,\mathrm{d}t. \tag{3.6.4}$$

Then from (3.6.1),

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(\mathrm{e}^{\mathrm{i}t})\frac{1 - |z|^2}{|1 - \overline{z}\mathrm{e}^{\mathrm{i}t}|^2}\,\mathrm{d}t.$$

Let

$$P(\zeta, z) = \frac{1 - |z|^2}{2\pi|1 - \overline{z}\zeta|^2} = \frac{1 - |z|^2}{2\pi|\zeta - z|^2},$$

known as the *Poisson kernel*. Then,

$$u(z) = \int_0^{2\pi} u(\zeta) P(\zeta, z) \, \mathrm{d}t, \tag{3.6.5}$$

where $\zeta = \mathrm{e}^{\mathrm{i}t}$. (3.6.5) is also known as the *Poisson Integral Formula*.

For all $z \in D(0, R)$, where $R > 0$, we can apply the transformation

$$\tilde{\varphi}_z(\zeta) = R\varphi_{\frac{z}{R}}\left(\frac{\zeta}{R}\right)$$

to extend the automorphism to $D(0, R)$. Let u instead be harmonic on $D(0, R)$ and continuous on $\overline{D(0, R)}$. Then,

$$u(0) = \frac{1}{2\pi} \int_0^{2\pi} u\big(R\mathrm{e}^{\mathrm{i}\psi}\big) \, \mathrm{d}\psi.$$

It follows that $u \circ \tilde{\varphi}_{-z}$ is also harmonic on $D(0, R)$ with

$$u(z) = u \circ \tilde{\varphi}_{-z}(0),$$

and from the bijectivity of $R\mathrm{e}^{\mathrm{i}\psi} = \tilde{\varphi}_z\big(R\mathrm{e}^{\mathrm{i}t}\big)$,

$$\begin{aligned}
\mathrm{d}\psi &= \frac{1 - \frac{|z|^2}{R^2}}{\big(1 - \frac{\overline{z}}{R}\mathrm{e}^{\mathrm{i}t}\big)^2} \mathrm{e}^{\mathrm{i}t} \mathrm{e}^{-\mathrm{i}\psi} \, \mathrm{d}t &(3.6.6)\\
&= \frac{1 - \frac{|z|^2}{R^2}}{\big(1 - \frac{\overline{z}}{R}\mathrm{e}^{\mathrm{i}t}\big)^2} \frac{1 - \frac{\overline{z}}{R}\mathrm{e}^{\mathrm{i}t}}{1 - \frac{z}{R}\mathrm{e}^{-\mathrm{i}t}} \, \mathrm{d}t &(3.6.7)\\
&= \frac{R^2 - |z|^2}{\left|R\mathrm{e}^{\mathrm{i}t} - z\right|^2} \, \mathrm{d}t. &(3.6.8)
\end{aligned}$$

Then because $\tilde{\varphi}_z^{-1} = \tilde{\varphi}_{-z}$,

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u\big(R\mathrm{e}^{\mathrm{i}t}\big) \frac{R^2 - |z|^2}{\left|R\mathrm{e}^{\mathrm{i}t} - z\right|^2} \, \mathrm{d}t.$$

The expression

$$P(\zeta, z) = \frac{|\zeta|^2 - |z|^2}{2\pi|\zeta - z|^2}$$

is a general form of the Poisson kernel. Then with $\zeta = R\mathrm{e}^{\mathrm{i}t}$,

$$u(z) = \int_0^{2\pi} u(\zeta) P(\zeta, z) \, \mathrm{d}t. \tag{3.6.9}$$

The Poisson kernel can also be rewritten as

$$P(\zeta, z) = \frac{|\zeta|^2 - |z|^2}{2\pi(\zeta - z)\left(\overline{\zeta} - \overline{z}\right)} \tag{3.6.10}$$

$$= \frac{1}{4\pi}\left(\frac{\zeta + z}{\zeta - z} + \frac{\overline{\zeta} + \overline{z}}{\overline{\zeta} - \overline{z}}\right) \tag{3.6.11}$$

$$= \frac{1}{2\pi}\,\mathfrak{Re}\left(\frac{\zeta + z}{\zeta - z}\right). \tag{3.6.12}$$

Thus, (3.6.9) is equivalent to

$$u(z) = \frac{1}{2\pi}\int_0^{2\pi} u(\zeta)\,\mathfrak{Re}\left(\frac{\zeta + z}{\zeta - z}\right)\mathrm{d}t.$$

Since $\mathrm{d}\zeta = \mathrm{i}\zeta\,\mathrm{d}t$, $\mathrm{d}t = \frac{\mathrm{d}\zeta}{\mathrm{i}\zeta}$, and

$$u(z) = \frac{1}{2\pi\mathrm{i}}\oint_{\partial D(0,R)} \frac{u(\zeta)}{\zeta}\,\mathfrak{Re}\left(\frac{\zeta + z}{\zeta - z}\right)\mathrm{d}\zeta = \mathfrak{Re}\left(\frac{1}{2\pi\mathrm{i}}\oint_{\partial D(0,R)} \frac{u(\zeta)}{\zeta}\frac{\zeta + z}{\zeta - z}\,\mathrm{d}\zeta\right),$$

where $z \in D(0, R)$. Since $R > 0$ and $\zeta - z \neq 0$ for all $\zeta \in \partial D(0, R)$ and $z \in D(0, R)$, the function

$$F(z) = \frac{1}{2\pi\mathrm{i}}\oint_{\partial D(0,R)} \frac{u(\zeta)}{\zeta}\frac{\zeta + z}{\zeta - z}\,\mathrm{d}\zeta$$

is holomorphic on $D(0, R)$: for each fixed $\zeta \in \partial D(0, R)$, the integrand is holomorphic in z, and on compact subsets of $D(0, R)$ we may differentiate under the integral sign. Therefore, $u(z)$ is the real part of a holomorphic function

$$f(z) = \frac{1}{2\pi\mathrm{i}}\oint_{\partial D(0,R)} \frac{u(\zeta)}{\zeta}\frac{\zeta + z}{\zeta - z}\,\mathrm{d}\zeta + \mathrm{i}c,$$

where $c \in \mathbb{R}$. Since $c \in \mathbb{R}$ is holomorphic, by Proposition 2.1.1, c is constant. For $f(z) = u(z) + \mathrm{i}v(z)$,

$$v(z) = c + \frac{1}{2\pi}\int_0^{2\pi} u(\zeta)\,\mathfrak{Im}\left(\frac{\zeta + z}{\zeta - z}\right)\mathrm{d}t. \tag{3.6.13}$$

Letting $z = 0$, the integral vanishes, and we obtain $c = v(0) = \mathfrak{Im}(f(0))$.

Define the *Schwarz kernel* to be

$$S(\zeta, z) = \frac{\zeta + z}{2\pi\mathrm{i}(\zeta - z)\zeta}.$$

Then for a holomorphic function f on $D(0, R)$ that is continuous on $\overline{D(0, R)}$, we obtain the *Schwarz Integral Formula*:

$$f(z) = \oint_{\partial D(0,R)} \mathfrak{Re}(f(\zeta))S(\zeta, z)\,\mathrm{d}\zeta + \mathrm{i}\,\mathfrak{Im}(f(0)). \tag{3.6.14}$$

The significance of this alternative formula implies that a holomorphic function can be recovered from the real part on the boundary of a disk and the imaginary part at a single point.

From (3.6.13), we can rewrite

$$\mathfrak{Im}\left(\frac{\zeta + z}{\zeta - z}\right) = \mathfrak{Im}\left(1 + \frac{2z}{\zeta - z}\right) \tag{3.6.15}$$

$$= \mathfrak{Im}\left(\frac{2z\left(\overline{\zeta} - \overline{z}\right)}{|\zeta - z|^2}\right) \tag{3.6.16}$$

$$= \frac{2\,\mathfrak{Im}\left(z\overline{\zeta}\right)}{|\zeta - z|^2}. \tag{3.6.17}$$

Let

$$Q(\zeta, z) = \frac{\mathfrak{Im}\left(z\overline{\zeta}\right)}{\pi|\zeta - z|^2},$$

which is known as the *conjugate Poisson kernel.* Then (3.6.13) yields yet another integral representation of harmonic functions:

$$v(z) = v(0) + \int_0^{2\pi} u(\zeta)Q(\zeta, z)\,\mathrm{d}t.$$

where $\zeta = R\mathrm{e}^{\mathrm{i}t}$. Two harmonic functions are said to be *conjugate* if they are the real and imaginary parts of a holomorphic function. As seen above, on open disks, any harmonic function will admit a unique conjugate, up to an additive constant $v(0)$. For a harmonic function u, we can construct its harmonic conjugate from (3.6.15).

The Poisson kernel is important in many branches of mathematics. We will introduce two of the important uses below.

Bibliography

[1] S. Gong and Y. Gong, *Concise Complex Analysis (Revised Edition).* World Scientific Publishing Company, 2007. [Online]. Available: https://books.google.com/books?id=xBE8DQAAQBAJ

[2] S. Gong and Y. Gong, *Concise Calculus.* World Scientific, 2017. [Online]. Available: https://books.google.com/books?id=uXb1QwAACAAJ

[3] R. Greene and S. Krantz, *Function Theory of One Complex Variable*, 3rd ed., vol. 40. in Graduate Studies in Mathematics, vol. 40. American Mathematical Society, 2006. [Online]. Available: https://books.google.com/books?id=u5vhseYCcqkC

[4] J. Conway, *Functions of One Complex Variable I.* in Functions of one complex variable / John B. Conway. Springer, 1978. [Online]. Available: https://books.google.com/books?id=9LtfZr1snG0C

[5] M. Bonk, "On Bloch's Constant," *Proceedings of the American Mathematical Society*, vol. 110, no. 4, pp. 889–894, 1990, [Online]. Available: http://www.jstor.org/stable/2047734

[6] Encyclopedia of Mathematics, "Landau theorems." [Online]. Available: https://encyclopediaofmath.org/wiki/Landau_theorems

[7] Encyclopedia of Mathematics, "Reinhardt domain." [Online]. Available: https://encyclopediaofmath.org/wiki/Reinhardt_domain

[8] L. V. Ahlfors, "An Extension of Schwarz's Lemma," *Transactions of the American Mathematical Society*, vol. 43, no. 3, pp. 359–364, 1938, [Online]. Available: http://www.jstor.org/stable/1990065

[9] T. W. Gamelin, *Complex Analysis.* in Undergraduate Texts in Mathematics. Springer, 2001. [Online]. Available: https://link.springer.com/book/10.1007/978-0-387-21607-2

[10] A. Holland and H. B, *Introduction to the Theory of Entire Functions*. in Educational Psychology. Academic Press, 1973. [Online]. Available: https://books.google.com/books?id=KXTZAQAACAAJ

[11] K. S. Charak, "Value Distribution Theory of Meromorphic Functions," *Mathematics Newsletter*, vol. 18, no. 4, 2009, [Online]. Available: https://www.isibang.ac.in/~sury/ramanujanews.pdf

[12] W. Hayman, *Meromorphic Functions*. in Oxford mathematical monographs. Clarendon Press, 1964. [Online]. Available: https://books.google.com/books?id=dTDvAAAAMAAJ

[13] H.-Y. Xu and H. Wang, "Nevanlinna's Five Values Theorem on Annuli," *Mathematics*, vol. 4, no. 1, p. 7, 2016, doi: 10.3390/math4010007.